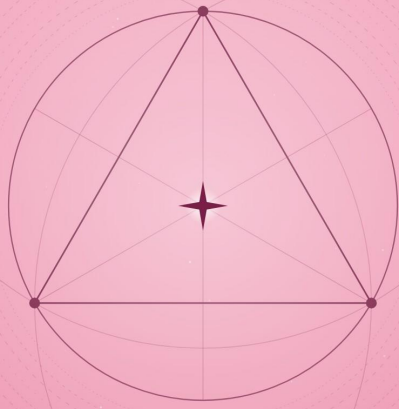
FIRST PRINCIPLES

About Thinking Differently



RON TAYLOR

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First Principles

52 Stories About Thinking Differently

Ron Taylor

An Executive Class Series

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First Principles
52 Stories About Thinking Differently

An Executive Class Series by Ron Taylor

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How to Use This Book

This book tells fifty-two true stories about people who solved a problem by thinking differently: an engineer who looked closely at a burr on his dog, a chemist who kept a glue that failed, a company that asked a better question than its rivals. Each story is short enough to read in one sitting and rich enough to discuss at work or in an interview.

How each story works

Documentation Level. Every story opens with a label. **Level 1, Directly Documented,** means the story is verified through primary sources, patents or independent accounts from the time. **Level 2** means the account rests mainly on the protagonist's own retelling, or the idea was documented but not used as first intended; a Level 2 marked *Mixed* contains some well-documented claims and some contested ones, and says which is which.

Executive Class Lens turns the story into a question about your own work. **Key Takeaways** and the **Summary** give you the main points in plain English.

Story Check separates what is *established*, what the *popular version adds*, and what remains *uncertain*. Many famous business stories improve with each retelling; this section shows you where.

Vocabulary & Phrasal Verbs and **Practice** give you the language to talk about the story. **Sources** are graded: **A** for primary sources such as patents, papers or the person's own writing; **B** for books and long-form reporting; **C** for reference works and press coverage; **D** for popular retellings.

This Connects To links each story to others in the book. **Interview & Workplace Practice** asks you first to retell the case, then to make it your own

story using the STAR method: the **Situation**, your **Task**, the **Action** you took and the **Result**. **Take the Idea Further** suggests where to go next.

PART I

Reframe the Problem

1

The Rocket That Was Too Expensive

DOCUMENTATION LEVEL 2 — SELF-REPORTED OR TRANSFERRED

The account rests mainly on the protagonist's own retelling, or the idea was documented but its original application was never adopted.

In 2001, Elon Musk wanted to send a small greenhouse to Mars — a symbolic project meant to reignite public interest in space exploration. He needed a rocket, so he flew to Moscow to buy one, second-hand, from the post-Soviet space industry. The trip did not go well. Russian officials quoted him a price that struck him as absurd for what the rocket actually was, then reportedly raised it further during the negotiation. Musk flew home without a rocket. Most people in that position would have treated the price as a fact about the world — rockets cost what rockets cost — and either paid it or given up. On the flight back, by his own later account, Musk did something different. He asked what a rocket actually consists of: aerospace-grade aluminum alloys, titanium, copper, carbon fibre. Then he asked a second question nobody in the room in Moscow had invited him to ask — what those raw materials were worth on the open commodity market.

The answer reframed the entire problem. The materials came to roughly two percent of what a finished rocket sold for. For comparison, raw materials typically run twenty to thirty percent of the sale price of a car. A rocket was not expensive because rocket-grade metal was expensive. It was expensive because of everything that happened to that metal between the mine and the launch pad — an industry built on layers of subcontractors, each adding margin, inside a customer base of governments not known for aggressive price negotiation. That distinction — between what a thing costs to make and what an industry currently charges for it — is the entire idea. A market price

is not a law of physics. It is the output of a system, and systems that nobody has recently questioned tend to carry costs that have nothing to do with the physics of the product. Musk chose the harder path available to him: instead of buying a rocket, he would build a company that manufactured its own, keeping the value chain in-house rather than paying a margin at every link.

SpaceX was founded in 2002. The first three launches of its Falcon 1 rocket failed. The fourth reached orbit in September 2008, months before the company would run out of money entirely. Within a decade, SpaceX was launching for a fraction of the price Musk had been quoted in Moscow — not because the physics of rockets had changed, but because the company had rebuilt the industry's cost structure from the ground up rather than accepting the one it inherited. The lesson is not that first-principles thinking makes hard problems easy. Getting Falcon 1 to orbit took four attempts and nearly bankrupted the company. The lesson is narrower and more useful: a price that everyone treats as fixed is often a description of a system, not a description of a material — and systems, unlike physics, can be redesigned.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which price in your world do you treat as a fact of nature that might actually just be a description of an unexamined process?*

KEY TAKEAWAYS

1. A market price reflects a system's current structure, not a fundamental limit — and systems can be redesigned.
2. Breaking a cost down into its physical components can reveal how much of the price is genuinely necessary.

3. First-principles thinking identifies the opportunity; it does not remove the difficulty of executing on it.

SUMMARY

Facing an inflated price for a used rocket, Musk broke the problem down to the actual cost of a rocket's raw materials and found they were roughly two percent of the sale price — revealing that the market price reflected the aerospace industry's structure, not the physics of building a rocket. He built SpaceX to manufacture rockets in-house rather than pay the existing industry's margins. The company nearly failed before its fourth launch finally succeeded.

STORY CHECK

ESTABLISHED

SpaceX was founded in 2002 after an unsuccessful attempt to buy a Russian rocket; the first three Falcon 1 launches failed; the fourth reached orbit in 2008.

POPULAR VERSION ADDS

The specific "two percent" figure and the account of the Moscow negotiation come from Musk's own retellings in interviews over many years, not from an independently documented record of the meeting.

UNCERTAIN

How much of SpaceX's eventual cost advantage came from the materials insight itself, versus reusability, vertical integration, and over a decade of iterative engineering that followed it.

VOCABULARY & PHRASAL VERBS

commodity (n.) a raw material or basic good bought and sold on an open market. Steel is traded as a commodity, with a price that moves daily.

margin (n.) the difference between what something costs to produce and what it sells for. Every subcontractor in the chain adds its own margin.

subcontractor (n.) a company hired by another company to do part of a larger job. Aerospace firms often outsource to subcontractors, who outsource again in turn.

to reframe (v.) to look at a problem in a fundamentally different way. He reframed the question from "what will they charge" to "what does it cost."

vertical integration (n.) owning multiple stages of a supply chain rather than buying from outside suppliers. Vertical integration let SpaceX avoid paying a margin at every step.

in-house (adj./adv.) done within an organization rather than outsourced. They decided to build the engines in-house.

run out of (phrasal verb) to have no more of something left. The company nearly ran out of money before the fourth launch.

fly home (phrasal verb) to travel back by air, often used here to mean returning without success. He flew home from Moscow without a rocket.

pay off (phrasal verb) to produce a good result after a period of risk or effort. The bet on in-house manufacturing eventually paid off.

PRACTICE

Where the Story Breaks Down

Musk had capital, technical background and a high tolerance for risk that most people attempting this kind of reframing do not have. Asking the right question is necessary but nowhere near sufficient.

Use the Language

Explain a cost in your own work that might be more about the system than the materials, using margin, subcontractor and reframe.

Discuss

What is a price or cost in your organization that nobody has questioned in years? What would "asking what it's actually made of" mean in that context?

Write

In 100 words, describe one system-level cost (not a materials cost) that might be driving up a price you currently accept as fixed.

SOURCES

B Vance, A., Elon Musk: Tesla, SpaceX, and the Quest for a Fantastic Future, Ecco (2015) — includes an account of the 2001 Moscow trip.

D Musk's "two percent" account of rocket material costs is drawn from multiple interviews, including Foundation podcast and Kimbal Musk's public retellings; no independently audited cost breakdown of that specific quote has been published.

A Federal Aviation Administration and SpaceX launch records confirming Falcon 1's failed 2006–2008 launches and its successful fourth flight on 28 September 2008.

THE CONTRAST

Story 1: The Rocket That Was Too Expensive vs. Story 39: Furniture Designed Around the Constraint (IKEA)

Story 1: The Rocket That Was Too Expensive: Reframing started by breaking a price down into its raw materials — asking what a thing is actually made of.

Story 39: Furniture Designed Around the Constraint (IKEA): Reframing started from a physical constraint — what fits in a car — and let that constraint drive the whole design.

Does reframing work better by breaking a problem down to its components, or by starting from a hard constraint and designing around it?

THIS CONNECTS TO

Story 32: Connect What Already Exists (Berners-Lee) SAME MECHANISM
Both cases show that a price or a barrier people treat as fixed is really just the current output of a system — one built by combining existing components differently, the other by rebuilding a supply chain from scratch.

Story 8: The Patent Clerk and the Train (Einstein) SAME MECHANISM
Both worked by refusing an inherited framing and reasoning from the underlying components instead of the accepted description.

Story 49: Invert the Problem (Munger) LIMITATION
Musk's reframe asked what something actually costs. Munger's habit asks a different kind of question entirely — what guarantees failure — showing that "question the frame" takes more than one form.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Elon Musk needed a rocket for a Mars project and was quoted a price for a used one that struck him as absurd.

T — Task

He needed to decide whether to pay the price or find another way to get a rocket.

A — Action

He broke the rocket down into its raw materials and found they cost a small fraction of the sale price, then built a company to manufacture rockets in-house instead.

R — Result

SpaceX was founded in 2002; after three failed launches, its fourth reached orbit in 2008, and the company went on to launch at a fraction of the price Musk had originally been quoted.

Level 2 — Make It Your Story

> *Tell me about a time when you broke a costly problem down into its actual components instead of accepting the price you were quoted.*

Use STAR to answer. Think about: a budget, a vendor quote, a process cost, a "that's just how much it costs" assumption you questioned.

TAKE THE IDEA FURTHER

> *What price, cost, or barrier in your work have you never actually broken down into what it's really made of?*

2

The Burrs on the Dog

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1941, the Swiss engineer George de Mestral returned from a walk in the countryside near Lausanne to find his trousers and his dog's fur covered in burdock burrs. Almost everyone who has walked a dog through late-summer countryside has had the same experience, and almost everyone responds to it the same way: they pull the burrs off, mildly annoyed, and think no more about it.

De Mestral did something different. He wanted to know why the burrs held on so stubbornly — hard enough to resist a light brushing, yet designed, presumably, to eventually let go once the animal carrying them had wandered far enough from the parent plant to spread its seeds. He took several burrs home and examined them under a microscope. What he found was not a sticky coating or an adhesive of any kind. The surface of each burr was covered in hundreds of tiny, curved hooks, shaped precisely to catch on the loops naturally present in woven fabric and tangled animal hair.

The burr was not glued to anything. It was mechanically interlocked — hooks finding loops — and it released the same way it caught, by force applied at the right angle. Recognizing the mechanism was the easy part. Reproducing it took de Mestral most of the next decade. His first attempts, worked out with a weaver in Lyon, used cotton, which caught well but wore out within a few uses. He eventually found that nylon, sewn under infrared light, formed tough, durable hooks with a kind of elastic memory that cotton lacked — solving one side of the fastener. The harder part was cutting them. Cutting

loops to the exact length and curve needed to grip reliably, at industrial scale, took years of failed machinery before de Mestral built a device — inspired, by his own account, by the mechanics of a barber's clippers — that could trim woven loops into consistent hooks.

By the time he filed his Swiss patent in 1951, roughly a decade had passed since the walk that started it. The patent was granted in 1955; the product, named Velcro from the French velours (velvet) and crochet (hook), reached the American market in 1957. It took another two decades before hook-and-loop fastening became the everyday object it is now, eventually finding uses from children's shoes to NASA spacesuits.

The lesson usually attached to this story is that nature inspired an invention, which is true but incomplete. Burrs stick to millions of dogs and trousers every autumn. What made de Mestral's walk different from everyone else's was not the burr. It was the decision to treat a minor irritation as a question worth a microscope, and then to spend ten years answering it.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What is a small, recurring annoyance in your work that you have always dismissed rather than actually investigated?*

KEY TAKEAWAYS

1. The burr's mechanism was structural, not chemical — a distinction that only became visible under close examination.

2. Recognizing a good idea and manufacturing it reliably are different problems, and the second one took roughly nine times longer than the first.
3. The habit that mattered was not cleverness at the moment of discovery, but the choice to investigate an ordinary irritation instead of dismissing it.

SUMMARY

After noticing burrs clinging stubbornly to his clothes and his dog after a countryside walk, de Mestral examined them under a microscope and found a mechanical hook structure rather than an adhesive. Reproducing that structure in fabric took him roughly a decade of experimentation, resulting in Velcro, patented in Switzerland in 1955. The insight came from observation; the value came from years of subsequent engineering.

STORY CHECK

ESTABLISHED

De Mestral filed his Swiss patent in 1951, was granted it in 1955, and the resulting hook-and-loop fastener was trademarked Velcro; nylon and infrared-shaped loops were part of the documented manufacturing solution.

POPULAR VERSION ADDS

The precise year of the walk varies between sources (1941 in most accounts, 1948 in at least one museum retelling), and the vivid "under a microscope that evening" detail is standard across tellings but not independently dated.

UNCERTAIN

The exact sequence and duration of the failed prototypes; most sources agree on "years," with figures for the mechanization process ranging from eight to ten years depending on which stage is being counted.

VOCABULARY & PHRASAL VERBS

burr (n.) a seed case covered in hooks or spines that catches on fur or fabric. Burdock burrs cling to almost anything they touch.

to interlock (v.) to fit together mechanically, each part gripping the other. The hooks and loops interlock rather than stick chemically.

prototype (n.) an early working version of a product, built to test an idea. His first prototype used cotton, which wore out quickly.

elastic memory (n.) a material's tendency to return to its original shape after being stretched or bent. Nylon's elastic memory made it hold its loop shape better than cotton.

at scale (phrase) done at a large, industrial volume rather than by hand. Cutting the hooks reliably at scale required a purpose-built machine.

to wear out (phrasal verb) to become damaged or unusable through repeated use. The cotton prototype wore out after a handful of uses.

to catch on (phrasal verb) here, to physically hook or snag onto something. The tiny hooks caught on the loops in the fabric.

pull off (phrasal verb) to remove something by pulling. Most people just pull the burrs off and move on.

spread out (phrasal verb) here, to extend over a period of time. The development process spread out over roughly a decade.

PRACTICE

Where the Story Breaks Down

De Mestral had engineering training and access to a skilled weaver's workshop; most people who notice an interesting irritation have neither the tools nor the specific expertise to act on it the way he did.

Use the Language

Describe an everyday annoyance and what might be discovered by examining it closely, using interlock, prototype and wear out.

Discuss

Think of something that irritates you regularly. What would "putting it under a microscope" mean for that specific irritation?

Write

In 100 words, describe the difference between recognizing a good mechanism and being able to manufacture it reliably, using an example from your own field.

SOURCES

A De Mestral, G., Swiss Patent No. 293,566, filed 1951, granted 1954; corresponding US Patent 2,717,437 (1955).

C Velcro Companies, official company history and timeline (velcro.com).

B Smithsonian Lemelson Center, "George de Mestral: Velcro Inventor" — invention.si.edu.

THIS CONNECTS TO

Story 11: The Five Whys SAME MECHANISM

Both cases treat a small, ordinary annoyance as worth serious investigation rather than a nuisance to shrug off.

Story 19: The Mold in the Laboratory (Fleming) SAME MECHANISM

Both discoveries came from close, patient examination of something almost anyone else would have brushed off or thrown away.

Story 23: The Stain That Wouldn't Wash Off (Scotchgard) LIMITATION

De Mestral was specifically looking for a mechanism when he examined the

burrs. Sherman and Smith weren't looking for anything — showing that close investigation and pure accident can both lead somewhere, but not by the same route.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

George de Mestral kept finding burrs stuck stubbornly to his dog's fur and his own clothes after walks.

T — Task

Instead of simply removing them, he wanted to understand exactly how they stuck so well.

A — Action

He examined the burrs under a microscope and discovered a mechanical hook-and-loop structure, then spent years developing a fabric that replicated it.

R — Result

The result, patented in the 1950s and trademarked Velcro, became a widely used fastening technology.

Level 2 — Make It Your Story

> *Tell me about a time when you investigated a small, recurring annoyance instead of just working around it.*

Use STAR to answer. Think about: a minor workplace irritation, a repeated small failure, something "everyone just deals with."

TAKE THE IDEA FURTHER

> *What small, recurring annoyance in your own work have you never actually stopped to investigate?*

3

The Bicycle Makers Who Wanted to Fly

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

By the turn of the twentieth century, powered flight had already destroyed several reputations. Serious, well-funded engineers had built machines, tested them publicly, and watched them crash. The most prominent of these failures, Samuel Langley's Aerodrome, was catapulted off a houseboat on the Potomac in December 1903 and immediately plunged into the river, days before two bicycle mechanics from Dayton, Ohio succeeded where a well-financed government program had just failed. Wilbur and Orville Wright had no formal engineering training and no government funding. What they had was a bicycle repair business, which had taught them something less glamorous than aerodynamics: how to build precisely, test repeatedly, and trust a measurement over a hunch. When their early glider designs performed worse than the published aeronautical tables predicted, most engineers in their position would have assumed they had built the glider wrong. The Wrights assumed the tables might be wrong instead — and set out to check.

They built a wind tunnel, a six-foot wooden box with a fan at one end, and spent the winter of 1901 testing more than two hundred wing shapes of their own design, measuring lift and drag directly rather than trusting figures other engineers had published decades earlier. The existing tables, it turned out, contained real errors. The Wrights' own measurements, taken with instruments they built themselves, became more reliable than the literature everyone else in aviation was working from.

They broke the overall problem of flight into three separate, solvable pieces: generating enough lift to leave the ground, maintaining balance in the air, and controlling the direction of flight once airborne. Balance and control turned out to be the piece nearly everyone else had underestimated. The brothers' key insight — warping the wingtips in opposite directions to bank and turn, an idea Wilbur reportedly traced to absentmindedly twisting a cardboard box at the bicycle shop — gave them a genuine mechanism for controlling roll, something no prior aircraft design had solved.

On 17 December 1903, at Kitty Hawk, North Carolina, Orville flew the Wright Flyer 120 feet in twelve seconds. It was not a dramatic distance. What made it different from Langley's failure days earlier, and from a decade of other failed attempts, was that it was the product of a genuinely tested machine rather than a theoretical one. The brothers had spent four years converting flight from a question answered by confidence into a question answered by data.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Where in your work are you trusting a published number, a competitor's benchmark, or "how it's always been measured" instead of testing the underlying assumption yourself?*

KEY TAKEAWAYS

1. When results contradict established data, the data itself is a valid thing to question, not only your own execution.
2. Breaking a large, undefined problem — "flight" — into separable sub-problems made each piece individually solvable.

3. Being outsiders to the aviation establishment meant the Wrights had no institutional stake in defending the existing aerodynamic tables.

SUMMARY

Facing published aerodynamic data that didn't match their glider's actual performance, the Wright brothers built their own wind tunnel and tested over two hundred wing shapes, discovering genuine errors in tables the rest of the field had relied on. They separated flight into lift, balance and control, solving the control problem with wing-warping — a mechanism no earlier design had achieved. Their December 1903 flight succeeded where better-funded contemporaries had just failed, largely because it was built on tested measurements rather than inherited assumptions.

STORY CHECK

ESTABLISHED

The Wrights built and used a wind tunnel in 1901; their December 1903 flight at Kitty Hawk is documented by contemporaneous witnesses, telegrams and photographs; Langley's Aerodrome failed publicly days earlier.

POPULAR VERSION ADDS

The detail that Wilbur's wing-warping idea came from twisting an empty cardboard inner-tube box comes from later Wright family retellings and cannot be independently corroborated to the moment.

UNCERTAIN

The precise number of wing shapes tested varies slightly between historical accounts (commonly cited as "over 200"), and exact wind-tunnel data logs have not all survived.

VOCABULARY & PHRASAL VERBS

aerodynamic (adj.) relating to how air moves around a solid object, especially in flight. Their aerodynamic tables turned out to contain real errors.

lift (n.) the upward force that allows an aircraft to leave the ground. Generating enough lift was the first of three separate problems.

to bank (v.) for an aircraft to tilt to one side, typically to turn. Wing-warping let the aircraft bank and turn.

literature (n.) here, the published body of research or data in a field. Their own measurements became more reliable than the existing literature.

institutional stake (n.) a vested interest an organization has in defending its existing position. As outsiders, they had no institutional stake in the old tables.

catapult (v.) to launch forcefully and suddenly. Langley's Aerodrome was catapulted off a houseboat.

take off (phrasal verb) for an aircraft to leave the ground and begin flying. The Flyer took off on 17 December 1903.

set out to (phrasal verb) to begin a task with a clear intention. They set out to check the published figures themselves.

build up (phrasal verb) to accumulate gradually, here referring to evidence or data. Four years of testing built up the evidence base that made 1903 possible.

PRACTICE

Where the Story Breaks Down

The Wrights had years of hands-on mechanical experience from bicycle manufacturing, which gave them fabrication skill few outsiders to a field possess; not every industry newcomer has an equivalent transferable craft.

Use the Language

Describe a time you distrusted a published benchmark or "industry standard" figure, using literature, institutional stake and set out to.

Discuss

What published number, benchmark or convention in your field have you never personally verified?

Write

In 100 words, describe how you would break a large, undefined problem in your own work into three separately solvable pieces, the way the Wrights split flight into lift, balance and control.

SOURCES

A Wright Brothers Aeroplane Company records and the Library of Congress Wright Brothers Papers, including 1901 wind tunnel data.

B Crouch, T. D., The Bishop's Boys: A Life of Wilbur and Orville Wright, W. W. Norton (1989).

C National Air and Space Museum, Smithsonian Institution, Wright Flyer exhibit documentation.

THE CONTRAST

Story 3: Wright Brothers vs. Story 5: The Glue That Wasn't Sticky Enough (Post-it Notes)

Story 3: Wright Brothers: Started with a clearly defined problem — powered flight — and built a systematic testing process to solve it.

Story 5: The Glue That Wasn't Sticky Enough (Post-it Notes): The useful discovery came first, unattached to any problem, and sat unused for years until someone else happened to need exactly that property.

Should innovation begin with a clear problem to solve — or can a useful discovery emerge before anyone has framed the problem it solves?

THIS CONNECTS TO

Story 9: The Planes That Didn't Return (Wald) SAME MECHANISM
Both distrusted a data set that looked authoritative — published aerodynamic tables, battle-damage diagrams — and built their own, more complete way of measuring rather than accepting it.

Story 36: Verify the Calculation (Katherine Johnson) SAME MECHANISM
Both used an independently built method — a wind tunnel, hand-calculated orbital mechanics — to check a result everyone else was relying on without direct verification.

Story 6: The Doctor Who Wasn't Supposed to Be Right (Semmelweis) LIMITATION
The Wrights convinced the world with one physical demonstration anyone could watch. Not every claim can be proven that cleanly — Semmelweis had a result just as real, but no equivalent single moment that would make it undeniable.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Published aerodynamic tables were the accepted standard, but the Wrights' gliders kept performing worse than those tables predicted.

T — Task

They needed to find out whether the problem was their design or the data everyone in the field was relying on.

A — Action

They built their own wind tunnel and tested over 200 wing shapes themselves, rather than trusting the published figures, and broke the flight problem into three separately solvable pieces.

R — Result

They discovered real errors in the existing tables, solved the control problem no one else had, and achieved the first powered flight in December 1903.

Level 2 — Make It Your Story

> *Tell me about a time when you tested several approaches before finding a solution.*

Use STAR to answer. Think about: work, university, a project, a team, a process you rebuilt from scratch, a "best practice" you questioned.

TAKE THE IDEA FURTHER

> *What published number, benchmark, or "best practice" in your field have you never personally tested?*

4

The Map That Challenged a Theory

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In late August 1854, cholera tore through the Soho district of London with terrifying speed. Within ten days, more than five hundred people in a few square blocks were dead. The dominant medical theory of the time held that cholera spread through "miasma" — bad air rising from filth and decay — a theory with no shortage of eminent supporters and centuries of institutional weight behind it. John Snow, a Soho physician, had spent years unconvinced by miasma theory, and had already published a paper on an alternative in 1849 to almost no response from the medical establishment. When the 1854 outbreak began, he had a specific hypothesis to test rather than a general suspicion: cholera, he believed, spread through contaminated water, not air. He began the unglamorous work of confirming it — walking the district, recording the address of every cholera death, and asking survivors and families exactly where each household drew its drinking water.

The pattern that emerged, once plotted onto a street map with a bar at each address showing the number of deaths there, clustered overwhelmingly around a single public water pump on Broad Street. Snow tested the pump water directly; the sample showed only mild organic contamination, not the clear-cut evidence he had hoped for. He presented his case to the local Board of Guardians anyway, based on the pattern of deaths rather than a chemical smoking gun, and on 8 September 1854 persuaded them to remove the pump's handle, making it unusable. The popular version of this story ends here, with cases collapsing dramatically the moment the handle came off. The

real sequence, which Snow documented candidly in his own writing, is less tidy and more instructive. The outbreak was already in steep decline by 8 September — many residents had already died or fled the district — and Snow himself acknowledged he could not be certain how much credit the pump handle deserved versus the outbreak simply running its course. His map had not even been drawn yet; it was compiled afterward, as part of the published report making his case.

Snow's stronger evidence came from a separate, quieter piece of work sometimes called his Grand Experiment: comparing cholera death rates among London households served by two different water companies, one of which drew water from a badly polluted stretch of the Thames and one from a cleaner source upstream. Streets and buildings were often identical in every respect except which pipe fed them — and death rates between the two groups differed by an order of magnitude. It took decades, and the eventual identification of the cholera bacterium by Robert Koch, for the medical establishment to fully accept what the map and the water-company comparison had already shown.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What belief in your organization is held mainly because of institutional authority rather than because anyone has recently checked the underlying data?*

KEY TAKEAWAYS

1. Snow tested a specific, falsifiable hypothesis rather than gathering evidence in support of a vague suspicion.

2. The map is remembered as the decisive proof, but Snow's own writing shows the pump-handle timing was ambiguous — the outbreak was already declining.
3. His stronger evidence, the natural comparison between two water companies' customers, receives far less popular attention than the more visually dramatic map.

SUMMARY

During the 1854 Soho cholera outbreak, John Snow mapped cholera deaths by address and traced a cluster to a single water pump, persuading authorities to disable it — though the outbreak was already declining by then, a fact Snow himself acknowledged. His stronger evidence came from a natural comparison of cholera rates between households served by two different water companies. It took decades and the later confirmation of the cholera bacterium for his water-borne theory to be widely accepted.

STORY CHECK

ESTABLISHED

Snow's 1855 report, the Broad Street death map, and the water-company comparison are all in the historical record in his own published writing.

POPULAR VERSION ADDS

The dramatic framing that removing the pump handle "stopped the outbreak" — Snow's own account states the outbreak was already declining and that he could not be certain how much the pump handle itself contributed.

UNCERTAIN

The exact chemical contamination in the original water sample was mild, not damning, which is often omitted from retellings that treat the case as immediately obvious.

VOCABULARY & PHRASAL VERBS

hypothesis (n.) a specific, testable proposed explanation. He had a hypothesis to test, not just a general suspicion.

miasma (n.) historically, a theory that disease spread through bad air. Miasma theory dominated medical thinking at the time.

falsifiable (adj.) capable of being proven wrong by evidence, a mark of a genuinely testable claim. A falsifiable hypothesis can actually be checked.

cluster (v./n.) to group closely together; a group formed this way. Deaths clustered around a single pump.

candidly (adv.) honestly and without concealing inconvenient details. Snow wrote about the timing candidly, including its ambiguity.

smoking gun (n.) a piece of evidence considered definitive proof. The water sample wasn't the smoking gun he'd hoped for.

run its course (phrase) for a process (often an illness or event) to continue naturally to its end. The outbreak may simply have been running its course.

trace to (phrasal verb) to follow evidence back to its origin. He traced the pattern of deaths to a single pump.

hold up (phrasal verb) to remain valid under scrutiny. His conclusion held up decades later, once germ theory was accepted.

PRACTICE

Where the Story Breaks Down

Snow had years of conviction and a published 1849 paper and a specific hypothesis already formed; most people investigating a problem for the first time don't have a five-year head start on what to look for.

Use the Language

Describe a claim in your work that would benefit from being made falsifiable, using hypothesis, cluster and hold up.

Discuss

What is your organization's version of "miasma theory" — a widely accepted explanation nobody has recently tested against real data?

Write

In 100 words, distinguish a dramatic-sounding piece of evidence from a quieter, more rigorous one in a decision you've seen made recently.

SOURCES

A Snow, J., On the Mode of Communication of Cholera, 2nd edition, John Churchill (1855).

A Snow, J., letter to the editor, Medical Times and Gazette (23 September 1854), acknowledging the outbreak's decline before the pump handle was removed.

B Johnson, S., The Ghost Map: The Story of London's Most Terrifying Epidemic, Riverhead Books (2006).

THE CONTRAST

Story 4: John Snow vs. Story 19: The Mold in the Laboratory (Fleming)

Story 4: John Snow: Started with a specific, falsifiable hypothesis and went looking for evidence that could prove it wrong.

Story 19: The Mold in the Laboratory (Fleming): Started with no hypothesis at all — the discovery arrived as an anomaly Fleming happened to notice and chose not to discard.

Is disciplined hypothesis-testing or open-minded noticing the more reliable route to a real discovery — and can you tell in advance which one a problem needs?

THIS CONNECTS TO

Story 6: The Doctor Who Wasn't Supposed to Be Right (Semmelweis)

SAME MECHANISM

Both physicians reached a conclusion about disease transmission that contradicted the dominant medical theory of their era, using evidence rather than accepted authority.

Story 9: The Planes That Didn't Return (Wald) SAME MECHANISM

Both built their case by mapping data spatially — a diagram of cholera deaths by pump, a diagram of bullet holes by location — and letting the pattern itself make the argument.

Story 35: The Diagram That Made Death Visible (Nightingale)

LIMITATION

Snow's map won the argument in part through its visual clarity. Nightingale's

chapter takes that one step further: sometimes the data existed all along, and only the presentation was ever the real barrier.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

A cholera outbreak was killing hundreds of people in Soho, London, and the dominant medical theory blamed bad air.

T — Task

Snow needed to test his alternative hypothesis — that cholera spread through contaminated water — against the accepted theory.

A — Action

He mapped cholera deaths by address and traced them to a single water pump, then had its handle removed.

R — Result

The pump was disabled as the outbreak was already declining; Snow's map became a foundational piece of evidence for what would later become germ theory and modern epidemiology.

Level 2 — Make It Your Story

> *Tell me about a time when you tested a specific hypothesis rather than just gathering general evidence for a hunch.*

Use STAR to answer. Think about: a diagnosis, a root-cause investigation, a customer complaint you traced to its source.

TAKE THE IDEA FURTHER

> *What belief in your organization is held mainly because of authority or habit, rather than because anyone has recently tested it?*

5

The Glue That Wasn't Sticky Enough

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1968, a chemist at 3M named Spencer Silver was trying to develop a stronger adhesive for the aerospace industry. What he produced instead was the opposite of what he was looking for: a light, pressure-sensitive adhesive made of tiny acrylic spheres that stuck to surfaces but could be peeled away cleanly, again and again, without leaving residue or tearing the material underneath. By the standard 3M was working to, it was a failure. A glue that doesn't hold permanently is not a stronger glue.

Silver did not discard the compound. He was convinced the property he had accidentally produced — light, repositionable adhesion — was valuable, even without a specific application in mind. For roughly five years, he presented the adhesive at internal 3M seminars, trying to interest colleagues in finding a use for it. Most were uninterested; a technology looking for a problem is a hard sell inside an organization built to solve problems people already have.

The breakthrough use came from a colleague in an unrelated department. Art Fry, an engineer who sang in his church choir, was frustrated by a specific and mundane annoyance: the paper bookmarks he used to mark hymns in his hymnal kept falling out. He had attended one of Silver's seminars years earlier and remembered the adhesive. Fry began experimenting on his own initiative, coating strips of paper with Silver's compound to make bookmarks that would stay in place yet lift off without damaging the page. Testing the idea inside 3M, Fry noticed something beyond the bookmark use. He wrote a note on one of the sticky strips, attached it to a report, and realized the note itself

— not just the bookmark — was the more valuable product. A piece of paper that could be written on, stuck anywhere, and removed without a trace solved a problem nobody had formally asked 3M to solve, because nobody had thought to frame "leaving a removable note" as a solvable problem at all.

3M's first test launch of the product, initially called "Press 'n Peel," performed poorly; customers given free samples in a 1977 test market in four cities didn't reorder, largely because they had never used the product and didn't understand what it was for. The company nearly shelved it. A second attempt in 1978, flooding a single market with free samples so people could actually experience how it worked, succeeded — customer reorder rates reportedly reached the high nineties. The product was renamed Post-it and launched nationally in 1980, twelve years after Silver's original "failed" adhesive.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What result in your organization has been filed away as a failure simply because it didn't achieve the specific goal it was originally measured against?*

KEY TAKEAWAYS

1. Silver's adhesive was a failure only relative to the specific goal — permanent bonding — it had been developed to achieve; the property itself had independent value.
2. It took a second person, in a different role and with a different problem, to recognize what the first person's result was actually good for.

3. Even after the right idea and the right person converged, the first market test still failed — because a genuinely new product category needs direct experience, not just distribution, to be understood.

SUMMARY

A 3M chemist trying to build a stronger adhesive produced instead a weak, repositionable one — a failure by the original goal, which he nonetheless kept promoting internally for years. A colleague later solved his own unrelated problem, unreliable hymnal bookmarks, using that same adhesive, then noticed the sticky note itself was more valuable than the bookmark. Even then, the first market launch failed, and it took a second, sample-heavy launch before the product succeeded as Post-it.

STORY CHECK

ESTABLISHED

Silver's 1968 adhesive development, Fry's bookmark application, and the 1977–1980 launch history are documented in 3M's own corporate history and multiple independent business retrospectives.

POPULAR VERSION ADDS

Many retellings compress the process into a single "aha" moment when Fry heard Silver's seminar and instantly invented Post-it notes; the real gap between the seminar and Fry's bookmark experiments was several years, and the first market launch still failed.

UNCERTAIN

The exact reorder percentages from the 1978 relaunch vary slightly between sources (commonly cited in the 90–95% range), and 3M has not published the original underlying sales data.

VOCABULARY & PHRASAL VERBS

pressure-sensitive (adj.) describing an adhesive activated by light pressure rather than heat or moisture. The adhesive was pressure-sensitive and repositionable.

repositionable (adj.) able to be moved and reattached without losing its function. A repositionable note could be stuck and restuck.

residue (n.) a substance left behind after something else is removed. The adhesive left no residue on the page.

shelve (v.) to postpone or abandon a project indefinitely. The company nearly shelved the product after the first launch failed.

converge (v.) for separate things to come together at one point. The right idea and the right person eventually converged.

reorder rate (n.) the percentage of customers who buy a product again after trying it. Reorder rates reached the high nineties in the second test.

fall out (phrasal verb) to become dislodged and drop out. His paper bookmarks kept falling out of the hymnal.

file away (phrasal verb) to set something aside as settled or finished, often prematurely. The adhesive was nearly filed away as a failed experiment.

flood the market (phrase) to distribute a product very widely, often via free samples. The second launch flooded one city with free samples.

PRACTICE

Where the Story Breaks Down

Silver worked inside a company with the internal culture, budget and patience to let an employee promote an unproven idea for five years; most failed

experiments in most organizations get discarded long before a second use is found.

Use the Language

Describe a "failed" result you've seen that might have value for a different purpose, using repositionable, shelve and converge.

Discuss

Has your organization ever launched something that failed the first time due to a lack of understanding rather than a lack of quality? What changed the second time?

Write

In 100 words, describe an internal seminar, side project or "useless" result in your own field that might be waiting for the right second person to find its use.

SOURCES

C 3M Company, official corporate history of the Post-it Note (3m.com).

B Nayak, P. R. & Ketteringham, J. M., Breakthroughs!, Rawson Associates (1986) — includes an early detailed account of the Post-it development timeline.

D Reorder-rate figures from the 1977–1978 test markets are commonly cited in business retrospectives without a single traceable primary source.

THIS CONNECTS TO

Story 3: The Bicycle Makers Who Wanted to Fly (Wright Brothers)

CONTRAST

See The Contrast in that chapter. In short: the Wrights tested toward a defined goal; Silver's adhesive found its value only after the goal was abandoned and someone else reframed the problem.

Story 9: The Planes That Didn't Return (Wald) LIMITATION

Wald's reframing was immediately actionable — once the military saw the argument, they acted on it. Silver's discovery sat unused for five years even after he had already reframed it as valuable, which shows that having the right insight is necessary but not sufficient; something else — patience, the right second person, organizational tolerance — still has to be present.

Story 25: The Glue Looking for a Problem (Super Glue)

SAME MECHANISM

Both compounds were developed for one purpose, judged a failure against that purpose, and only became valuable once someone unrelated to the original project found a different use for the exact property that had made them a failure.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

A 3M chemist trying to invent a stronger adhesive produced a weak, repositionable one instead — a failure by the original goal.

T — Task

He believed the property itself was valuable and needed to find someone who could use it, even without a defined application.

A — Action

He promoted it internally for five years. A colleague eventually used it to solve an unrelated problem — bookmarks falling out of a hymnal — and then noticed the note itself, not the bookmark, was the real product.

R — Result

Even then, the first market launch failed. A second, sample-heavy launch succeeded, and the product became Post-it.

Level 2 — Make It Your Story

> *Tell me about a time when something that initially failed became useful.*

Use STAR to answer. Think about: a project that missed its original goal, a discarded idea, a "failed" test, a pivot.

TAKE THE IDEA FURTHER

> *What "failed" result in your own work might actually be a solution waiting for the right problem?*

6

The Doctor Who Wasn't Supposed to Be Right

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In the 1840s, giving birth in a hospital was, counterintuitively, more dangerous than giving birth at home or being attended by a midwife. At Vienna General Hospital, two maternity wards operated side by side under nearly identical conditions, with one crucial difference: the First Clinic was staffed by physicians and medical students, who also performed autopsies as part of their training; the Second Clinic was staffed entirely by midwives, who did not. Mortality from puerperal fever — a bloodstream infection following childbirth — ran three to five times higher in the doctors' clinic than in the midwives' clinic, a difference so well known that women in labour sometimes begged to be admitted to the midwives' ward instead, or gave birth in the street rather than risk the First Clinic.

Ignaz Semmelweis, a Hungarian physician appointed assistant at the First Clinic in 1846, spent months testing possible explanations. He changed the position of delivery beds. He altered ventilation. He even changed the route by which a priest walked to deliver last rites to dying patients, on the theory that fear itself might be a contributing cause. None of it moved the mortality rate.

The insight arrived through tragedy rather than experiment. In early 1847, a colleague of Semmelweis's, Jakob Kolletschka, died after accidentally cutting his finger during an autopsy. Kolletschka's own autopsy showed pathological

changes strikingly similar to those found in the women dying of puerperal fever. Semmelweis reasoned that "cadaverous particles" — some invisible contaminating agent, in an era before germ theory existed — were being carried on the hands of doctors and students directly from the dissection room to the delivery room, and from the delivery room to the next patient.

He ordered every doctor and student to wash their hands in a chlorinated lime solution before examining any patient — chosen because it was the most effective substance he found at removing the smell of decaying tissue from skin. The results were immediate and dramatic: mortality in the First Clinic fell from around 18 percent in April 1847 to roughly 2 percent within a few months of the practice being adopted in May, matching the midwives' ward for the first time in the hospital's history.

Rather than being celebrated, Semmelweis was largely dismissed. He had no accepted theoretical mechanism to offer — germ theory was decades away from Louis Pasteur — only a correlation and a horrified colleague's autopsy. Established physicians resented the implication that they themselves were the vector of the disease they were failing to explain. Semmelweis's professional relationships deteriorated; his increasingly furious public letters accusing prominent obstetricians of being "murderers" further isolated him from a medical establishment that might otherwise have taken years to come around regardless. In 1865, exhausted and reportedly showing signs of mental decline, he was committed to an asylum, where he died just weeks later of sepsis — the same category of infection he had spent his career fighting, in an outcome historians believe may have followed a severe beating by asylum guards.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What result have you seen dismissed by an organization because nobody could yet explain why it worked, even though the evidence was strong?*

KEY TAKEAWAYS

1. A clear, dramatic result — an 18% to 2% drop in mortality — was not sufficient on its own to overturn consensus without an accepted explanation for the mechanism behind it.
2. The insight came from a single tragic data point (a colleague's death) that resembled a much larger pattern Semmelweis had already been struggling to explain.
3. Being right did not protect Semmelweis professionally or personally; the resistance he faced eventually cost him his career, his relationships and, indirectly, his life.

SUMMARY

Semmelweis discovered that mortality from childbed fever was far higher in a doctor-staffed maternity ward than a midwife-staffed one, and traced the cause to doctors carrying invisible contamination from autopsies to delivery patients, after a colleague died from an infection with strikingly similar autopsy findings. Ordering handwashing with chlorinated lime cut mortality from roughly 18% to 2% within months. The medical establishment rejected his conclusion for lack of a theoretical mechanism, and Semmelweis died in an asylum less than two decades later, before germ theory vindicated him.

STORY CHECK

ESTABLISHED

The mortality figures (roughly 18% before, roughly 2% after) and the timeline of the intervention (May 1847) are documented in Semmelweis's own published statistics and corroborated by hospital records.

POPULAR VERSION ADDS

The story is sometimes simplified as an immediate triumph; in reality, the resistance persisted for decades, and Semmelweis's own conduct — increasingly hostile public accusations against fellow physicians — likely contributed to his professional isolation alongside the establishment's stubbornness.

UNCERTAIN

The precise cause of death and the extent of any beating by asylum guards remain debated among historians, though sepsis from an infected wound is well documented.

VOCABULARY & PHRASAL VERBS

puerperal fever (n.) a bacterial infection following childbirth, also called childbed fever. Puerperal fever killed a significant share of new mothers in the 1840s.

mortality rate (n.) the proportion of a population that dies from a given cause. The mortality rate fell from 18% to 2%.

vector (n.) a carrier that transmits something harmful, especially disease. Doctors themselves were the vector of the infection.

correlation (n.) a statistical relationship between two things, without necessarily proving cause. He had a strong correlation but no accepted mechanism.

germ theory (n.) the theory that microorganisms cause disease, not widely accepted until decades later. Germ theory was decades away from acceptance.

vindicate (v.) to prove someone was right after they were previously doubted or dismissed. Germ theory eventually vindicated Semmelweis.

come around (phrasal verb) to eventually change one's opinion or position. The establishment might have come around regardless, given more time.

isolate (v.) to separate someone from support or community, often socially. His accusations isolated him further from his colleagues.

overturn (v.) to reverse an established decision, belief or consensus. A clear result wasn't enough to overturn consensus by itself.

PRACTICE

Where the Story Breaks Down

Semmelweis had clear, dramatic before-and-after numbers on his side; most people challenging an entrenched consensus have far murkier evidence and no equivalent tragedy to sharpen the case.

Use the Language

Describe a case where a strong correlation wasn't enough to change minds without an accepted explanation, using correlation, vindicate and come around.

Discuss

Is it possible to be right and still lose, professionally or socially, because you can't yet explain why you're right? What would you do differently than Semmelweis did?

Write

In 100 words, describe how you would present a result you couldn't yet fully explain to a skeptical audience, without alienating them the way Semmelweis did.

SOURCES

A Semmelweis, I., Die Ätiologie, der Begriff und die Prophylaxis des Kindbettfiebers [The Etiology, Concept and Prophylaxis of Childbed Fever] (1861), containing his original mortality statistics.

B Carter, K. C. & Carter, B. R., Childbed Fever: A Scientific Biography of Ignaz Semmelweis, Transaction Publishers (2005).

C Vienna General Hospital historical mortality records, First and Second Obstetrical Clinics, 1841–1849.

THE CONTRAST

Story 6: Semmelweis vs. Thomas Kuhn

Thomas Kuhn was a 20th-century historian of science, known for the idea that scientific communities don't abandon an established framework just because contrary evidence appears — they tend to keep the framework until enough anomalies force a wholesale change in how the field thinks, a shift he called a "paradigm shift."

Story 6: Semmelweis: Evidence can reveal that accepted practice is wrong, and a clear enough result should be able to speak for itself.

Thomas Kuhn: An established framework shapes what a field is even willing to count as evidence — a result that doesn't fit the current framework can be dismissed rather than absorbed, however strong it is.

Why is good evidence sometimes not enough, on its own, to change a system?

THIS CONNECTS TO

Story 9: The Planes That Didn't Return (Wald) SAME MECHANISM

Both reached a conclusion that contradicted what everyone else was reading directly off the same visible evidence.

Story 3: The Bicycle Makers Who Wanted to Fly (Wright Brothers) LIMITATION

The Wrights convinced the world with a single physical demonstration anyone could watch. Semmelweis had a result just as real but nothing that could be demonstrated in a moment — acceptance depended on a mechanism (germ theory) that didn't exist yet, and took decades longer.

Story 36: Verify the Calculation (Katherine Johnson) LIMITATION

Johnson's verification worked partly because it plugged into a method — orbital mechanics — the whole field already trusted. Semmelweis had no accepted method to plug his result into, which is a large part of why his evidence, unlike hers, wasn't enough by itself.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

A Vienna maternity ward run by doctors had a mortality rate three to five times higher than the midwife-run ward next door.

T — Task

Semmelweis needed to find the cause of the difference and stop it, without any accepted theory of disease to guide him.

A — Action

After a colleague died from an infection resembling the one killing new

mothers, Semmelweis concluded doctors were carrying invisible contamination from autopsies to patients, and ordered mandatory handwashing in chlorinated lime.

R — Result

Mortality fell from roughly 18% to roughly 2% within months — but the medical establishment rejected the explanation for decades, and it cost Semmelweis his career and, ultimately, his life.

Level 2 — Make It Your Story

> *Tell me about a time when you had evidence that others did not initially accept.*

Use STAR to answer. Think about: work, university, a project, a team decision, a proposal you had to defend, a time you were right too early.

TAKE THE IDEA FURTHER

> *Where in your work might you need more than good evidence — you might need an accepted explanation — before people will act on it?*

7

The Telescope That Moved the Earth

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Galileo Galilei did not invent the telescope; Dutch spectacle-makers had already built working versions by 1608. What Galileo did, working from secondhand descriptions of the Dutch device in 1609, was rapidly improve the design — reaching roughly 20x to 30x magnification within months, far beyond the 3x instruments then circulating — and then do something almost nobody with access to a telescope had yet done: point it at the sky and look, carefully and repeatedly, rather than simply marvelling at it as a novelty for viewing distant ships.

What he found directly contradicted the accepted cosmology of his time. The prevailing Aristotelian model held that celestial bodies were perfect, unchanging spheres, and that everything in the heavens orbited a stationary Earth. Galileo's telescope showed the Moon was not a smooth sphere but a rough, cratered, mountainous world, its terminator line uneven exactly the way a mountainous surface would produce shadows. It showed Jupiter was orbited by four moons of its own — a miniature system of bodies clearly not circling the Earth, which meant Earth was demonstrably not the single centre of all orbital motion in the universe. Later observations of Venus showed the planet went through a full set of phases, similar to the Moon's, which was only possible if Venus orbited the Sun rather than the Earth.

None of this alone proved the Copernican model — that the Earth orbits the Sun — correct. What it did was make the entrenched geocentric model increasingly difficult to defend on its own terms, using evidence anyone with

a good telescope and clear skies could, in principle, verify for themselves. Galileo published his findings in 1610 in a short work called Sidereus Nuncius (The Starry Messenger), largely image-based, so that readers could see the crater shadows and the changing positions of Jupiter's moons from night to night rather than simply take his word for it.

The response from parts of the astronomical and religious establishment was not primarily "your telescope is broken" — although some contemporaries did initially doubt the instrument's reliability — but something closer to "even if what you see is real, it does not require us to abandon the Earth-centred model." Galileo's later, more explicit and more combative advocacy for Copernican heliocentrism, particularly in his 1632 Dialogue Concerning the Two Chief World Systems, led to his trial by the Roman Inquisition in 1633, a formal finding of suspected heresy, and house arrest for the remainder of his life. He was not condemned for building a telescope or for the moons of Jupiter. He was condemned for insisting, publicly and unyieldingly, that the evidence required a specific and religiously contentious conclusion — that the Earth itself was in motion.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What conclusion in your work have you avoided stating clearly, even though your own evidence points there, because the implication would be unwelcome to people with authority over you?*

KEY TAKEAWAYS

1. Galileo's contribution was not the telescope itself but the discipline of pointing it carefully at a specific question and publishing verifiable, reproducible observations.
2. Evidence that undermines a model does not automatically force acceptance of the alternative model — the geocentric establishment absorbed the moons of Jupiter without abandoning geocentrism.
3. The conflict that ended Galileo's freedom was less about the observations themselves than about his insistence on a specific, unwelcome interpretation of what they meant.

SUMMARY

Galileo rapidly improved the newly invented telescope and used it to observe lunar craters, the four largest moons of Jupiter, and the phases of Venus — findings that directly undermined the accepted Earth-centred model of the universe. He published his observations so others could verify them for themselves. The evidence alone did not force acceptance of a Sun-centred model, and Galileo's later, forceful public advocacy for that conclusion led to his trial and house arrest in 1633.

STORY CHECK

ESTABLISHED

Galileo's 1609–1610 telescopic observations, his publication of Sidereus Nuncius, and his 1633 trial and condemnation are all extensively documented in primary sources, including his own writings and Inquisition records.

POPULAR VERSION ADDS

The observations themselves are often presented as immediately and universally convincing; in reality, several contemporaries questioned the telescope's reliability, and the broader astronomical community took decades to shift, with some serious astronomers proposing hybrid geocentric-heliocentric models as a middle path.

UNCERTAIN

Popular retellings sometimes conflate "being condemned for the discoveries" with the more accurate "being condemned for the later, explicit advocacy" — a meaningful distinction historians are careful to maintain.

VOCABULARY & PHRASAL VERBS

magnification (n.) the degree to which an optical instrument enlarges an object's apparent size. Galileo achieved far greater magnification than earlier telescopes.

terminator (n.) the line dividing the illuminated and dark sides of a moon or planet. The Moon's terminator line was uneven, suggesting mountains.

geocentric (adj.) describing a model with Earth at the centre. The geocentric model held that everything orbited the Earth.

heliocentric (adj.) describing a model with the Sun at the centre. Copernicus had proposed a heliocentric model decades earlier.

entrenched (adj.) firmly established and resistant to change. The entrenched geocentric model absorbed the new evidence slowly.

reproducible (adj.) capable of being repeated and verified independently by others. He published reproducible, image-based observations.

unyielding (adj.) refusing to change position despite pressure. His unyielding public advocacy led to his trial.

hold water (phrase) for an argument or claim to be logically sound. The old model no longer held water once the moons of Jupiter were observed.

come under fire (phrasal verb) to be strongly criticized or attacked. Galileo came under fire not for his telescope but for his conclusions.

PRACTICE

Where the Story Breaks Down

Galileo's evidence was visual and directly reproducible by anyone with a good telescope; most modern disagreements involve evidence that is statistical, indirect or contested in its interpretation, making "just look for yourself" much harder to apply.

Use the Language

Describe a case where evidence alone wasn't enough to change an entrenched view, using entrenched, hold water and come under fire.

Discuss

Is there a difference, in your own experience, between presenting evidence and insisting on its unwelcome interpretation? Which is riskier?

Write

In 100 words, describe a piece of evidence in your field that people can verify for themselves, the way Galileo's readers could look through a telescope.

SOURCES

A Galilei, G., Sidereus Nuncius [The Starry Messenger], Venice (1610).

A Records of the Roman Inquisition trial of Galileo Galilei (1633), Vatican Apostolic Archive.

B Heilbron, J. L., Galileo, Oxford University Press (2010).

THE CONTRAST

Story 7: Galileo vs. Story 6: The Doctor Who Wasn't Supposed to Be Right (Semmelweis)

Story 7: Galileo: Had evidence and chose to publish the conclusion clearly, accepting the professional and personal risk of stating it plainly.

Story 6: The Doctor Who Wasn't Supposed to Be Right (Semmelweis): Had evidence and stated the conclusion just as plainly — and paid a comparable price for it.

If speaking the conclusion clearly costs you professionally either way, does it matter how carefully you word it?

THIS CONNECTS TO

Story 4: The Map That Challenged a Theory (John Snow)

SAME MECHANISM

Both built a case through careful, direct observation, then stated the conclusion the observation pointed to, rather than the conclusion that was comfortable.

Story 51: Predict From Similar Projects (Flyvbjerg) SAME MECHANISM

Both involve stating an unwelcome conclusion that the data clearly supports, in the face of institutions that would rather not hear it.

Story 43: Seeing the Smartphone Too Early (Nokia) LIMITATION

Galileo's chapter is about the cost of stating a true conclusion. Nokia's is almost the mirror image — the cost of having a true signal internally and choosing not to state it at all.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Galileo built an improved telescope and turned it toward the night sky at a time when the prevailing theory held that the Earth was the fixed centre of the universe.

T — Task

He needed to decide what to do with observations that clearly contradicted that theory.

A — Action

He recorded and published his observations — including moons orbiting Jupiter, which could not be circling the Earth — and stated the implication clearly rather than hedging it.

R — Result

His observations became key evidence for a heliocentric model of the solar system, and the conflict with religious and academic authority ultimately led to his trial and house arrest.

Level 2 — Make It Your Story

> *Tell me about a time when you stated a conclusion clearly, even though you knew it would be unwelcome.*

Use STAR to answer. Think about: a report, a recommendation, a piece of feedback, a finding you didn't soften.

TAKE THE IDEA FURTHER

> *What conclusion in your own work have you avoided stating clearly, even though your own evidence points there?*

8

The Patent Clerk and the Train

> **DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED**
> Verified through primary sources, patents, or independent contemporaneous accounts.

In 1905, Albert Einstein was not a professor. He was a twenty-six-year-old technical examiner, third class, at the Swiss patent office in Bern, evaluating applications for mechanical devices — a number of which, notably, concerned the newly urgent problem of synchronizing clocks across railway networks and telegraph lines, a technical detail some historians believe sharpened his thinking on the exact question he would soon revolutionize: what does it actually mean for two events to happen "at the same time"? Einstein did not have access to a research laboratory, an experimental particle accelerator, or any of the equipment that physics would later depend on. What he had was a habit of constructing precise imaginary scenarios and following their logical consequences rigorously, regardless of how strange the destination. One of his most famous, developed and refined over several years, imagined a person riding alongside a beam of light, moving at the same speed. What would that light look like, from that rider's perspective? Classical physics offered a strange answer: a beam of light that appeared frozen, stationary, neither moving nor oscillating — a description that violated everything electromagnetic theory said light should be.

Rather than dismiss the paradox as too strange to be meaningful, Einstein pursued its implications directly. If light could not consistently appear stationary regardless of the observer's own speed, then perhaps the assumption doing the damage was not about light at all, but about time and simultaneity — the unexamined assumption that two events happening "at

the same moment" means the same thing to every observer, regardless of how fast they are moving relative to each other. In his 1905 paper on special relativity, published in the journal Annalen der Physik while he still held his patent office job, Einstein showed that simultaneity is relative: two events that appear simultaneous to one observer can appear to happen at different times to an observer moving at a different velocity, and that the speed of light itself, not the flow of time, is the constant that stays the same for everyone.

The thought experiment did not replace mathematics or experimental verification — Einstein's paper included rigorous derivations, and the theory's predictions were tested and confirmed by astronomers observing starlight bending during a solar eclipse in 1919, the result that made Einstein a global public figure almost overnight. What the thought experiment provided was something a chalkboard of equations alone rarely supplies: a concrete, almost physically felt scenario that let Einstein and, later, his readers actually locate the place where an old assumption was quietly doing work nobody had thought to question.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What "obviously true" assumption in your field have you never actually tested by imagining, concretely, what would happen if it turned out to be false?*

KEY TAKEAWAYS

1. A rigorous imagined scenario, followed to its logical conclusion, can locate a hidden assumption that abstract argument alone tends to leave undisturbed.

2. Einstein's day job evaluating clock-synchronization patents may have sharpened, rather than distracted from, the exact question his physics would answer.
3. The thought experiment was the starting point for locating the problem, not a substitute for the mathematical derivation and experimental confirmation that followed it.

SUMMARY

Working outside academia as a patent clerk, Einstein used a thought experiment — imagining travel alongside a beam of light — to expose a hidden assumption in classical physics: that simultaneity is the same for every observer regardless of speed. Following that imagined scenario's logical consequences led to his 1905 theory of special relativity, a theory he extended into general relativity in 1915, whose prediction of light bending was confirmed during the 1919 solar eclipse. The thought experiment located the flawed assumption; rigorous mathematics and experimental testing did the rest.

STORY CHECK

ESTABLISHED

Einstein's employment at the Swiss patent office in Bern from 1902–1909 and his 1905 paper "On the Electrodynamics of Moving Bodies" are extensively documented; the 1919 eclipse expedition confirming light-bending is independently recorded by the Royal Astronomical Society.

POPULAR VERSION ADDS

The specific "riding alongside a beam of light" thought experiment is often dated precisely to his teenage years (commonly cited as age sixteen) based on

Einstein's own later autobiographical writing — a retrospective account written decades after the fact, not a contemporaneous record.

UNCERTAIN

How directly Einstein's patent-office work on clock synchronization influenced his 1905 paper is debated among historians; the connection is plausible and widely discussed but not something Einstein explicitly confirmed in writing.

VOCABULARY & PHRASAL VERBS

thought experiment (n.) an imagined scenario used to explore the logical consequences of an idea. His thought experiment exposed a hidden assumption.

simultaneity (n.) the property of two or more events happening at the same time. Relativity showed that simultaneity is relative, not absolute.

derivation (n.) a step-by-step mathematical proof leading to a conclusion. The paper included rigorous derivations, not just the thought experiment.

velocity (n.) speed in a specified direction. Observers moving at different velocities can disagree about simultaneity.

oscillate (v.) to move or vary repeatedly between two states. Light was supposed to oscillate, not appear frozen.

paradox (n.) a statement or situation that seems to contradict itself yet may be true. He treated the paradox as worth pursuing, not dismissing.

hold up (phrasal verb) to remain true or valid under scrutiny. The old assumption didn't hold up once he followed it through.

carry through (phrasal verb) to complete something by following it to its logical end. He carried the scenario through to its uncomfortable conclusion.

locate (v.) to identify the specific position or source of something. The thought experiment helped him locate the flawed assumption.

PRACTICE

Where the Story Breaks Down

Einstein's thought experiment eventually required rigorous mathematics most readers cannot independently reproduce; an intuitive scenario is a starting point for insight, not proof on its own.

Use the Language

Describe an "obviously true" assumption in your field and a thought experiment that might test it, using simultaneity (or a suitable substitute term), hold up and locate.

Discuss

Has an outsider's perspective — someone without formal standing in your field — ever seen something insiders had missed? Why might that happen?

Write

In 100 words, construct a concrete thought experiment (imagine a specific scenario and follow it through) that tests an assumption in your own work.

SOURCES

A Einstein, A., "Zur Elektrodynamik bewegter Körper" [On the Electrodynamics of Moving Bodies], Annalen der Physik 17 (1905).

A Dyson, F. W., Eddington, A. S. & Davidson, C., "A Determination of the Deflection of Light by the Sun's Gravitational Field," Philosophical Transactions of the Royal Society A (1920), reporting the 1919 eclipse results.

B Isaacson, W., Einstein: His Life and Universe, Simon & Schuster (2007).

THE CONTRAST

Story 8: Einstein vs. Story 49: Invert the Problem (Munger)

Story 8: Einstein: Used a rigorous imagined scenario — chasing a beam of light — followed all the way to its logical conclusion, to find a hidden assumption.

Story 49: Invert the Problem (Munger): Used a different kind of imagined scenario — deliberately imagining failure — to find hidden risks rather than hidden assumptions.

Are thought experiments better used to test whether an assumption is true, or to test what would happen if a plan went wrong?

THIS CONNECTS TO

Story 1: The Rocket That Was Too Expensive (Musk) SAME MECHANISM
Both worked by refusing an inherited framing and reasoning from first principles rather than the accepted description of how things work.

Story 51: Predict From Similar Projects (Flyvbjerg) LIMITATION
Einstein's thought experiment worked from pure reasoning, with no data at all. Flyvbjerg's method is the opposite — pure data, no imagination — which shows there's more than one legitimate route to correcting an assumption everyone else accepts.

Story 46: Science With Almost Nothing (Curie) SAME MECHANISM
Both worked at the edge of what their era's instruments and institutions could support, and got somewhere real anyway through sheer rigour.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

As a young patent clerk with no university position, Einstein kept returning to a specific imagined scenario: what would it look like to travel alongside a beam of light.

T — Task

He needed to work out what that scenario implied about the nature of time and space, since the answer seemed to contradict the physics everyone accepted.

A — Action

He followed the thought experiment rigorously to its logical conclusion, rather than dismissing the strange implication.

R — Result

The result was his 1905 special theory of relativity, developed outside the traditional academic system, extended into general relativity in 1915, whose prediction of light bending was confirmed during the 1919 solar eclipse.

Level 2 — Make It Your Story

> *Tell me about a time when you followed an assumption to its logical conclusion and found it didn't hold up.*

Use STAR to answer. Think about: a plan, a policy, an "obviously true" rule you actually tested by imagining it through.

TAKE THE IDEA FURTHER

What "obviously true" assumption in your field have you never actually tested by imagining, concretely, what would happen if it turned out false?

9

The Planes That Didn't Return

DOCUMENTATION LEVEL 2 — SELF-REPORTED OR TRANSFERRED

The account rests mainly on the protagonist's own retelling, or the idea was documented but its original application was never adopted.

By 1943, the United States Army Air Forces had a straightforward engineering problem and, they believed, a straightforward way to solve it. Bombers were returning from missions over Europe covered in bullet and flak damage, and commanders wanted to add extra armour plating to protect the aircraft better. Armour is heavy, and a plane can only carry so much of it before the added weight costs more in fuel and maneuverability than the protection is worth. So the natural approach was to study exactly where the returning planes had been hit, and reinforce those specific locations.

The data, once mapped onto a diagram of a bomber's outline, was unambiguous. Damage clustered heavily on the wings, the fuselage, and around the tail gunner's position. The engines and cockpit, showed comparatively little damage. The engineering conclusion seemed obvious: armour the wings and the tail, since that is where the aircraft was demonstrably getting hit. The military's Statistical Research Group, based at Columbia University and staffed with mathematicians and statisticians pulled from academia for the war effort, was asked to review the analysis before it became policy. Among them was Abraham Wald, a Hungarian-born mathematician who had fled Nazi-annexed Austria in 1938 and rebuilt his career in the United States. Wald looked at the same diagram everyone else had studied and reached the opposite conclusion.

The flaw, as Wald identified it, was in what the sample actually contained. Every data point on that diagram came from a bomber that had survived its

mission and made it back to base to be examined. A bullet hole in the wing was evidence that a bomber could take a hit there and still fly home — which meant the wing, whatever the raw hole-count suggested, was not the area where damage was fatal. The truly dangerous locations were the ones that were nearly absent from the data: the engines, the cockpit, the fuel system.

Wald's reasoning was that planes hit in those places did not return at all, and so contributed zero data points to a chart built entirely from survivors. The visible pattern was not showing where bombers were vulnerable. It was showing, almost exactly, where they were not. Accounts say the military shifted armour toward the engines and fuel systems rather than the wings and tail. Wald did not deliver this as a single dramatic insight; his actual contribution was a series of statistical memoranda developing a mathematical method for estimating the survival probability of a hit in a given location — a far more technical and less quotable piece of work than the version usually told.

The vivid line most often attributed to him — "the armour doesn't go where the bullet holes are; it goes where the bullet holes are not" — captures the idea precisely but does not appear verbatim in his surviving technical papers; it is almost certainly a later paraphrase that has hardened, through repetition, into a direct quotation. What makes the reasoning durable well beyond its wartime origin is that the trap it identifies — studying only what survived a selection process and drawing conclusions from a sample that has already been silently filtered — appears constantly outside aviation, in business, medicine and everyday judgment, wherever the failures are invisible and only the successes get counted.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What in your organization do you study by looking only at the "survivors" — the completed projects, the retained customers, the successful hires — while the failures leave no data at all?*

KEY TAKEAWAYS

1. A data set built only from survivors is systematically biased in a specific, predictable direction: it underrepresents whatever caused failure.
2. The absence of damage in the data was more informative than its presence, once the selection process producing the data was properly understood.
3. Wald's actual contribution was a mathematical model for estimating survival probability, not a single intuitive observation — the popular one-line version simplifies a technical body of work.

SUMMARY

Asked to help decide where to add armour to bombers, based on damage patterns from planes that returned, Abraham Wald recognized that the data was drawn entirely from survivors and that damage-free areas on returning planes likely marked the truly fatal hit locations, since planes struck there never made it back to be counted. The military reinforced the engines and fuel systems rather than the wings and tail. The famous summary quotation attributed to Wald is a widely repeated paraphrase rather than a verified direct quote.

STORY CHECK

ESTABLISHED

Wald worked at the Statistical Research Group at Columbia University during WWII and produced technical memoranda on aircraft vulnerability and survival-probability estimation; accounts say the military shifted its armour placement after SRG recommendations, though the record of implementation is thin.

POPULAR VERSION ADDS

The pithy "goes where the bullet holes are not" quotation is not found verbatim in Wald's surviving papers and is treated by historians as a later paraphrase of his argument rather than a direct quote.

UNCERTAIN

Precisely how many lives or aircraft the change in armour placement saved; no rigorous wartime count exists, and the number circulated in retellings is an estimate, not a documented figure.

VOCABULARY & PHRASAL VERBS

fuselage (n.) the main body of an aircraft, excluding wings and tail. Damage clustered on the fuselage and wings.

flak (n.) anti-aircraft artillery fire, or the shrapnel it produces. The bombers returned covered in flak damage.

sample (n.) a subset of data used to represent a larger group. The sample contained only aircraft that had survived.

selection process (n.) a mechanism that filters which items or cases end up in a data set. Returning safely was itself a selection process.

to filter out (phrasal verb) to remove certain items from a set, often silently or automatically. The missing planes had been filtered out of the sample before anyone saw the data.

counterintuitive (adj.) the opposite of what common sense would predict. Wald's recommendation was counterintuitive: armour where the damage wasn't.

paraphrase (n./v.) a restatement of something in different words, not a direct quotation. The famous line is widely treated as a later paraphrase.

to harden into (phrasal verb) for a claim to become fixed and unquestioned through repetition. The paraphrase hardened into a direct quote after decades of retelling.

make it back (phrasal verb) to return successfully, especially from danger. Planes hit in the engine rarely made it back to base.

PRACTICE

Where the Story Breaks Down

Wald's mathematics required a specific, well-defined population (all aircraft sent on a mission) and clear knowledge of what counted as "not returning." Most real-world "missing data" problems are far murkier about what the full population even was.

Use the Language

Describe a situation where you only see the "survivors," using sample, selection process and filter out.

Discuss

Where in your own field do you only ever see the successes — the retained customers, the completed projects, the people who stayed? What would the "missing" cases look like if you could see them?

Write

In 100 words, describe a real or plausible example of survivorship bias in your own work, being specific about what the "invisible failures" would be.

SOURCES

A Wald, A., "A Method of Estimating Plane Vulnerability Based on Damage of Survivors," Statistical Research Group, Columbia University, Memorandum CRC 432 (1943), reissued by the Center for Naval Analyses (1980).

B Mangel, M. & Samaniego, F. J., "Abraham Wald's Work on Aircraft Survivability," Journal of the American Statistical Association, 79(386), 259–267 (1984) — the standard scholarly source distinguishing Wald's actual mathematics from the popularized retelling.

D The widely circulated "armour doesn't go where the bullet holes are" quotation appears in numerous secondary retellings without a verified primary citation.

THE CONTRAST

Story 9: Abraham Wald vs. Daniel Kahneman

Story 9: Abraham Wald: Ask what information is missing. The danger is in the data you never see.

Daniel Kahneman: Be careful about how your mind interprets the information you already have. The danger is in how you process what's right in front of you.

Is a poor decision more likely to result from missing information — or from biased interpretation of the information you have?

THIS CONNECTS TO

Story 36: Verify the Calculation (Katherine Johnson) SAME MECHANISM
Both cases show the danger of trusting information without asking what has not been checked or what might be absent.

Story 4: The Map That Challenged a Theory (John Snow)

SAME MECHANISM

Both cases depend on reaching a conclusion by looking beyond the obvious pattern and examining the evidence — or its absence — carefully.

Story 6: The Doctor Who Wasn't Supposed to Be Right (Semmelweis)

LIMITATION

Wald changed a conclusion by reframing what the data showed. Semmelweis shows that even an airtight reframing doesn't guarantee the system will accept it — being right is not the same as being believed.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

The U.S. military wanted to add armour to its aircraft. Engineers studied returning planes and found bullet holes concentrated in certain areas.

T — Task

They needed to decide where additional armour would provide the greatest protection without making the aircraft unnecessarily heavy.

A — Action

Abraham Wald challenged the assumption that the visible damage showed where aircraft were most vulnerable. He considered the missing data — the planes that had been hit and never returned.

R — Result

The analysis showed that armour was needed in the areas where returning aircraft showed little or no damage, including the engines and cockpit.

Level 2 — Make It Your Story

> *Tell me about a time when you realized you were working with incomplete information.*

Use STAR to answer. Think about: work, university, a project, a team decision, a customer, a problem you misunderstood, a mistake you corrected.

TAKE THE IDEA FURTHER

> *Where in your life or work might you currently be seeing only the "survivors"?*

PART II

Test, Don't Assume

10

The O-Ring in the Ice Water

> **DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED**
> Verified through primary sources, patents, or independent contemporaneous accounts.

On 28 January 1986, the Space Shuttle Challenger broke apart 73 seconds after launch, killing all seven crew members. The immediate cause, investigators would eventually confirm, was the failure of a rubber O-ring seal in one of the solid rocket boosters — a seal that failed to expand properly and contain hot gas because the temperature at launch, 36°F, was far colder than any previous shuttle launch.

Physicist Richard Feynman, appointed to the presidential Rogers Commission investigating the disaster, became famous for a single televised moment during a February 1986 hearing: he clamped a sample of O-ring material, submerged it in a glass of ice water, then released the clamp on camera and showed that the compressed rubber did not spring back to its original shape — visual, immediate proof that the seal material became dangerously rigid in cold conditions. The demonstration cut through days of dense technical testimony and became the moment the investigation is remembered for.

The version of this story usually told treats it as a spontaneous flash of genius — Feynman, alone in the room, seeing what nobody else could see. The documented sequence is different, and involves a name that stayed out of the public record for over two decades. Fellow commission member General Donald Kutyna, working on a car in his garage the night before a key hearing, mentioned to Feynman, almost in passing, that cold weather made rubber engine seals stiffen. Feynman recognized the parallel to the O-ring question

immediately. What Kutyna did not reveal at the time was where his own suspicion about cold and Orings had originated: a wordless exchange with fellow commissioner and astronaut Sally Ride, who had silently passed him a NASA document showing O-ring resiliency dropping sharply at low temperatures — data Ride had access to through internal channels but could not risk revealing directly herself without jeopardizing colleagues at NASA who had shared it with her.

Feynman's demonstration itself was also more deliberate than its "spontaneous genius" reputation suggests. He requested a glass of ice water in advance of the hearing and took a piece of O-ring from the joint model on the table and used a C-clamp he had bought, purchased at a hardware store, specifically prepared to make the point visible on camera at the moment testimony turned to booster seal performance. Kutyna kept Ride's role private for years, revealing it publicly only after her death in 2012 — meaning the popular story of a lone scientist's improvised insight quietly erased the two people, one of them a NASA astronaut protecting her own colleagues, whose contributions made the demonstration possible in the first place.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Whose contribution to a "breakthrough" moment in your organization has gone uncredited because they couldn't safely take credit for it themselves?*

KEY TAKEAWAYS

1. Feynman's demonstration was planned and prepared in advance, not a spontaneous improvisation, despite its reputation.

2. The insight that made the demonstration possible originated with Sally Ride, who could not reveal it directly without risk to colleagues, and reached Feynman only through an intermediary.
3. A vivid, visual demonstration of a single critical variable — temperature — cut through technical testimony in a way that data tables alone had not.

SUMMARY

Feynman's ice-water demonstration during the Challenger hearings became the iconic moment proving O-ring seals failed in cold temperatures, but the tip that led to it originated with astronaut Sally Ride, who passed critical NASA data to General Kutyna in secret to protect colleagues who had shared it with her; Kutyna in turn steered Feynman toward the insight. The demonstration itself was carefully prepared rather than spontaneous. Ride's role remained unknown to the public until Kutyna revealed it after her death in 2012.

STORY CHECK

ESTABLISHED

The ice-water demonstration, the 36°F launch temperature, and the O-ring failure as root cause are all confirmed in the Rogers Commission's official report; Kutyna's later public account of Ride's role is on record in multiple interviews he gave after her death.

POPULAR VERSION ADDS

The demonstration is frequently retold as spontaneous, unplanned genius; documented accounts confirm Feynman requested ice water and brought his materials in advance.

UNCERTAIN

The exact private conversation between Ride and Kutyna is known only through Kutyna's own later recollection, since Ride never spoke publicly about it before her death.

VOCABULARY & PHRASAL VERBS

resiliency (n.) the ability of a material to return to its original shape after being deformed. O-ring resiliency dropped sharply in cold temperatures.

intermediary (n.) a person who passes information or acts between two other parties. Ride's information reached Feynman through an intermediary.

jeopardize (v.) to put something at risk of harm or loss. Ride couldn't reveal the data without jeopardizing her colleagues.

root cause (n.) the fundamental underlying reason for a problem, as opposed to a symptom. The O-ring failure was identified as the root cause.

on the record (phrase) officially documented or stated for attribution. Kutyna's account is on the record in several interviews.

cut through (phrasal verb) to overcome complexity or confusion decisively. The demonstration cut through days of dense testimony.

pass on (phrasal verb) to transfer information from one person to another. Ride passed the data on to Kutyna without a word.

stay out of (phrasal verb) to remain absent or excluded from something, often deliberately. Ride's name stayed out of the public record for over two decades.

come to light (phrase) to become known or revealed, often after being hidden. Ride's role only came to light after her death.

PRACTICE

Where the Story Breaks Down

Feynman had the personal fame, credibility and platform to make a single demonstration land publicly; Ride had the same technical insight earlier but could not deliver it the same way without professional risk — a difference in position, not in ability.

Use the Language

Describe a situation where someone's real contribution was hidden or indirect, using intermediary, jeopardize and come to light.

Discuss

Why might someone with critical information choose not to reveal it directly, even at real cost to their own credit?

Write

In 100 words, describe a "critical variable" in a current problem you face — the one factor that, if isolated and demonstrated clearly, would make the issue obvious to others.

SOURCES

A Rogers Commission, Report of the Presidential Commission on the Space Shuttle Challenger Accident, especially Feynman's personal appendix, "Personal Observations on Reliability of Shuttle" (1986).

B Kutyna, D. J., interviews given after 2012 confirming Sally Ride's role, including accounts republished by Popular Mechanics and NPR.

B Feynman, R. P. (with Ralph Leighton), What Do You Care What Other People Think?, W. W. Norton (1988).

THE CONTRAST

Story 10: Feynman vs. Story 14: The Moth in the Machine (Grace Hopper's story)

Story 10: Feynman: A dramatic, memorable public demonstration turns out to have been carefully planned in advance, not spontaneous — the popular version overstates the improvisation.

Story 14: The Moth in the Machine (Grace Hopper's story): A dramatic, memorable origin story turns out to be largely inaccurate — the popular version overstates a different person's role entirely.

When a famous demonstration or origin story turns out to be simplified, does correcting it diminish the achievement, or protect it?

THIS CONNECTS TO

Story 44: Build Learning Into the System (Kaizen) LIMITATION
Feynman's demonstration exposed a single dramatic failure point. Kaizen addresses a different problem — building the ongoing habit of surfacing small issues before they ever reach that scale.

Story 33: The Checklist (Gawande) SAME MECHANISM
Both address failures that came from institutional pressure and routine overconfidence overriding a known, specific technical risk.

Story 43: Seeing the Smartphone Too Early (Nokia) CONTRAST
Feynman's chapter is about a warning that existed but wasn't acted on in time to prevent disaster. Nokia's is about a signal that existed even earlier, with time to act — and still wasn't.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

The Space Shuttle Challenger disaster was traced to an O-ring seal that failed in cold temperatures, and the presidential commission's investigation risked becoming a bureaucratic, inconclusive process.

T — Task

Feynman needed a way to demonstrate the flaw's mechanism clearly, in a public hearing, in terms anyone could understand.

A — Action

He clamped a piece of O-ring material and dipped it in ice water during a televised hearing to show it lost its resilience in cold conditions.

R — Result

The demonstration made the technical cause immediately visible to the public and the committee, though it had in fact been prepared in advance with help from an astronaut and a general.

Level 2 — Make It Your Story

> *Tell me about a time when you had to make a technical or complex problem immediately understandable to people without the technical background.*

Use STAR to answer. Think about: a presentation, a demo, a report you had to simplify without losing the substance.

TAKE THE IDEA FURTHER

> *Whose contribution to a "breakthrough" moment in your organization has gone uncredited because they couldn't safely take credit for it themselves?*

11

The Five Whys

> **DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED**
> Verified through primary sources, patents, or independent contemporaneous accounts.

Taiichi Ohno joined Toyota in 1943 and spent the following decades building what would eventually be formalized as the Toyota Production System — a set of manufacturing practices credited with transforming Toyota from a struggling domestic automaker into a global benchmark for efficiency and quality. Among the simplest and most widely copied of his contributions was not a machine or a formula, but a discipline of asking a single question, repeatedly, until a superficial problem statement turned into an actual, actionable cause.

Ohno's own frequently cited example concerns a machine on the assembly line that stopped working. The first "why" — why did the machine stop? — produces an easy, mechanical answer: the circuit overloaded, tripping a fuse. A team satisfied with that answer replaces the fuse and moves on, and the machine stops again days or weeks later. Ohno's method insists on continuing. Why did the circuit overload? Because the bearing wasn't sufficiently lubricated. Why wasn't it lubricated? Because the lubrication pump wasn't circulating enough oil. Why wasn't it circulating enough? Because the pump's intake was clogged with metal shavings.

And why was it clogged with shavings? Because the pump had no filter to catch debris in the first place. Five questions in, the "problem" has moved from a blown fuse — a symptom, fixable in minutes and certain to recur — to a missing filter, a design flaw that, once corrected, prevents the entire chain of failures from happening again. The number five is not a strict rule; Ohno

himself described it as a rough guideline, and some problems resolve to a genuine root cause in three iterations, others in seven or eight. The discipline is not counting to five. It is refusing to accept the first mechanically true answer as the actual explanation, because the first answer is almost always a symptom of something further upstream.

The technique's power lies partly in what it prevents. Organizations under time pressure have a strong incentive to stop at the first plausible explanation, because the first explanation is usually the fastest one to act on and the easiest one to present as resolved. A blown fuse is a five-minute fix that looks, on a status report, exactly like a solved problem. A missing filter requires a design change, a supplier conversation, and possibly a retrofit across every machine on the line — a genuinely larger undertaking that only becomes visible if someone keeps asking why after the convenient answer has already arrived.

Toyota's broader production system, of which the Five Whys was one component among many — including jidoka (the practice of stopping production the instant a defect is detected) and just-in-time inventory — became, by the 1980s, the subject of intense study by American and European manufacturers struggling to understand how a company with far less capital had overtaken them on both cost and quality. The Five Whys survived that transplant more easily than almost any other Toyota practice, in part because it requires no capital investment and no factory redesign — only the discipline to keep asking one more question after the room is ready to move on.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *In your organization's last "root cause" fix, how many "whys" deep did the investigation actually go before everyone agreed the problem was solved?*

KEY TAKEAWAYS

1. The first mechanically true explanation for a failure is usually a symptom, not the actual root cause.
2. Five is a guideline, not a rule — the discipline is continuing past the first convenient stopping point, however many iterations that takes.
3. Organizations under time pressure are structurally biased toward stopping at the fast, visible fix rather than the slower, structural one.

SUMMARY

Taiichi Ohno's Five Whys technique, developed as part of the Toyota Production System, asks "why" repeatedly after a failure until a superficial symptom — a blown fuse — is traced back to its actual structural cause — a missing filter. The number five is a rough guideline rather than a strict rule; the point is resisting the pressure to stop at the first plausible, fast-to-fix answer. The technique spread more widely than most other Toyota practices because it requires no capital investment, only sustained discipline.

STORY CHECK

ESTABLISHED

Ohno's development of the Toyota Production System from the 1940s onward, and his own published account of the Five Whys machine example, are documented in his own writing and Toyota's official corporate history.

POPULAR VERSION ADDS

The technique is sometimes presented as though "five" is a fixed, universal rule; Ohno's own description treats it as an illustrative guideline, with the real number of iterations varying by problem.

UNCERTAIN

Precisely how and when the Five Whys was first formalized as a named technique, versus simply being part of Ohno's general management practice, is not pinned to a single documented moment.

VOCABULARY & PHRASAL VERBS

root cause (n.) the fundamental, structural reason behind a problem, as opposed to its visible symptom. The blown fuse was a symptom, not the root cause.

symptom (n.) a visible sign of an underlying problem, not the problem itself. A tripped fuse is a symptom of a deeper issue.

upstream (adv./adj.) earlier in a process or causal chain. The real cause was further upstream than anyone had checked.

retrofit (n./v.) a modification added to existing equipment after the fact. Fixing it properly required a retrofit across every machine.

jidoka (n.) a Toyota Production System principle of stopping production immediately when a defect is detected. Jidoka complemented the Five Whys as part of the same system.

transplant (n., figurative) here, the adoption of a practice or system into a new organizational context. The Five Whys survived the transplant to Western manufacturers more easily than most Toyota practices.

status report (n.) a summary document showing the current state of a project or issue. A five-minute fix looks resolved on a status report.

move on (phrasal verb) to stop addressing something and proceed to the next task. A team satisfied with the first answer replaces the fuse and moves on.

dig into (phrasal verb) to investigate something thoroughly and persistently. Digging into the problem meant asking one more "why."

PRACTICE

Where the Story Breaks Down

Some failures genuinely don't have a clean five-step causal chain — multiple contributing causes can combine, and forcing a single linear "why" chain onto a genuinely tangled problem can produce a false sense of resolution.

Use the Language

Describe a recent fix at your work that addressed a symptom rather than the root cause, using root cause, symptom and upstream.

Discuss

What pressure in your organization pushes teams to stop at the first plausible explanation for a failure?

Write

In 100 words, apply the Five Whys to a real recurring problem you face, writing out each "why" and each answer.

SOURCES

A Ohno, T., Toyota Production System: Beyond Large-Scale Production, Productivity Press (1988; English translation) — contains Ohno's own machine/fuse example.

B Liker, J. K., The Toyota Way, McGraw-Hill (2004).

C Toyota Motor Corporation, official corporate history of the Toyota Production System (toyota-global.com).

THE CONTRAST

Story 11: The Five Whys vs. Story 16: The Factory That Could Stop Itself (Jidoka)

Story 11: The Five Whys: Finds the root cause after a problem has already occurred, by repeatedly asking why.

Story 16: The Factory That Could Stop Itself (Jidoka): Stops a problem the instant it occurs, before it can even become something to investigate afterward.

Is it more valuable to get better at diagnosing problems after they happen, or to build a system that catches them the moment they start?

THIS CONNECTS TO

Story 2: The Burrs on the Dog (Velcro) SAME MECHANISM
Both treat a small, ordinary irritation as worth serious investigation rather than a nuisance to shrug off.

Story 44: Build Learning Into the System (Kaizen) SAME MECHANISM
Both are Toyota Production System tools built on the same premise — that the first visible explanation for a problem is rarely the real one, and structured, repeated inquiry gets closer to it.

Story 33: The Checklist (Gawande) LIMITATION

Five Whys finds the deep cause of a specific failure. A checklist doesn't find causes at all — it prevents a known failure from recurring regardless of its root cause, which is a different kind of fix for a different kind of problem.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

A specific failure occurred, and the first explanation offered accounted for what happened but not why it kept happening.

T — Task

The team needed to find the actual root cause, not just the most visible mechanical explanation.

A — Action

They asked "why" repeatedly, each time pushing past the previous answer, until they reached a cause that was structural rather than incidental.

R — Result

Addressing that deeper cause, rather than the first symptom identified, prevented the problem from recurring in a way a surface fix would not have.

Level 2 — Make It Your Story

> *Tell me about a time when your team's first explanation for a problem turned out not to be the real cause.*

Use STAR to answer. Think about: a bug, a customer complaint, a process failure, a recurring mistake.

TAKE THE IDEA FURTHER

> *In your organization's last "root cause" fix, how many "whys" deep did the investigation actually go before everyone agreed the problem was solved?*

12

The Plane Built in a Barn

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In June 1943, the United States Army Air Forces gave Lockheed an urgent, narrow problem: German jet fighters were entering service, and the Allies had nothing comparable in the air. Lockheed was asked to design and deliver a working jet fighter prototype — the aircraft that would become the P-80 Shooting Star — in 180 days, roughly a third of the time a conventional aircraft development program of that era typically required.

Clarence "Kelly" Johnson, the engineer Lockheed assigned to lead the project, made an unusual request before agreeing to the deadline: he wanted the team physically separated from Lockheed's main engineering bureaucracy, with the authority to make engineering decisions without routing them through the company's standard multi-layered approval chain. Space was tight at the Burbank plant, so the team was set up in a rented circus tent adjacent to a plastics factory, whose fumes reportedly filled the workspace badly enough that Johnson's team nicknamed the operation "Skunk Works," after a foul-smelling still from the Li'l Abner comic strip popular at the time.

The physical arrangement was almost incidental. What actually mattered was the organizational one. Johnson operated with a small, deliberately co-located team of engineers and machinists working in close physical proximity, so that a design question could be answered by walking twenty feet rather than filing a request and waiting days for a committee review. Decisions that would normally require sign-off from several departments were made on the spot by people with the authority and the technical knowledge to make them.

Johnson later formalized this experience into a set of operating rules — commonly known as Kelly's Fourteen Rules — that explicitly limited the number of people who needed to approve a decision, minimized formal reporting in favour of direct verbal communication, and gave the program manager broad authority over both technical and budget decisions without needing to escalate routine choices upward.

The P-80 prototype, the XP-80, was completed in 143 days and first flew on 8 January 1944, ahead of the already aggressive 180-day target. It did not see combat in World War II, arriving too late to meaningfully affect the war's outcome, but it became the foundation of Lockheed's jet aircraft program and validated an organizational model — a small, empowered, physically co-located team insulated from routine bureaucracy — that the Skunk Works division went on to use for decades of subsequent classified aircraft programs, including the U-2 and SR-71 Blackbird.

The lesson usually drawn from this story is about speed, but the more precise lesson is about what actually consumed the time in conventional development: not engineering difficulty, but the accumulated delay of routing every decision through people who were not close enough to the problem to decide it quickly, and who bore no direct consequence for the days their signature took to obtain.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Which decisions in your organization take the longest not because they're hard, but because they have to pass through people several steps removed from the actual problem?*

KEY TAKEAWAYS

1. Johnson's request was organizational, not technical: fewer approval layers, not smarter engineers.
2. Physical co-location reduced the time cost of a decision from days of routed approval to a twenty-foot walk.
3. The speed gain came from removing friction in the decision-making structure itself, not from working the team harder on the underlying engineering.

SUMMARY

Given 180 days to design and build a prototype jet fighter, Kelly Johnson insisted on physically and organizationally separating his team from Lockheed's standard bureaucracy, working in a rented tent with a small, co-located group empowered to make engineering decisions on the spot. The prototype flew in 143 days, ahead of schedule. Johnson later formalized the approach into a set of operating rules that shaped Lockheed's Skunk Works division for decades of subsequent aircraft programs.

STORY CHECK

ESTABLISHED

The 180-day contract, the 143-day delivery of the XP-80, and Kelly's Fourteen Rules are documented in Lockheed Martin's own corporate archives and Johnson's published memoir.

POPULAR VERSION ADDS

The "circus tent next to a foul-smelling plastics factory" detail is often told as a colourful, almost accidental origin story; the organizational structure —

small team, reduced approval layers, co-location — was the deliberate and replicable part, and mattered far more than the tent itself.

UNCERTAIN

The exact origin of the "Skunk Works" nickname (from the Li'l Abner comic strip) is consistently reported across Lockheed's own sources, though the precise moment it was first used is not independently pinned down.

VOCABULARY & PHRASAL VERBS

co-located (adj.) physically positioned together in the same space. A co-located team could resolve questions without formal routing.

sign-off (n.) formal approval required before a decision or action can proceed. Decisions normally requiring sign-off from several departments were made on the spot.

escalate (v.) to pass a decision or issue up to a higher level of authority. Routine choices didn't need to escalate upward.

insulate (v.) to protect or separate something from an external influence. The team was insulated from routine bureaucracy.

on the spot (phrase) immediately, at the location where a decision is needed. Decisions were made on the spot.

friction (n., figurative) resistance or delay within a process or system. The real target was removing organizational friction, not working harder.

route through (phrasal verb) to send something via an official channel or process. Design questions no longer had to be routed through committees.

ahead of schedule (phrase) earlier than planned or required. The prototype flew ahead of schedule.

bear the consequence (phrase) to be personally affected by the results of a decision or delay. The approvers bore no direct consequence for the delay their signature caused.

PRACTICE

Where the Story Breaks Down

Johnson had an Army contract with wartime urgency behind it, which gave him unusual leverage to demand structural independence; most internal teams asking for the same autonomy have far less institutional leverage to secure it.

Use the Language

Describe a decision in your organization that could move faster with fewer approval layers, using sign-off, escalate and friction.

Discuss

What would a "Skunk Works" version of your own team or department look like — smaller, co-located, and empowered to decide without routing upward?

Write

In 100 words, identify one specific approval step in a process you know well, and argue for who should be allowed to make that decision on the spot instead.

SOURCES

A Lockheed Martin Corporation, Skunk Works official history and "Kelly's 14 Rules" (lockheedmartin.com).

B Johnson, C. L. "Kelly" (with Maggie Smith), Kelly: More Than My Share of It All, Smithsonian Institution Press (1985).

B Rich, B. R. & Janos, L., Skunk Works: A Personal Memoir of My Years at Lockheed, Little, Brown (1994).

THE CONTRAST

Story 12: Skunk Works vs. Story 28: A Different Domain Solves the Problem (Hedy Lamarr)

Story 12: Skunk Works: Solved an urgent problem by removing organizational obstacles — approval layers, distance from decision-makers — around a team that already had the technical skill.

Story 28: A Different Domain Solves the Problem (Hedy Lamarr): Solved an urgent problem by importing a solution from a completely unrelated field, where the technical skill already existed elsewhere.

When a problem is urgent, is it faster to strip the organization down around your own experts, or to look outside your field for an answer that already exists?

THIS CONNECTS TO

Story 17: The Factory That Started Moving (Ford) SAME MECHANISM
Both achieved dramatic speed gains by redesigning how work and decisions moved through the organization, not by asking individuals to work harder.

Story 34: Remove the Friction (Amazon) SAME MECHANISM
Both identify organizational or procedural steps — approval layers, checkout fields — as the real source of delay, not the underlying technical work.

Story 43: Seeing the Smartphone Too Early (Nokia) LIMITATION

Skunk Works shows what a team can do once bureaucratic layers are removed. Nokia's chapter shows the opposite failure mode: even with the right internal team and the right technology, keeping the ordinary approval structure in place can be enough to lose the outcome anyway.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

The U.S. military needed a jet fighter to match new German aircraft, on an urgent wartime timeline.

T — Task

Lockheed engineer Kelly Johnson needed to deliver a working design far faster than the normal defence procurement process allowed.

A — Action

He requested, and was given, a small team working with minimal management layers and direct authority to make decisions, operating out of a rented circus tent rather than a conventional facility.

R — Result

The team delivered a working prototype in 143 days, and the approach became the template for Lockheed's long-running Skunk Works division.

Level 2 — Make It Your Story

> *Tell me about a time when removing organizational layers, rather than adding people or skill, was the thing that actually sped up a project.*

Use STAR to answer. Think about: an approval process, a decision that took too many steps, a small team you protected from bureaucracy.

TAKE THE IDEA FURTHER

> *Which decisions in your organization take the longest not because they're hard, but because they have to pass through people several steps removed from the actual problem?*

13

Solve Backward

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED

Verified through primary sources, patents, or independent contemporaneous accounts.

George Pólya spent much of his career as a mathematics professor, first in Europe and later at Stanford, but his most widely read work was not a proof or a theorem. It was a short 1945 book called How to Solve It, written for students who could follow a completed mathematical argument perfectly well but had no idea how to begin generating one themselves. Pólya had noticed a specific, recurring gap: mathematics was taught almost entirely through finished results, polished and forward-facing, with every trace of the false starts, dead ends and reversed reasoning that actually produced them carefully erased.

Among the handful of heuristics Pólya collected and popularized, one in particular has outlived its original mathematical context by decades: working backward from the goal. Rather than starting from what is currently known and asking what follows from it, the method starts from the desired end state and asks what condition would have to be true immediately before it, then what condition would have to be true before that, continuing the chain backward until it connects to something already known or achievable. A geometry student trying to prove two lines are parallel might struggle for an hour pushing forward from the given information. Asked instead, "what would have to be true for these lines to be parallel?", and then, "what would have to be true for that condition to hold?", the same student often reaches the answer in minutes — because the backward chain is frequently far shorter and more constrained than the forward search.

Pólya was explicit that this was not a novel invention of his own. He was documenting and formalizing a heuristic that skilled mathematicians, going back at least to the ancient Greek geometers, had used instinctively without ever writing it down as a teachable method. His contribution was making the implicit explicit: naming the move, breaking it into repeatable steps, and packaging it so a student without natural mathematical intuition could deliberately apply a technique that talented problem-solvers had previously only stumbled into by feel.

The method's reach beyond mathematics turned out to be considerable. Detective fiction's classic structure — a body is found, and the story works backward through means, motive and opportunity to the killer — is functionally the same manoeuvre applied to narrative rather than proof. Project planners routinely build schedules by starting from a fixed deadline and working backward through the milestones that must precede it, rather than starting from today and hoping forward planning happens to land on the deadline. In each case, the appeal is the same one Pólya identified in 1945: the space of possible starting points is enormous and underconstrained, while the space of conditions that must precede a specific, fixed goal is usually much smaller and easier to search.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What goal are you currently trying to reach by planning forward from today, when working backward from the deadline or outcome might reveal a far more direct path?*

KEY TAKEAWAYS

1. Working backward from a fixed goal narrows the search space more than working forward from an open starting point.
2. Pólya's contribution was not inventing the technique but formalizing an instinct skilled problem-solvers already used, making it teachable to people who didn't have the instinct naturally.
3. The method generalizes well beyond mathematics, to any situation with a clearly defined end state and an unclear path toward it.

SUMMARY

Mathematician George Pólya's 1945 book How to Solve It formalized a set of problem-solving heuristics used by skilled mathematicians for centuries, most notably working backward from a desired goal to determine what conditions must precede it, rather than searching forward from the current state. The backward chain is often far shorter than the forward search because it is more tightly constrained. The technique has since been widely applied outside mathematics, from narrative structure to project planning.

STORY CHECK

ESTABLISHED

Pólya's 1945 publication of How to Solve It and its content, including the working-backward heuristic, are directly documented in the book itself.

POPULAR VERSION ADDS

The technique is sometimes credited to Pólya as an original invention; Pólya's own text explicitly frames it as a formalization of ancient and widely used mathematical practice, not a new discovery.

UNCERTAIN

The precise origin of working-backward reasoning in mathematics predates any single documented source; Pólya himself traces its use to antiquity without citing a specific first instance.

VOCABULARY & PHRASAL VERBS

heuristic (n.) a practical, general problem-solving technique, not guaranteed to work but usually helpful. Pólya collected a set of heuristics for mathematical problem-solving.

dead end (n.) a path or approach that leads nowhere useful. Forward reasoning often leads to dead ends.

constrained (adj.) limited or restricted in scope. The backward search space is far more constrained.

milestone (n.) a significant point or event marking progress toward a goal. Planners work backward through the milestones before a deadline.

formalize (v.) to give a clear, structured, teachable form to something previously informal. Pólya formalized an instinct into a repeatable method.

land on (phrasal verb) to arrive at a particular outcome, often by chance. Forward planning doesn't always land on the deadline.

stumble into (phrasal verb) to arrive at something by accident rather than deliberate method. Skilled mathematicians had stumbled into the technique by feel.

push forward (phrasal verb) to continue advancing with an approach, often with effort. The student pushed forward from the given information without success.

narrow down (phrasal verb) to reduce a range of possibilities to a smaller set. Working backward narrows down the search space considerably.

PRACTICE

Where the Story Breaks Down

Working backward requires a genuinely fixed, well-defined goal; many real-world problems have vague or shifting end states, which makes the backward chain much harder to construct than in a geometry proof.

Use the Language

Describe a project you planned forward from today rather than backward from the deadline, using milestone, constrained and narrow down.

Discuss

What is a current goal in your work with a genuinely fixed end state? What would working backward from it reveal?

Write

In 100 words, work backward from a specific goal you have, listing the conditions that would have to be true immediately before it, and before that.

SOURCES

A Pólya, G., How to Solve It: A New Aspect of Mathematical Method, Princeton University Press (1945).

B Schoenfeld, A. H., "Pólya, Problem Solving, and Education," Mathematics Magazine 60(5), 283–291 (1987).

THE CONTRAST

Story 13: Solve Backward (Pólya) vs. Story 49: Invert the Problem (Munger)

Story 13: Solve Backward (Pólya): Works backward from a fixed, desired goal to find the path that reaches it.

Story 49: Invert the Problem (Munger): Works backward from a fixed, undesired outcome — failure — to find the paths that must be avoided.

Is it more useful to work backward from where you want to end up, or backward from where you don't want to end up?

THIS CONNECTS TO

Story 48: Sacrificing the Present for the Future (Bezos)

SAME MECHANISM

Both start from a fixed future point — a solved problem, a long-term market position — and let that fixed point determine what to do now, rather than extrapolating forward from the present.

Story 45: The Goal That Had to Change (Shackleton) LIMITATION

Pólya's method assumes the goal stays fixed while you work backward from it. Shackleton's chapter shows what happens when the goal itself has to change mid-plan — working backward from a goal that no longer applies can be actively dangerous.

Story 51: Predict From Similar Projects (Flyvbjerg) SAME MECHANISM

Both are disciplined methods for narrowing an otherwise open-ended planning problem, rather than trusting an unstructured forecast.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Mathematics students commonly got stuck on problems by starting from the given information and searching forward without a clear direction.

T — Task

Pólya wanted a general, teachable method for approaching problems that didn't rely on flashes of insight.

A — Action

He proposed starting from the goal and reasoning backward: what would you need immediately before the goal, and what would you need before that.

R — Result

His 1945 book How to Solve It became one of the most widely read texts on problem-solving, and the backward-reasoning method it popularized is now used well outside mathematics.

Level 2 — Make It Your Story

> *Tell me about a time when working backward from your goal or deadline revealed a path forward that planning forward from today had missed.*

Use STAR to answer. Think about: a project deadline, a target outcome, a plan you reversed.

TAKE THE IDEA FURTHER

> *What goal are you currently trying to reach by planning forward from today, when working backward from the deadline or outcome might reveal a shorter path?*

14

The Moth in the Machine

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED

Verified through primary sources, patents, or independent contemporaneous accounts.

On 9 September 1947, engineers working on the Harvard Mark II computer traced a persistent malfunction to Relay #70 on Panel F, where they found the source of the problem: an actual moth, physically trapped between the relay's contacts, interrupting the electrical circuit. Someone on the team taped the moth's remains into the machine's logbook, beside the handwritten note: "First actual case of bug being found."

The story, as it is almost universally told today, credits Grace Hopper — the mathematician and Navy officer working on the Mark II project, who later became one of computing's most influential and celebrated pioneers — with finding the moth, taping it into the log, and coining the terms "bug" and "debug" in that moment. Nearly every element of that popular version is either wrong or unverifiable. The Smithsonian, which now holds the actual logbook page, states that the log book was "probably not Hopper's". "Bug," as a term for a mechanical or technical fault, had already been in engineering use for decades before 1947 — Thomas Edison used it in written correspondence as far back as the 1870s to describe flaws in his own inventions.

Hopper was present on the Mark II project at the time, and did not deny the association during her lifetime, but the specific acts of finding the moth and writing the log entry are not attributed to her in the primary documentation. What Hopper actually did, and did remarkably well, was something less mythologized but arguably more consequential: for the following four and a half decades, in speeches, lectures and interviews, she told the moth story so

vividly, so often, and with such evident delight that it became fixed in the profession's collective memory as the origin of "debugging" — regardless of whether she personally performed the actions the story credits her with.

She became, in effect, the story's most effective narrator rather than its central actor, and the story's staying power owes as much to her gift for telling it as to the moth itself. This is worth taking seriously rather than treating as a minor footnote, because Hopper's actual documented contributions to computing are substantial and don't need the moth story to establish her importance. She developed one of the first compilers, which translated human-readable code into machine instructions, and was a driving force behind the eventual development of COBOL, one of the earliest high-level programming languages, both achievements resting on far stronger documentation than the log book page ever offered. The moth story persisted not because her real achievements needed padding, but because a specific, vivid, well-told anecdote is simply more memorable and more repeatable than "she helped develop early compiler technology" — a pattern that recurs constantly in how technical fields construct their own folklore.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What story does your organization tell about its own founding or its own key breakthrough that has drifted from what actually happened, because the drifted version is simply a better story?*

KEY TAKEAWAYS

1. The term "bug" for a technical fault predates the 1947 moth incident by decades and was already established engineering slang.

2. The primary documentation does not support Hopper personally finding the moth or writing the logbook entry; it was likely another team member.
3. Hopper's genuine, well-documented contributions to computing — early compiler work and her role in developing COBOL — are independent of the moth story and don't require it for significance.

SUMMARY

The famous "first computer bug" — a moth found jamming a relay in the Harvard Mark II in 1947 — is commonly credited to Grace Hopper as the person who found it, logged it, and coined the term "bug." The logbook entry was likely written by someone else, and "bug" as a term for a technical fault was already decades old by 1947. Hopper's real contribution was retelling the story vividly for 45 years, embedding it in computing folklore, independent of her separately well-documented achievements in compiler and programming-language development.

STORY CHECK

ESTABLISHED

The moth, the logbook entry, and the date (9 September 1947) are physically preserved and displayed by the Smithsonian; the pre-existing use of "bug" as engineering slang, including by Edison, is separately documented in his own correspondence.

POPULAR VERSION ADDS

The claim that Hopper personally found the moth, wrote the entry, and coined the terminology in that moment is widely repeated but not supported by the primary artifact, which the Smithsonian itself notes was "probably not Hopper's".

UNCERTAIN

Exactly which team member did write the log entry has not been definitively established in the historical record.

VOCABULARY & PHRASAL VERBS

relay (n.) an electromechanical switch used in early computers to control circuits. The moth was trapped in a relay on Panel F.

misattribution (n.) the incorrect assignment of credit for something to the wrong person or source. The moth story is a case of persistent misattribution.

folklore (n.) traditional stories and beliefs passed down within a group or profession. The story became part of computing's folklore.

compiler (n.) a program that translates human-readable code into machine-executable instructions. Hopper developed one of the first compilers.

staying power (n.) the ability of something to remain relevant or remembered over time. The story's staying power exceeded its accuracy.

pad (v., figurative) to add unnecessary material to make something seem more substantial. Her real achievements didn't need padding from the moth story.

fix in (phrasal verb) to establish something firmly, especially in memory. The story fixed itself in the profession's collective memory.

drift from (phrasal verb) to gradually move away from an original, accurate version. The story drifted from what actually happened.

hold up (phrasal verb) to remain valid or supported under examination. The popular version doesn't hold up against the primary documentation.

PRACTICE

Where the Story Breaks Down

Correcting a beloved origin story risks being read as diminishing a genuinely important historical figure; the corrective needs care to separate "this specific anecdote is inaccurate" from "this person's contributions were not significant," which are entirely different claims.

Use the Language

Describe a founding story or legend in your organization that might not hold up to scrutiny, using misattribution, folklore and drift from.

Discuss

Why might a vivid, well-told but inaccurate story survive longer than a dry, accurate account of the same events?

Write

In 100 words, describe the difference between correcting a myth about someone's achievement and diminishing the achievement itself.

SOURCES

A Smithsonian National Museum of American History, "Log Book With Computer Bug," object record and catalogue notes.

A Hopper, G. M., "The First Bug," Annals of the History of Computing 3(3), 285–286 (1981).

B Beyer, K., Grace Hopper and the Invention of the Information Age, MIT Press (2009).

THE CONTRAST

Story 36: Verify the Calculation (Katherine Johnson) vs. Story 14: The Moth in the Machine

Story 36: Verify the Calculation (Katherine Johnson): Verify the calculation — check the result itself, independently, by a genuinely different method.

Story 14: The Moth in the Machine: Verify the story — question whether the popular account of what happened, and who gets credit, is actually accurate.

What's the difference between checking a calculation and checking the narrative that later grows up around it?

THIS CONNECTS TO

Story 10: The O-Ring in the Ice Water (Feynman) SAME MECHANISM

Both popular versions overstate one individual's spontaneous, singular role in a moment that was actually more collective or more prepared than the story suggests.

Story 31: The Company That Killed Its Own Business (Netflix) SAME MECHANISM

Both organizations' own founding or breakthrough stories turned out, under scrutiny, to be tidier and more flattering than the documented record supports.

Story 27: What a Machine Could Become (Ada Lovelace) LIMITATION

Hopper's chapter is about credit wrongly given. Lovelace's is close to the

opposite problem — a real, foundational contribution that took over a century to be properly recognized at all.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

A persistent malfunction in the Harvard Mark II computer was traced to a moth physically trapped in a relay, and someone taped it into the logbook.

T — Task

The popular retelling needed to be checked against the primary documentation, since it credited Grace Hopper specifically with finding the moth and coining the term "bug."

A — Action

Historians examined the original logbook and the documented history of the word "bug," which predates the 1947 incident by decades as existing engineering slang.

R — Result

The moth and the log entry are real and physically preserved, but Hopper's personal role in finding it and the claim that she coined "bug" and "debug" are not supported by the primary record.

Level 2 — Make It Your Story

> *Tell me about a time when you looked into a team or company story more carefully and found the popular version wasn't quite accurate.*

Use STAR to answer. Think about: a founding story, a legend about a past project, a widely repeated explanation you fact-checked.

TAKE THE IDEA FURTHER

What story does your organization tell about its own founding or its own key breakthrough that has drifted from what actually happened, because the drifted version is simply more satisfying to repeat?

15

5,127 Prototypes

> **DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED**
> Verified through primary sources, patents, or independent contemporaneous accounts.

In the late 1970s, James Dyson, an English industrial designer working on an improved wheelbarrow, became irritated by a much smaller domestic problem: his vacuum cleaner kept losing suction as its bag filled with dust, a limitation shared by essentially every consumer vacuum cleaner on the market at the time, all of which relied on the same basic bag-filtration principle. Dyson had noticed a similar airflow phenomenon at an industrial sawmill, where a cyclone system separated sawdust from air using centrifugal force rather than a filter. He wondered whether the same principle could be applied at domestic scale, replacing the vacuum bag entirely.

The idea was straightforward to state and extraordinarily difficult to execute. Dyson built a prototype, tested it, found it didn't separate dust effectively enough, adjusted the design, and tested again — a cycle he repeated, by his own later account, 5,126 times before arriving at a version, prototype 5,127, that worked reliably. The process took approximately five years, during which Dyson had no income from the eventual product and financed the work through his family's savings, reportedly bringing the household close to bankruptcy before a workable design emerged.

What makes the number worth dwelling on is not its size alone but what each failed prototype actually represented. Dyson has described the process not as 5,126 failures followed by one success, but as 5,127 learning steps, each one eliminating a specific configuration that didn't work and narrowing the remaining possibility space by a small, concrete amount. A vacuum with the

cyclone chamber too narrow clogged; one with it too wide lost separating force; incremental adjustments to each failed version taught him something specific enough to inform the next attempt, rather than requiring him to guess randomly across the entire space of possible designs each time.

Even after achieving a working design, commercial success did not follow immediately. British vacuum manufacturers, whose existing business models depended heavily on the recurring sale of replacement bags, showed little interest in a bagless design that would eliminate that revenue stream — a dynamic that meant the established industry had a structural disincentive to adopt Dyson's innovation even if it worked. Dyson eventually licensed the technology in Japan, where it succeeded commercially, before returning to build his own manufacturing and sales operation in the UK. The Dyson company he founded became, decades later, a large and internationally recognized manufacturer of bagless vacuum cleaners and other appliances built on variations of the same cyclonic principle.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What worthwhile idea have you abandoned after a handful of failed attempts, when the actual obstacle was simply not yet having iterated enough times to learn what the working version requires?*

KEY TAKEAWAYS

1. Each failed prototype was treated as a specific, informative data point that narrowed the remaining design space, not merely a repeated failure.
2. The project's greatest obstacle after achieving a technical solution was not engineering but the existing industry's financial disincentive to adopt it.

3. Five years and thousands of iterations, financed entirely out of personal savings, is a genuinely large and risky commitment — the story is remarkable partly because of how rarely that level of sustained investment in iteration actually happens.

SUMMARY

James Dyson built and tested 5,127 prototypes over roughly five years to develop a bagless vacuum cleaner using cyclonic separation, financing the process through his own savings and nearly exhausting them before succeeding. Each prototype was a targeted refinement rather than a random guess, progressively narrowing the working design. Even after achieving a working product, established vacuum manufacturers, whose business depended on replacement-bag sales, showed little interest, and Dyson found his first commercial success licensing the technology in Japan.

STORY CHECK

ESTABLISHED

The 5,127 prototype figure, the roughly five-year development period, and the financial strain on Dyson's household are consistently reported across Dyson's own public statements and independent business retrospectives.

POPULAR VERSION ADDS

The number "5,127" is frequently repeated as a precise, almost talismanic figure; Dyson himself has used it consistently in interviews, though the exact count depends on what counts as a distinct "prototype" versus an incremental adjustment to an existing one.

UNCERTAIN

The precise commercial terms of the early Japanese licensing deal are not fully public.

VOCABULARY & PHRASAL VERBS

suction (n.) the force created by removing air, used to draw in dust and debris. The bag design caused a steady loss of suction as it filled.

centrifugal force (n.) the outward force experienced by objects moving in a circular path, used here to separate dust from air. A cyclone separates dust using centrifugal force.

possibility space (n.) the full range of potential solutions or configurations to a problem. Each failed prototype narrowed the possibility space.

disincentive (n.) a factor that discourages a particular action or choice. Replacement-bag revenue created a disincentive to adopt his design.

license (v.) to grant another party legal permission to manufacture or sell a technology. He licensed the technology to a Japanese company first.

narrow down (phrasal verb) to reduce a set of possibilities to a smaller, more specific set. Each prototype narrowed down the working design.

run close to (phrase) to come near to a negative outcome, such as bankruptcy. The project ran close to bankrupting the family.

hold out for (phrasal verb) to persist in waiting for a specific, better outcome rather than settling. He held out for a workable cyclonic design rather than compromising.

pay off (phrasal verb) to produce a beneficial result after a period of risk or investment. Five years of iteration eventually paid off.

PRACTICE

Where the Story Breaks Down

Dyson had savings to draw on and a working spouse whose income kept the family afloat during the development period; the same level of sustained,

unpaid iteration is simply not available to everyone facing an equivalent technical problem.

Use the Language

Describe an idea you gave up on too early, using possibility space, narrow down and pay off.

Discuss

What is the difference between an organization that treats failed attempts as information and one that treats them as simply failures?

Write

In 100 words, describe a current problem you face and estimate how many genuinely distinct iterations you have actually tried, versus how many you assume would be required.

SOURCES

C Dyson company official history (dyson.com), including founder's own account of the prototype development process.

B Dyson, J., Against the Odds: An Autobiography, Texere (1997).

D The precise "5,127" figure is drawn from Dyson's own repeated public statements rather than an independently audited prototype log.

THE CONTRAST

Story 15: 5,127 Prototypes (Dyson) vs. Story 51: Predict From Similar Projects (Flyvbjerg)

Story 15: 5,127 Prototypes (Dyson): Treats a huge number of iterations as the reliable path to a working solution — keep testing until it works.

Story 51: Predict From Similar Projects (Flyvbjerg): Treats a large number of past projects as the reliable check on a plan — look at what similar efforts actually took, before committing.

Is a large number more useful as a source of iteration, or as a source of realistic forecasting?

THIS CONNECTS TO

Story 21: The Fibre That Wasn't Supposed to Work (Kwolek)

LIMITATION

Dyson's persistence was about volume — trying enough variations to find the one that worked. Kwolek's insight was about noticing a single unusual result and insisting it be taken seriously, which is a different kind of persistence.

Story 39: Furniture Designed Around the Constraint (IKEA)

SAME MECHANISM

Both treat a large number of small, specific attempts — prototypes, delivery problems — as informative data points that narrow the design space, rather than as simple repeated failure.

Story 46: Science With Almost Nothing (Curie) SAME MECHANISM

Both required sustained, unglamorous repetition over years before the result the person was actually looking for appeared.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

James Dyson believed conventional vacuum cleaners lost suction because their bags clogged with dust, and wanted to solve that problem with a bagless design.

T — Task

He needed a working cyclonic separation mechanism that could maintain suction without a bag.

A — Action

He built and tested thousands of prototypes over several years, treating each failure as data that narrowed down what the right design might be, rather than simply repeated failure.

R — Result

After 5,127 prototypes, he arrived at a working design, which became the basis for his company's vacuum cleaners.

Level 2 — Make It Your Story

> *Tell me about a time when you kept iterating on something long after most people would have given up, and what you learned from each attempt.*

Use STAR to answer. Think about: a design, a piece of writing, a process, a product you refined through many failed versions.

TAKE THE IDEA FURTHER

> *What worthwhile idea have you abandoned after a handful of failed attempts, when the actual obstacle was simply not yet having iterated enough?*

16

The Factory That Could Stop Itself

> **DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED**
> Verified through primary sources, patents, or independent contemporaneous accounts.

Sakichi Toyoda, born in rural Japan in 1867, spent much of his early career designing improved looms for the textile industry his family and community depended on. One recurring problem preoccupied him: when a single thread broke during weaving, an ordinary automatic loom had no way to detect the fault and kept operating, producing yards of defective fabric before a human worker happened to notice and stop the machine. The defect rate this produced, and the manual inspection labour required to catch it, was a persistent drag on both quality and cost.

Toyoda's solution, patented in 1924 as part of his Type G automatic loom, was a simple mechanical device that detected a broken thread and immediately halted the machine — converting an automatic process that ran blindly until interrupted by a human into one that stopped itself the instant something went wrong. The Japanese term later applied to this principle, jidoka, is sometimes translated as "automation," but a more precise rendering, favoured within Toyota itself, is closer to "automation with a human touch" or "intelligent automation" — the emphasis is not on machines running unattended, but on machines equipped to recognize their own abnormal conditions and stop rather than continue producing defects.

The principle migrated from looms to automobiles when Sakichi's son, Kiichiro Toyoda, founded Toyota Motor Corporation, carrying his father's core insight into an entirely different manufacturing context. On Toyota's assembly lines, jidoka took the form of the andon cord — a cord any worker

on the line could pull the instant they noticed a defect or an abnormal condition, immediately stopping that section of the line rather than allowing the faulty unit to continue downstream, where the same defect would be far more expensive and difficult to correct once buried under subsequent assembly steps.

This ran directly against the conventional manufacturing wisdom of the era, particularly as practiced in large-scale American assembly-line production, where stopping the line was treated as an emergency to be avoided at nearly any cost, and defects were more commonly caught and corrected in a separate inspection and rework stage after the fact. Toyota's bet was that catching a defect the instant it occurred, even at the cost of frequent small stoppages, was cheaper and more effective than building a large downstream correction process to catch what an earlier stoppage would have prevented entirely. The empirical case for that bet built over subsequent decades, as Toyota's build quality and manufacturing efficiency became benchmarks the rest of the global auto industry spent years trying to understand and replicate — jidoka, alongside just-in-time inventory management, forming two of the central pillars later formalized as the Toyota Production System.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Does your organization catch defects the instant they occur, or does it let them continue downstream to be caught and corrected later, at greater cost, in a separate inspection stage?*

KEY TAKEAWAYS

1. Jidoka's core innovation was equipping a process to detect its own abnormal conditions and stop automatically, rather than requiring continuous human monitoring to catch every fault.
2. The principle transferred directly from textile looms to automobile assembly lines across two generations of the same family, despite the enormous difference in manufacturing context.
3. Stopping a process immediately at the point of defect ran against conventional manufacturing wisdom, which favoured continuous operation and downstream correction.

SUMMARY

Sakichi Toyoda's 1924 automatic loom introduced a mechanism that stopped the machine the instant a thread broke, rather than continuing to produce defective fabric — the origin of the principle later called jidoka. His son Kiichiro carried the same insight into Toyota's automobile manufacturing as the andon cord, letting any worker stop the assembly line the moment they spotted a defect. This ran against conventional manufacturing practice, which favoured continuous operation and downstream correction, but became one of the two central pillars of the Toyota Production System.

STORY CHECK

ESTABLISHED

Sakichi Toyoda's 1924 Type G loom patent and its automatic stop mechanism are documented in Toyota Group's own corporate history; the andon cord and its role in the Toyota Production System are extensively documented in manufacturing literature.

POPULAR VERSION ADDS

"Jidoka" is frequently translated simply as "automation," which loses the specific nuance Toyota itself emphasizes — automation with built-in defect detection and human-centred stoppage, not unattended automatic operation.

UNCERTAIN

The precise early adoption timeline of the andon cord specifically (as distinct from jidoka as a general principle) within Toyota's automobile plants is less precisely documented than the original loom patent.

VOCABULARY & PHRASAL VERBS

loom (n.) a machine used for weaving fabric from thread. Toyoda designed improved automatic looms.

defect (n.) a flaw or fault in a manufactured product. An undetected broken thread produced yards of defective fabric.

jidoka (n.) a Toyota Production System principle of automation that detects abnormalities and stops itself. Jidoka means automation with a human touch.

andon cord (n.) a physical cord on a Toyota assembly line that any worker can pull to stop production. Pulling the andon cord halted that section of the line immediately.

downstream (adv./adj.) later in a process or sequence. Catching the defect early avoided costly downstream correction.

rework (n.) the process of correcting a defective product after the fact. Many manufacturers relied on a separate rework stage.

run against (phrasal verb) to conflict with or oppose an established idea. The approach ran against conventional manufacturing wisdom.

carry into (phrasal verb) to bring a principle or practice from one context into another. Kiichiro carried the insight into automobile manufacturing.

catch up with (phrasal verb) to reach the same level as something or someone ahead. Competitors spent years trying to catch up with Toyota's quality.

PRACTICE

Where the Story Breaks Down

Frequent line stoppages are only affordable if the organization has the capacity and cultural buy-in to treat a stoppage as valuable information rather than a crisis; adopting the andon cord without that underlying cultural shift often fails.

Use the Language

Describe a process in your organization that would benefit from stopping immediately at the point of defect, using defect, downstream and rework.

Discuss

What would an "andon cord" look like in your own work — a way for anyone to halt a process the instant something goes wrong?

Write

In 100 words, compare the cost of catching a problem immediately versus catching it several steps downstream, using a specific example from your own field.

SOURCES

C Toyota Motor Corporation, official corporate history of the Toyota Production System and Sakichi Toyoda's automatic loom (toyota-global.com).

B Ohno, T., Toyota Production System: Beyond Large-Scale Production, Productivity Press (1988).

B Liker, J. K., The Toyota Way, McGraw-Hill (2004).

THIS CONNECTS TO

Story 11: The Five Whys CONTRAST

See The Contrast in that chapter. In short: Five Whys finds the root cause after a problem has already happened; jidoka stops the problem the instant it occurs, before there's anything to diagnose.

Story 44: Build Learning Into the System (Kaizen) SAME MECHANISM

Both are pillars of the Toyota Production System — jidoka stops defects at the moment they occur, while kaizen builds the ongoing habit of improving the process itself.

Story 33: The Checklist (Gawande) LIMITATION

Jidoka lets a machine or process stop itself automatically. A checklist depends on a person following it correctly — a more fragile kind of safeguard, useful precisely in situations where automatic detection isn't possible.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Sakichi Toyoda's textile looms would keep weaving even after a thread broke, producing large amounts of flawed fabric before anyone noticed.

T — Task

He needed the machine itself to detect the abnormal condition and stop, rather than relying on a worker to catch every break in time.

A — Action

He designed a mechanism that automatically halted the loom the instant a thread broke.

R — Result

The principle, later named jidoka, became one of the two pillars of the Toyota Production System, extended from looms to automated defect-detection across manufacturing.

Level 2 — Make It Your Story

> *Tell me about a time when you built a check into a process itself, rather than relying on someone to catch a problem manually.*

Use STAR to answer. Think about: an automated alert, a safeguard you added, a process redesigned to catch its own errors.

TAKE THE IDEA FURTHER

> *Does your organization catch defects the instant they occur, or does it let them continue downstream to be caught and corrected later, at greater cost?*

17

The Factory That Started Moving

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Before 1913, building an automobile meant assembling it in one place. A team of workers gathered around a stationary chassis, brought parts to it, and performed every task — installing the engine, the wheels, the body — at that single fixed location before moving on to the next vehicle. At Ford's Piquette Avenue plant, producing a single Model T this way took roughly twelve and a half hours of labour per car, and the process was fundamentally limited by how many teams could physically work around one vehicle at once without getting in each other's way.

Henry Ford and his production engineers, drawing partly on ideas already in use in meatpacking plants — where carcasses moved along an overhead rail past stationary workers, each performing one specific cutting task rather than following the animal through the entire process — inverted the arrangement for automobile manufacturing. Instead of workers moving to a stationary car, the car itself would move, carried along a belt or chain past workers who each remained in one place and performed a single, repeated task as each successive vehicle passed their station. Ford's team tested and refined the approach through 1913 at the new Highland Park plant, experimenting with belt speed, station spacing and task division before settling on a configuration that reduced the time to assemble a chassis from roughly twelve and a half hours to about ninety minutes — an order-of-magnitude improvement achieved not through faster individual workers or better tools, but through a fundamental redesign of how the work itself was sequenced and moved.

The productivity gain let Ford dramatically lower the Model T's price even while raising wages — the famous 1914 announcement of a five-dollar daily wage, roughly double the prevailing rate for factory labour, was financed in significant part by the efficiency the moving line had unlocked, and had the additional effect of reducing Ford's punishing employee turnover, which had itself been a major hidden cost of the old, more physically brutal production process. The Model T's price fell from around $850 in 1908 to under $300 by the mid-1920s, placing automobile ownership within reach of a genuinely mass market for the first time. The lesson usually drawn from this story is about the assembly line specifically, but the more general and more transferable insight is about the kind of redesign Ford's team performed: rather than trying to make each worker faster at the existing arrangement of tasks, they questioned the arrangement itself, borrowing a structural idea from an entirely unrelated industry and adapting it to a problem nobody in automobile manufacturing had previously framed as a sequencing question at all.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Is your organization trying to make people work faster within an existing process, when the process's underlying arrangement — not the people's speed — is what's actually limiting throughput?*

KEY TAKEAWAYS

1. The productivity gain came from redesigning how work was sequenced and moved, not from individual workers moving faster.

2. The core idea — moving the product past stationary workers rather than workers to a stationary product — was borrowed from an unrelated industry, meatpacking, and adapted to a new context.
3. The efficiency gain financed higher wages, which in turn reduced costly employee turnover, showing how a single process change can compound into benefits beyond the one originally targeted.

SUMMARY

Ford's production team redesigned automobile assembly around a moving line, inverting the previous arrangement of stationary vehicles and mobile worker teams, borrowing the underlying concept from meatpacking plants. The change cut chassis assembly time from roughly twelve and a half hours to about ninety minutes. The resulting efficiency gains financed both lower Model T prices and higher factory wages, dramatically expanding automobile ownership and reducing costly worker turnover.

STORY CHECK

ESTABLISHED

The Highland Park moving assembly line, the 1913 timeline, the reduction in chassis assembly time, and the 1914 five-dollar day are all extensively documented in Ford Motor Company's own archives and independent business history.

POPULAR VERSION ADDS

Ford is often credited as the sole inventor of the assembly line concept; historians generally treat the moving line as an adaptation and refinement of ideas already circulating in other industries, including meatpacking's disassembly lines, rather than a wholly original invention.

UNCERTAIN

The exact sequence of internal experiments at Highland Park through 1913 that led to the final configuration is not fully documented step by step.

VOCABULARY & PHRASAL VERBS

chassis (n.) the base frame of a vehicle, onto which other components are mounted. Assembling a chassis took twelve and a half hours before the redesign.

stationary (adj.) not moving; fixed in one place. Workers previously gathered around a stationary vehicle.

throughput (n.) the amount of material or product processed in a given time. Redesigning the sequence increased throughput dramatically.

turnover (n.) the rate at which employees leave and are replaced. High turnover was a hidden cost of the old process.

invert (v.) to reverse the position or order of something. Ford's team inverted the traditional arrangement.

mass market (n.) a very large market of ordinary consumers, as opposed to a small specialized one. Lower prices brought the car within reach of a mass market.

draw on (phrasal verb) to use an existing idea or resource as a source. Ford's engineers drew on ideas from meatpacking.

settle on (phrasal verb) to choose and finalize an option after considering alternatives. They settled on a configuration after months of testing.

unlock (v., figurative) to make something possible or available that was previously inaccessible. The efficiency gain unlocked higher wages.

PRACTICE

Where the Story Breaks Down

Ford's redesign was applicable because automobile assembly could genuinely be broken into single, repeatable tasks; not every kind of work decomposes as cleanly, and forcing an assembly-line structure onto genuinely variable or judgment-heavy work can backfire.

Use the Language

Describe a process in your organization that might benefit from questioning its sequence rather than speeding up its people, using throughput, invert and settle on.

Discuss

What unrelated industry might contain a structural idea worth borrowing for a problem in your own field?

Write

In 100 words, describe a process where the people are working hard but the process's arrangement, not their effort, is the real limit on output.

SOURCES

A Ford Motor Company archival records, Highland Park production data (1913–1914), The Henry Ford Museum and Benson Ford Research Center.

B Hounshell, D. A., From the American System to Mass Production, 1800–1932, Johns Hopkins University Press (1984).

C Ford Motor Company official corporate history (corporate.ford.com).

THE CONTRAST

Story 17: The Assembly Line (Ford) vs. Story 37: The Network That Made Overnight Delivery Possible (FedEx)

Story 17: The Assembly Line (Ford): Sped up production by bringing the work to a stationary worker on a moving line.

Story 37: The Network That Made Overnight Delivery Possible (FedEx): Sped up delivery by routing everything through a central hub, making any single package's individual journey longer, not shorter.

Is the fastest system always the one that looks most direct, or can a structurally indirect design actually be faster overall?

THIS CONNECTS TO

Story 16: The Factory That Could Stop Itself (Jidoka)

SAME MECHANISM

Both are foundational manufacturing innovations that changed how a factory's physical workflow was structured, not how hard individual workers worked.

Story 12: The Plane Built in a Barn (Skunk Works) LIMITATION

Ford's gain came from redesigning the physical flow of work. Skunk Works' gain came from redesigning the organizational flow of decisions — a different kind of bottleneck, solved a different way.

Story 34: Remove the Friction (Amazon) SAME MECHANISM

Both identify that the sequence and arrangement of a process — not the effort of the people in it — was the real lever on speed.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Before 1913, building a car meant assembling it in one place, with workers moving to the vehicle to perform each task in turn.

T — Task

Ford needed to dramatically increase the speed and lower the cost of automobile production to reach a mass market.

A — Action

He redesigned the process so the vehicle chassis moved along a line while workers stayed in place, each performing one specialized task repeatedly.

R — Result

The moving assembly line cut the time to build a car from over twelve hours to about ninety minutes, and became the model for mass manufacturing worldwide.

Level 2 — Make It Your Story

> *Tell me about a time when redesigning how work was sequenced, rather than asking people to work faster, solved a speed problem.*

Use STAR to answer. Think about: a workflow, a process bottleneck, a handoff between people or teams you restructured.

TAKE THE IDEA FURTHER

> *Is your organization trying to make people work faster within an existing process, when the process's underlying arrangement — not the people — is actually the limit?*

18

The Square Peg and the Round Hole

> **DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED**
> Verified through primary sources, patents, or independent contemporaneous accounts.

On 13 April 1970, an oxygen tank exploded aboard Apollo 13, roughly 200,000 miles from Earth, crippling the spacecraft's main systems and forcing the three-man crew to abandon their lunar landing and retreat into the lunar module, a much smaller craft designed as a temporary shuttle for two astronauts descending to the Moon's surface, never intended to sustain three people for the days required to return to Earth.

The lunar module's carbon dioxide scrubbers — canisters of lithium hydroxide that absorbed the CO2 astronauts exhaled, preventing it from building up to dangerous concentrations in the sealed cabin — were designed for two occupants over roughly two days. With three astronauts now breathing in that space for the four days the return trip would require, CO2 levels began climbing toward levels that would eventually impair judgment and, left unaddressed, prove fatal. The command module carried plenty of spare scrubber canisters. They were the wrong physical shape: square, designed for the command module's square receptacles, while the lunar module's system was built for cylindrical canisters.

Mission Control assembled a team of engineers on the ground and gave them an unusual, precisely bounded constraint: design a way to fit the square canisters into the round openings, using only the materials the astronauts already had on board the spacecraft — no additional parts could be delivered, no substitution of unavailable materials was possible. The team laid out, on a table, an actual inventory of what was physically present in the capsule: plastic

bags, cardboard from a procedure manual cover, duct tape, and the hoses from the crew's spacesuits. Working within exactly those constraints, the ground team designed an improvised adapter — a cardboard-and-plastic funnel taped together, connected to a suit hose to force air through the mismatched canister — and then relayed the assembly instructions to the crew verbally, step by step, over the radio, since no other communication method existed to convey a physical design across the distance. The astronauts, following instructions from people who could not see what they were doing and had no way to demonstrate the assembly directly, successfully built the device in real time, using it to keep carbon dioxide levels within survivable limits for the remainder of the flight. All three astronauts returned safely to Earth on 17 April 1970.

The engineering team's discipline throughout is worth naming precisely, because it is easy to lose in the retelling: they did not search for an ideal solution and then wish for materials that didn't exist. They inventoried exactly what was physically present and designed exclusively within that inventory, treating the constraint not as an obstacle to route around but as the entire boundary of the actual design problem.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Are you currently trying to solve a problem by wishing for resources you don't have, rather than designing the best possible solution using precisely what's already available to you?*

KEY TAKEAWAYS

1. The engineering team began by inventorying exactly what materials were physically present, rather than starting from an ideal solution and identifying a gap.
2. The constraint (only using available materials) was treated as the actual boundary of the problem, not an obstacle to be argued around.
3. The solution had to be communicable by voice alone, adding a second, equally strict constraint on top of the materials limitation.

SUMMARY

After an oxygen tank explosion crippled Apollo 13, the crew's carbon dioxide scrubbers were the wrong shape for the lunar module they were forced to shelter in. Ground engineers, working exclusively with an inventory of materials already on board — plastic bags, cardboard, duct tape and suit hoses — designed an improvised adapter and relayed the assembly instructions to the crew by voice alone. The astronauts built the device and survived the return trip using only what they already had.

STORY CHECK

ESTABLISHED

The oxygen tank explosion, the CO2 scrubber mismatch, and the improvised adapter (informally nicknamed "the mailbox" by the crew) are extensively documented in NASA's own mission records and post-flight reports.

POPULAR VERSION ADDS

The dramatic film depiction of the "dump the parts on the table" scene is a compressed and somewhat theatrical version of a real engineering process that took longer and involved more iteration than the single scene suggests.

UNCERTAIN

The exact real-time sequence and duration of the ground team's design process before the solution was relayed to the crew is not as precisely documented minute-by-minute as the final design itself.

VOCABULARY & PHRASAL VERBS

scrubber (n.) a device that removes a specific substance (here, carbon dioxide) from air. The CO2 scrubbers were designed for two occupants, not three.

receptacle (n.) an opening or container designed to hold something else. The lunar module's receptacles were round, not square.

inventory (n./v.) a complete list of items available; to make such a list. The team began by taking an inventory of everything on board.

improvise (v.) to create something using only materials immediately available, without advance preparation. They had to improvise an adapter from spare parts.

relay (v.) to pass on information or instructions from one party to another. Instructions were relayed to the crew by radio.

route around (phrasal verb, figurative) to avoid or bypass an obstacle rather than confront it directly. They didn't try to route around the constraint; they designed within it.

lay out (phrasal verb) to arrange items or ideas clearly for examination. They laid out every available material on a table.

fit into (phrasal verb) to be made to match the size or shape of something else. The square canisters had to fit into round openings.

within limits (phrase) inside an acceptable or survivable range. CO2 levels were kept within survivable limits.

PRACTICE

Where the Story Breaks Down

The Apollo 13 team had extremely precise, verifiable information about exactly what materials existed on board; many real-world "improvise with what you have" situations involve much less certainty about what's genuinely available or usable.

Use the Language

Describe a problem you solved using only what was already available, using inventory, improvise and fit into.

Discuss

What resources do you currently have access to that you haven't fully inventoried or considered using for a current problem?

Write

In 100 words, describe a current constraint you're treating as an obstacle to work around, and consider whether treating it as the actual boundary of the problem changes your approach.

SOURCES

A NASA, Apollo 13 Mission Report, MSC-02680 (1970).

B Lovell, J. & Kluger, J., Lost Moon: The Perilous Voyage of Apollo 13, Houghton Mifflin (1994).

C NASA official Apollo 13 mission history (nasa.gov).

THE CONTRAST

Story 18: The Square Peg and the Round Hole (Apollo 13) vs. Story 1: The Rocket That Was Too Expensive (Musk)

Story 18: The Square Peg and the Round Hole (Apollo 13): Started from a fixed, severely limited inventory of physical materials and designed the best possible solution using only what was actually on board.

Story 1: The Rocket That Was Too Expensive (Musk): Started from an unlimited hypothetical — what would this cost if you built it from scratch — with no material constraint at all.

Does genuine constraint make creative problem-solving sharper, or does it just make it harder?

THIS CONNECTS TO

Story 39: Furniture Designed Around the Constraint (IKEA)

SAME MECHANISM

Both treat a hard physical constraint — what's on board, what fits in a car — as the design requirement itself, rather than an obstacle to work around.

Story 40: The Arm That Couldn't Grab (Canadarm) SAME MECHANISM

Both solved an engineering problem by inventorying exactly what was physically possible in the actual environment, rather than starting from an ideal, unconstrained design.

Story 49: Invert the Problem (Munger) LIMITATION

The Apollo 13 team solved a single acute, survival-critical constraint under

time pressure. Munger's method is a slower, standing habit for finding risk before a crisis — useful for very different situations.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

After an oxygen tank explosion crippled Apollo 13, the crew's carbon dioxide scrubbers — designed for the command module — didn't fit the lifeboat lunar module's differently shaped receptacles.

T — Task

Engineers on the ground needed to build a working adapter using only materials the astronauts actually had on board.

A — Action

They inventoried exactly what was physically present in the spacecraft and designed a solution using only those materials — plastic bags, cardboard, and tape — rather than the ideal component that didn't exist.

R — Result

The improvised adapter worked, carbon dioxide levels dropped to safe levels, and the crew returned safely to Earth.

Level 2 — Make It Your Story

> *Tell me about a time when you had to solve a problem using only the resources you actually had, rather than the resources you wished you had.*

Use STAR to answer. Think about: a limited budget, a missing tool, a deadline that ruled out the ideal solution.

TAKE THE IDEA FURTHER

> *Are you currently trying to solve a problem by wishing for resources you don't have, rather than designing the best possible solution using what's actually available?*

PART III

Follow the Anomaly

19

The Mold in the Laboratory

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED

Verified through primary sources, patents, or independent contemporaneous accounts.

In September 1928, Alexander Fleming returned from a two-week holiday to his notoriously cluttered laboratory at St Mary's Hospital in London to find, among the piled-up petri dishes he had left out rather than properly sterilized, one plate that had gone unusually wrong. A colony of mould — later identified as Penicillium notatum, likely drifted in from a lab downstairs or through a window — had contaminated a dish of staphylococcus bacteria. That much would ordinarily mean a ruined, useless experiment, fit only for the bin. What made Fleming's dish different was what he noticed before discarding it: a clear ring surrounding the mould colony, where the bacteria had failed to grow at all, as though the mould itself were producing something actively hostile to the bacteria's survival. Fleming isolated the mould, confirmed the effect was reproducible, and published his observation in 1929, naming the bacteria-killing substance penicillin after the mould's genus.

This is where the popularly told version of the story typically ends, on a note of triumphant discovery. What actually followed is the part that gives the story its real weight, and it is routinely left out. Fleming's 1929 paper attracted almost no attention. He lacked the chemistry expertise to purify or stabilize the substance into something usable as a medicine, tried periodically over the following years to interest chemists in the problem, and largely moved on to other research. For eleven years, penicillin existed as a

documented laboratory curiosity, sitting in the published literature, unused, while people continued dying of infections it might have treated.

The substance became an actual medicine because of a separate team, working more than a decade later and initially unaware their work would connect to Fleming's forgotten paper at all: Howard Florey, Ernst Chain and colleagues at Oxford, who in 1939 began systematically investigating antibacterial compounds and rediscovered Fleming's 1929 findings in the literature. Their team solved the purification and mass-production problems Fleming himself had never been equipped to solve, ran the first controlled animal trials in 1940 — eight mice, half treated with penicillin and half untreated, all deliberately infected with a lethal bacterial dose — and conducted the first human trials in 1941. Fleming, Florey and Chain shared the 1945 Nobel Prize in Physiology or Medicine, a three-way award that itself quietly acknowledges what the popular one-man version of the story omits: the discovery and the medicine were two different, decade-separated achievements, accomplished by different people with different, complementary skills.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What genuinely valuable observation or result in your organization has been published, noted, or filed away, and then simply sat unused for years because nobody with the right complementary skill picked it up?*

KEY TAKEAWAYS

1. Fleming's real contribution was noticing and correctly interpreting an anomaly he could easily have discarded as a ruined experiment.
2. The observation alone did not become a medicine; it required a second team, with different and complementary expertise, working over a decade later.
3. The three-way 1945 Nobel Prize is itself evidence that the standard "lone genius" version of the story compresses two separate, sequential achievements into one.

SUMMARY

Fleming's famous 1928 observation — a mould contaminating a bacterial culture and killing the surrounding bacteria — was a genuine act of noticing and correctly interpreting an anomaly rather than discarding it. But his 1929 paper was largely ignored for over a decade, and it took a separate Oxford team led by Howard Florey and Ernst Chain, starting in 1939, to purify the substance, run the first trials, and turn the observation into an actual usable medicine. Fleming, Florey and Chain shared the 1945 Nobel Prize, reflecting two distinct achievements separated by more than ten years.

STORY CHECK

ESTABLISHED

Fleming's 1929 publication, the eleven-year gap before serious follow-up, and the Oxford team's 1939–1941 work culminating in human trials are all extensively documented in the scientific literature and Nobel Prize records.

POPULAR VERSION ADDS

The story is very commonly told as ending at Fleming's discovery, implying a direct line from his 1928 observation to penicillin as a usable drug; the decade-plus gap and the essential role of a second team are frequently omitted entirely.

UNCERTAIN

The exact route by which the mould spore reached Fleming's petri dish (through a window versus up a stairwell from a lab below) is debated among historians and cannot be established with certainty.

VOCABULARY & PHRASAL VERBS

contaminate (v.) to make something impure or unsuitable by introducing an unwanted substance. A mould spore had contaminated the bacterial culture.

reproducible (adj.) capable of being repeated with the same result. He confirmed the effect was reproducible before publishing.

purify (v.) to remove impurities from a substance, making it usable. Fleming lacked the expertise to purify the substance himself.

stabilize (v.) to make a substance or process consistent and reliable over time. Turning the mould into medicine required stabilizing the compound.

curiosity (n., figurative) an interesting but practically unused observation or object. Penicillin existed for over a decade as a laboratory curiosity.

pick up (phrasal verb) to take on and continue work that someone else began or set aside. The Oxford team picked up Fleming's forgotten observation.

move on (phrasal verb) to stop working on something and begin something else. Fleming largely moved on to other research.

sit unused (phrase) to remain available but not applied or acted upon. The finding sat unused in the published literature.

give credit (phrase) to acknowledge someone's contribution to an achievement. The shared Nobel Prize gives credit to both discovery and development.

PRACTICE

Where the Story Breaks Down

Fleming's observation was visually clear and immediately reproducible once someone looked for it; many valuable but overlooked results are far more ambiguous and require deliberate, skilled follow-up investigation before anyone can even tell they're worth pursuing.

Use the Language

Describe a case where an initial observation needed a second, differently skilled team to become genuinely useful, using reproducible, pick up and sit unused.

Discuss

What separates "noticing an anomaly" from "turning an anomaly into something usable"? Are they the same skill?

Write

In 100 words, describe an observation or result in your own work that might be sitting unused, waiting for someone with a different, complementary skill to pick it up.

SOURCES

A Fleming, A., "On the Antibacterial Action of Cultures of a Penicillium," British Journal of Experimental Pathology 10, 226–236 (1929).

A Chain, E., Florey, H. W. et al., "Penicillin as a Chemotherapeutic Agent," The Lancet 236(6104), 226–228 (1940), reporting the initial animal trials.

B Lax, E., The Mold in Dr. Florey's Coat: The Story of the Penicillin Miracle, Henry Holt (2004).

THIS CONNECTS TO

Story 4: The Map That Challenged a Theory (John Snow) CONTRAST
See The Contrast in that chapter. In short: Snow started from a specific, falsifiable hypothesis; Fleming started from no hypothesis at all, and the discovery arrived as an anomaly he chose not to discard.

Story 20: The Candy That Melted (Spencer) SAME MECHANISM
Both treated an inconvenient, easily dismissed observation — a ruined culture, a melted chocolate bar — as worth deliberate investigation rather than an embarrassment to clean up and forget.

Story 25: The Glue Looking for a Problem (Super Glue) LIMITATION
Fleming noticed the anomaly immediately and pursued it. Coover's compound was noticed, set aside, and only recognized as valuable years later on a second, unrelated project — noticing isn't always enough on its own.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Alexander Fleming returned from vacation to find mould contaminating one of his bacterial culture dishes, with a clear zone where bacteria had failed to grow around it.

T — Task

He needed to decide whether to discard the ruined dish as a failed experiment or investigate the anomaly.

A — Action

He examined the mould closely and identified that it was producing a substance that killed the surrounding bacteria.

R — Result

The mould, Penicillium, led to the development of penicillin, one of the most significant antibiotics in medical history.

Level 2 — Make It Your Story

> *Tell me about a time when you noticed something odd or inconvenient in your work and chose to investigate it rather than clean it up and move on.*

Use STAR to answer. Think about: a failed test, an unexpected result, a "ruined" experiment or process you looked at more closely.

TAKE THE IDEA FURTHER

> *What genuinely valuable observation or result in your organization has been noted, filed away, and simply sat unused for lack of follow-up?*

20

The Candy That Melted

DOCUMENTATION LEVEL 2 — SELF-REPORTED OR TRANSFERRED

The account rests mainly on the protagonist's own retelling, or the idea was documented but its original application was never adopted.

Percy Spencer left school after the fifth grade, was orphaned young, and taught himself electrical engineering largely through correspondence courses and hands-on naval work, eventually becoming a self-taught expert in magnetron technology at Raytheon during the Second World War — magnetrons being the vacuum-tube devices that generated the microwave radiation used in radar systems, a wartime technology of enormous strategic importance. In 1945, while working near an active magnetron, Spencer noticed something odd: a chocolate bar in his pocket had melted, despite the room not being particularly warm and Spencer having done nothing that would normally cause it. Many engineers working around the same equipment would likely have noticed the same phenomenon at some point and dismissed it as a minor, uninteresting curiosity, or simply an embarrassing accident unrelated to the work at hand. Spencer's response was different: he treated the melted chocolate as data worth investigating rather than an inconvenience worth ignoring.

He began deliberately testing the effect. He placed popcorn kernels near the magnetron; they popped. He tried an egg; by most accounts, it exploded, having built up steam pressure inside its shell with no way to release it gradually the way conventional heating would allow. Spencer had identified, through this sequence of small, deliberate tests, that the microwave radiation his magnetrons produced was causing rapid internal heating in food — a fundamentally different heating mechanism from a conventional oven, which

heats food from the outside in via conducted heat, rather than exciting water molecules throughout the food's interior more or less simultaneously.

Raytheon filed a patent for a microwave cooking process in October 1945, and the company's first commercial microwave oven, the Radarange, reached the market in 1947. It was a substantial piece of equipment — roughly the size of a refrigerator, requiring plumbing for water cooling, and priced well beyond most household budgets — intended initially for restaurants, railway dining cars and other commercial kitchens rather than home use. It took until the 1960s and 1970s, as the technology miniaturized and prices fell substantially, for the microwave oven to become the compact, affordable countertop appliance now standard in most household kitchens, more than two decades after Spencer's chocolate bar first melted.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What small, odd, easily dismissed anomaly have you personally observed in your work recently, and did you investigate it or simply find it mildly annoying and move on?*

KEY TAKEAWAYS

1. Spencer's key move was treating an inconvenient, easily dismissed observation as worth deliberate investigation, rather than an embarrassing accident to ignore.
2. He tested the anomaly systematically — popcorn, then an egg — rather than drawing a conclusion from a single data point.

3. The commercial product that resulted took over two decades to reach the affordable, compact form now considered standard, illustrating how far an initial discovery can be from its eventual mainstream application.

SUMMARY

While working with radar magnetrons at Raytheon in 1945, Percy Spencer noticed a chocolate bar in his pocket had melted near the equipment and, rather than dismissing the observation, deliberately tested the effect further with popcorn and an egg, confirming that microwave radiation rapidly heated food from within. Raytheon patented the process later that year and released the first commercial microwave oven, the Radarange, in 1947 — a large, expensive commercial appliance that took another two to three decades to become the compact, affordable household product familiar today.

STORY CHECK

ESTABLISHED

Spencer's employment at Raytheon, the 1945 patent filing, and the 1947 Radarange release are documented in Raytheon's own corporate history and US patent records.

POPULAR VERSION ADDS

The chocolate-bar anecdote is Spencer's own account, retold consistently over decades but not independently witnessed or documented at the moment it happened; some retellings embellish the exact sequence of the popcorn and egg tests.

UNCERTAIN

Whether the egg specifically "exploded" in front of Spencer or a colleague, and the precise order of the follow-up tests, vary slightly between different secondary retellings.

VOCABULARY & PHRASAL VERBS

magnetron (n.) a vacuum-tube device that generates microwave radiation. Spencer was a self-taught expert in magnetron technology.

anomaly (n.) something that deviates from what is expected or normal. He treated the melted chocolate as an anomaly worth investigating.

kernel (n.) a seed or grain, especially of corn. He placed popcorn kernels near the magnetron to test the effect.

conduct (v., physics) to transfer heat or electricity through direct contact. Conventional ovens heat food by conducted heat from the outside in.

excite (v., physics) to increase the energy state of particles, here water molecules. Microwaves excite water molecules throughout the food.

miniaturize (v.) to make something significantly smaller while retaining its function. The technology took decades to miniaturize.

dismiss (v.) to decide something is not worth serious attention. Most engineers would have dismissed the melted chocolate.

file a patent (phrase) to formally register a claim to an invention. Raytheon filed a patent within months of the discovery.

reach the market (phrase) for a product to become commercially available. The Radarange reached the market in 1947.

PRACTICE

Where the Story Breaks Down

Spencer had deep, specific technical expertise in exactly the equipment producing the anomaly, which let him correctly interpret what he was seeing; the same observation made by someone without that background might have been noticed but not correctly understood.

Use the Language

Describe a small anomaly you've noticed recently in your own work, using anomaly, dismiss and investigate (or a suitable synonym).

Discuss

What is the difference between an anomaly worth investigating and one that really is just noise? How do you tell them apart in practice?

Write

In 100 words, describe a deliberate, systematic way you could test a small anomaly you've noticed, the way Spencer moved from chocolate to popcorn to an egg.

SOURCES

A US Patent 2,495,429, "Method of Treating Foodstuffs," filed by Percy Spencer for Raytheon (1945).

C RTX (Raytheon), official corporate history of the Radarange and Percy Spencer (raytheon.com).

B Osepchuk, J. M., "A History of Microwave Heating Applications," IEEE Transactions on Microwave Theory and Techniques 32(9), 1200–1224 (1984).

THIS CONNECTS TO

Story 19: The Mold in the Laboratory (Fleming) SAME MECHANISM

Both treated an inconvenient, easily dismissed observation as worth deliberate investigation, rather than an embarrassing accident to clean up and forget.

Story 22: The Wrong Resistor SAME MECHANISM

Both discoveries began with a result that didn't match what was expected — and the key move was asking what the unexpected result was actually good for, instead of simply correcting it.

Story 2: The Burrs on the Dog (Velcro) LIMITATION

Both started from personal, physical observation. But de Mestral was specifically examining something that puzzled him; Spencer's discovery began with something as incidental as a snack melting in his pocket — showing how differently "paying attention" can look.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

While working near an active magnetron during radar research, Percy Spencer noticed a chocolate bar in his pocket had melted.

T — Task

He needed to figure out whether this was a coincidence or a real effect of the microwave radiation the equipment produced.

A — Action

Rather than dismissing it, he deliberately tested the effect on other foods, including popcorn kernels, to see if the heating effect was real and reproducible.

R — Result

The observation led directly to the development of the microwave oven.

Level 2 — Make It Your Story

> *Tell me about a time when a small, odd observation in your work turned out to be worth taking seriously.*

Use STAR to answer. Think about: an unexpected side effect, a strange result, something that happened "by accident" that you followed up on.

TAKE THE IDEA FURTHER

> *What small, odd, easily dismissed anomaly have you personally observed in your work recently, and did you investigate it or simply find it mildly annoying?*

21

The Fibre That Wasn't Supposed to Work

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1965, Stephanie Kwolek was a chemist at DuPont's research division, working on a project searching for a new generation of lightweight, high-strength synthetic fibres, motivated partly by the anticipated need for lighter materials as gasoline prices and automotive fuel-efficiency concerns began drawing attention. Kwolek was experimenting with long-chain molecules called polyamides, dissolving them into a solution intended to eventually be spun into fibre, when one of her preparations produced something that, by the normal standards of the lab, looked wrong. The solution came out cloudy and thin, almost watery, rather than the clear, thick, syrup-like consistency that polymer solutions were expected to have before spinning. By the conventional judgment of the time, a solution with this appearance would typically be discarded as a failed, unusable preparation — contaminated, degraded, or simply not properly formed. Kwolek, rather than disposing of it, asked the lab technician operating the spinning equipment to run the unusual solution through the spinneret anyway, despite the technician's own reluctance, reportedly out of concern that the odd solution might damage the expensive equipment.

The resulting fibre was extraordinary. It was, by weight, several times stronger than steel, remarkably stiff, and resistant to heat in a way no synthetic fibre previously produced had matched. Kwolek recognized immediately that the cloudy, "wrong-looking" solution had actually produced a fundamentally different and more valuable molecular structure than the clear solutions her

colleagues considered normal and correct — the polymer chains, it later became clear, had aligned into a highly ordered, liquid-crystalline arrangement that gave the resulting fibre its exceptional strength, an arrangement the "correct-looking" clear solutions never produced.

DuPont patented the material and brought it to market under the name Kevlar in the early 1970s. Its applications expanded well beyond the tire-reinforcement purpose Kwolek's original project had targeted, into bulletproof vests, protective gloves, aircraft components, and a wide range of applications requiring an exceptional strength-to-weight ratio. Kevlar-based body armour is now credited with saving thousands of lives among police officers and soldiers, an outcome with no connection to the original tire-fibre research goal that had prompted the experiment in the first place. Kwolek herself was reportedly characteristically modest about the discovery in later interviews, attributing much of it to being willing to actually test a result that conventional lab judgment said should have been thrown away.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What result in your work looked "wrong" by your team's usual standards, and was discarded without being tested, simply because it didn't match what a correct result was supposed to look like?*

KEY TAKEAWAYS

1. The unusual, "wrong-looking" solution was discarded by conventional judgment before Kwolek insisted on testing it anyway.

2. The property that made the fibre exceptional — its molecular alignment — was directly caused by the same unusual characteristic that made the solution look like a failure.
3. The eventual dominant application, bulletproof protective equipment, had no connection to the original research goal of lightweight tire reinforcement.

SUMMARY

While researching lightweight synthetic fibres at DuPont in 1965, Stephanie Kwolek produced an unusually cloudy, thin polymer solution that would conventionally have been discarded as a failed preparation. She insisted on testing it anyway, and the resulting fibre proved several times stronger than steel by weight — the cloudiness reflected a highly ordered molecular alignment the "correct-looking" clear solutions never achieved. DuPont commercialized the material as Kevlar, which found its most significant application in bulletproof protective equipment, entirely unconnected to the original tire-reinforcement research goal.

STORY CHECK

ESTABLISHED

Kwolek's 1965 discovery, the unusual solution characteristics, and DuPont's subsequent Kevlar patent and commercialization are documented in DuPont's own corporate history and Kwolek's own published interviews and technical papers.

POPULAR VERSION ADDS

Some retellings compress the technician's reluctance and the testing process into a single dramatic moment; Kwolek's own accounts describe a more

gradual process of insisting on testing, followed by further characterization work to understand why the fibre was so strong.

UNCERTAIN

The exact wording of the lab technician's initial reluctance is drawn from later interviews and retellings rather than a contemporaneous record.

VOCABULARY & PHRASAL VERBS

polymer (n.) a large molecule made of repeating chemical units, the basis of many synthetic materials. Kwolek was working with long-chain polymer molecules.

spinneret (n.) a device used to extrude polymer solution into fine fibres. The unusual solution was run through the spinneret anyway.

liquid-crystalline (adj.) describing a molecular arrangement with a degree of order between a liquid and a solid crystal. The polymer chains formed a liquid-crystalline arrangement.

strength-to-weight ratio (n.) a measure of a material's strength relative to how much it weighs. Kevlar's strength-to-weight ratio made it valuable across many industries.

discard (v.) to throw away or dispose of something considered useless. The solution would normally have been discarded.

run through (phrasal verb) to pass a material through a process or piece of equipment. She asked the technician to run the solution through the spinneret.

bring to market (phrase) to make a product commercially available for the first time. DuPont brought Kevlar to market in the early 1970s.

credited with (phrase) formally acknowledged as responsible for a positive outcome. Kevlar is credited with saving thousands of lives.

branch out (phrasal verb) to expand into new areas beyond an original purpose. The material's applications branched out well beyond tire reinforcement.

PRACTICE

Where the Story Breaks Down

Kwolek had the standing and credibility within DuPont's lab to insist on testing a result a technician was reluctant to run; someone earlier in their career or with less institutional standing might not have been able to override that reluctance as easily.

Use the Language

Describe a result that looked wrong by conventional standards but might be worth testing anyway, using discard, run through and branch out.

Discuss

What "correct-looking" standard in your own field might be quietly filtering out valuable but unconventional results before anyone tests them?

Write

In 100 words, describe how you would advocate for testing an unusual result that colleagues wanted to discard, drawing on how Kwolek handled the reluctant technician.

SOURCES

A Kwolek, S. L., US Patent 3,671,542, "Wholly Aromatic Carbocyclic Polycarbonamide Fibre Having Orientation Angle of Less Than About 45°" (1972).

C DuPont, official corporate history of Kevlar and Stephanie Kwolek (dupont.com).

B Hermes, M. E., Enough for One Lifetime: Wallace Carothers, Inventor of Nylon, American Chemical Society (1996) — provides broader context on DuPont's polymer fibre research culture that Kwolek's work extended.

THE CONTRAST

Story 21: Kevlar (Kwolek) vs. Story 15: 5,127 Prototypes (Dyson)

Story 21: Kevlar (Kwolek): Persistence meant insisting a single unusual result be taken seriously and tested properly, against colleagues' initial dismissal.

Story 15: 5,127 Prototypes (Dyson): Persistence meant generating enough volume of attempts — thousands of them — to eventually land on a working design.

Is persistence more valuable as insisting on one overlooked result, or as sheer volume of iteration?

THIS CONNECTS TO

Story 22: The Wrong Resistor SAME MECHANISM

Both cases hinge on the difference between a result that looks wrong by conventional standards and a result that is actually useless — the distinction mattered enormously in each case.

Story 28: A Different Domain Solves the Problem (Hedy Lamarr) LIMITATION

Kwolek's insight came from refusing to discard an unusual in-house result.

Lamarr's came from importing a solution from an entirely unrelated field — both defeated a stuck problem, but from opposite directions.

Story 47: The Salad Oil Decision (AmEx/Buffett) LIMITATION
Kwolek trusted her own direct observation against her colleagues' skepticism. Buffett's chapter is almost the opposite skill — trusting a careful separation of risks against the market's undifferentiated panic.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation
Stephanie Kwolek, a chemist at DuPont, was searching for a new generation of lightweight, high-strength synthetic fibres, and produced a solution that behaved unlike anything her colleagues expected — thin, cloudy, and unusual rather than thick and clear.

T — Task
Conventional judgment said the strange-looking solution should be discarded as a failed batch, but Kwolek believed it was worth testing anyway.

A — Action
She insisted the lab technician spin the unusual solution into fibre despite his reluctance, and tested the resulting fibre's strength.

R — Result
The fibre turned out to be exceptionally strong, becoming the basis for Kevlar, now used in protective equipment including bulletproof vests.

Level 2 — Make It Your Story

> *Tell me about a time when a result looked wrong by your team's usual standards, and you tested it anyway instead of discarding it.*

Use STAR to answer. Think about: an unusual output, a test result nobody expected, something you insisted on checking despite skepticism.

TAKE THE IDEA FURTHER

> *What result in your work looked "wrong" by your team's usual standards, and was discarded without being tested, simply because it didn't match expectations?*

22

The Wrong Resistor

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1956, Wilson Greatbatch was an assistant professor of electrical engineering at the University of Buffalo, working in a barn behind his home on a device intended to record the electrical sounds of a heartbeat. Reaching into a box of spare parts, he grabbed a resistor and soldered it into his circuit — and only later realized, by its colour coding, that it was the wrong value, roughly a hundred times higher resistance than the one his design called for.

The circuit did not fail the way a simple wiring error usually does. Instead of behaving unpredictably or not working at all, it began pulsing in a strikingly regular pattern: a burst of current for 1.8 milliseconds, followed by a pause of about one second, repeating steadily. Greatbatch recognized the rhythm immediately. It was, unmistakably, the timing of a healthy human heartbeat. "I stared at the thing in disbelief," he later said. Most engineers, discovering their circuit was producing the wrong output because of a parts-bin mistake, would have corrected the error and moved on with the original recording device. Greatbatch instead asked a different question: what would this accidental pulse generator be good for, if not for what he'd originally intended?

The answer, given his awareness of a specific unmet medical need, arrived almost immediately. Patients with "heart block" — a condition in which the heart's own electrical signals fail to properly trigger a regular beat — were being treated at the time with external pacemaker equipment that was bulky, painful, tethered to a wall outlet, and required the patient to remain near it. A

pulse generator small enough to implant inside the body, delivering the same steady rhythm Greatbatch's mis-wired circuit had accidentally produced, could free those patients from that dependency entirely. Turning the accident into a viable medical device took years of further work, much of it unglamorous.

Greatbatch partnered with a local surgeon, William Chardack, and their first attempts at sealing the electronics against the body's fluids failed within hours — a unit wrapped in electrical tape lasted only four hours after implantation in a dog before body fluid found its way in and shorted the circuit. Switching to a solid casting of epoxy resin solved the sealing problem, extending device life to months rather than hours. The first human implant took place in 1960, in an elderly patient with severe heart block; the device worked, and Greatbatch's company went on to develop the corrosion-resistant lithium battery that eventually extended pacemaker lifespan from about two years to a full decade. What began as an error corrected in a box of mismatched parts became a device now implanted in millions of people worldwide.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What result at work recently didn't do what you wanted it to do — and did you correct it and move on, or did you ask what it might be good for instead?*

KEY TAKEAWAYS

1. The resistor error produced a genuinely different output, not a broken one — the distinction between "wrong" and "useless" mattered enormously here.

2. Recognizing the pulse's value required Greatbatch to already be aware of a specific, unmet medical need his accidental circuit happened to match.
3. Turning the insight into an actual medical device took years of separate, unglamorous engineering work — sealing, testing, partnering with a surgeon — well beyond the original accidental moment.

SUMMARY

While building a heart-sound recording device in 1956, Wilson Greatbatch accidentally installed the wrong resistor, producing a circuit that pulsed in the exact rhythm of a healthy heartbeat rather than failing outright. Recognizing the match to an unmet medical need — patients with heart block relying on bulky external equipment — he spent years developing the accident into the first implantable pacemaker, solving serious sealing and reliability problems along the way. The first human implant succeeded in 1960.

STORY CHECK

ESTABLISHED

The 1956 resistor error, the 1.8-millisecond pulse pattern, the partnership with surgeon William Chardack, and the 1960 first human implant are all documented in Greatbatch's own accounts and independent medical history sources.

POPULAR VERSION ADDS

The story is sometimes told as though the accidental circuit was immediately implantable; in reality, the sealing and reliability problems took considerable further engineering, including an initial failed attempt that lasted only four hours in a test implant.

UNCERTAIN

The exact wording of Greatbatch's "I stared at the thing in disbelief" quotation comes from his own later retelling rather than a contemporaneous written record.

VOCABULARY & PHRASAL VERBS

resistor (n.) an electronic component that limits the flow of electrical current. He grabbed the wrong resistor from a box of spare parts.

heart block (n.) a medical condition in which the heart's electrical signals fail to trigger a regular beat. Patients with heart block relied on bulky external equipment.

implantable (adj.) designed to be placed permanently inside the body. He envisioned an implantable version of the device.

tethered (adj.) connected by a cord or cable, limiting movement. The old equipment kept patients tethered to a wall outlet.

short (a circuit) (v.) to create an unintended electrical connection that disrupts normal function. Body fluid shorted the circuit within hours.

casting (n.) a solid material formed by pouring liquid into a mould, here used to seal electronics. A solid epoxy casting solved the sealing problem.

stumble on (phrasal verb) to discover something by chance rather than deliberate search. He stumbled on the pulse pattern by accident.

free from (phrasal verb) to release someone from a limitation or dependency. The implantable device could free patients from tethered equipment.

extend (v.) to make something last longer or reach further. The lithium battery extended pacemaker lifespan to a decade.

PRACTICE

Where the Story Breaks Down

Greatbatch happened to already know about heart block and its treatment limitations when the accident occurred; recognizing the match required specific prior knowledge most people encountering the same accidental result wouldn't have had.

Use the Language

Describe an error at work that produced something unexpectedly useful, using stumble on, casting (or a suitable substitute) and extend.

Discuss

What prior knowledge would you need to recognize the value in an unexpected result in your own field?

Write

In 100 words, describe a recent mistake or unexpected result at work, and consider what specific problem it might accidentally solve.

SOURCES

B The Lancet, obituary and career summary of Wilson Greatbatch (2011).

C University at Buffalo Jacobs School of Medicine, official account of Greatbatch's pacemaker development.

C Smithsonian Lemelson Center, "Innovative Lives: Making Hearts Beat: Wilson Greatbatch."

THIS CONNECTS TO

Story 21: The Fibre That Wasn't Supposed to Work (Kwolek)

SAME MECHANISM

Both cases hinge on the difference between a result that is wrong by conventional standards and a result that is actually useless — the distinction mattered enormously in each case.

Story 20: The Candy That Melted (Spencer) SAME MECHANISM

Both discoveries began with a result that didn't match what was expected — and the key move was asking what the unexpected result was actually good for, instead of simply correcting it.

Story 25: The Glue Looking for a Problem (Super Glue) LIMITATION

Both involve an error producing a genuinely different, unplanned output. Coover's error produced a whole new product category; this chapter's error produced a life-saving medical device — not every productive mistake scales the same way.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

An engineer built a circuit using the wrong-value resistor, expecting the mistake to simply produce a broken or malfunctioning result.

T — Task

They needed to decide whether to treat the unexpected output as a defect to fix immediately, or as something worth understanding first.

A — Action

Rather than correcting the error right away, they examined what the mismatched resistor had actually produced.

R — Result

The output turned out to be different, not broken — a genuinely useful effect the original design had never intended to create.

Level 2 — Make It Your Story

> *Tell me about a time when a mistake produced a genuinely different result rather than simply a broken one, and what you did with that result.*

Use STAR to answer. Think about: a coding error, a miscalibrated setting, a process deviation that produced something usable.

TAKE THE IDEA FURTHER

> *What result at work recently didn't do what you wanted it to do — and did you correct it and move on, or did you ask what it might be good for instead?*

23

The Stain That Wouldn't Wash Off

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED

Verified through primary sources, patents, or independent contemporaneous accounts.

In 1953, Patsy Sherman and Samuel Smith, chemists at 3M, were assigned a specific and unglamorous task: develop a synthetic rubber compound resistant enough to jet fuel to be used in aircraft fuel line hoses. It was applied research aimed at a narrow industrial specification, with no connection whatsoever to consumer products or household goods. During the course of that work, a lab assistant accidentally dropped a bottle of one of Sherman's experimental fluorochemical compounds. Some of it splashed onto the assistant's white canvas tennis shoes. What happened next was the entire discovery: the spot where the compound landed would not come clean. Every solvent and cleaning method they tried failed to remove it — but more strikingly, the spot itself stayed remarkably clean even as the rest of the shoe accumulated ordinary dirt and grime over the following days. The compound was not merely stubborn. It was actively repelling the substances that were staining the rest of the shoe.

A less curious pair of chemists might have treated the ruined tennis shoe as an annoying lab accident, cleaned up as best they could, and returned attention to the jet fuel hose project they were actually being paid to complete. Sherman and Smith instead recognized that the spilled compound was demonstrating a genuinely valuable property that had nothing to do with fuel resistance: an ability to make fabric resist both water-based and oil-based stains simultaneously, a combination that was unusual, since most substances that repelled water absorbed oil, and vice versa.

Turning that observation into a commercial product took roughly three more years of research, refining the fluorochemical polymer until it could be reliably applied to wool fabric without altering its look or feel. The resulting product, launched under the name Scotchgard in 1956, became one of 3M's most commercially significant consumer product lines, eventually expanding to roughly forty variants covering different fabric types and applications. Sherman herself held thirteen joint patents over her career and was later inducted into the National Inventors Hall of Fame. Her own advice to aspiring inventors, delivered often in later interviews, captured the entire episode precisely: "Keep your eyes and mind open, and don't ignore something that doesn't come out the way you expect it to."

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What side effect or unintended result have you observed recently in your own work that was outside the scope of what you were actually assigned to do?*

KEY TAKEAWAYS

1. The valuable discovery was entirely unrelated to the original research goal — a jet fuel hose project produced a fabric stain repellent by accident.
2. Recognizing the value required noticing not just that the compound resisted cleaning, but that it actively repelled subsequent staining, a subtler and more specific observation than simply "this won't come off."
3. It took roughly three more years of dedicated development to turn the accidental observation into a reliable commercial product.

SUMMARY

While developing a jet-fuel-resistant rubber compound at 3M in 1953, chemists Patsy Sherman and Samuel Smith noticed that an accidentally spilled sample on a lab assistant's tennis shoe would not wash off and actively repelled further staining — an unrelated but valuable property. Rather than dismissing the accident, they spent three years developing the observation into Scotchgard, launched in 1956, which became one of 3M's most significant consumer product lines.

STORY CHECK

ESTABLISHED

The 1953 lab accident, the jet-fuel-hose research context, and the 1956 Scotchgard launch are documented in 3M's own corporate history and Sherman's biographical records from the National Inventors Hall of Fame.

POPULAR VERSION ADDS

Some retellings identify the lab assistant by name (Joan Mullan, per some secondary sources) while others leave the assistant anonymous; the underlying spill-on-tennis-shoes event itself is consistently corroborated across sources.

UNCERTAIN

The precise chemical mechanism Sherman and Smith initially hypothesized, versus what they later confirmed through further research, is not always clearly distinguished in secondary retellings.

VOCABULARY & PHRASAL VERBS

fluorochemical (n./adj.) a class of chemical compounds containing fluorine, often used for their repellent or resistant properties. The spilled fluorochemical compound resisted cleaning.

repel (v.) to prevent something (such as water or oil) from penetrating or adhering. The compound repelled both water-based and oil-based stains.

specification (n.) a detailed technical requirement a product or material must meet. The original project had a narrow industrial specification.

applied research (n.) research aimed at solving a specific practical problem, as opposed to general scientific inquiry. It was applied research aimed at jet fuel hoses.

grime (n.) dirt that has accumulated on a surface over time. The rest of the shoe accumulated ordinary grime.

come off (phrasal verb) to be removed, typically through cleaning or washing. The stain would not come off no matter what they tried.

stumble upon (phrasal verb) to encounter or discover something unexpectedly. They stumbled upon a valuable property by accident.

look into (phrasal verb) to investigate a matter further. They looked into the odd result rather than dismissing it.

open-minded (adj.) willing to consider new or unexpected ideas. Sherman credited her open-minded approach to unexpected results.

PRACTICE

Where the Story Breaks Down

Sherman and Smith had the institutional standing and the internal culture at 3M to pursue an unplanned discovery for three years, well outside their original assignment; not every organization tolerates that kind of redirected effort.

Use the Language

Describe an unintended result from a project with a different original purpose, using repel (or a suitable substitute verb), stumble upon and look into.

Discuss

What internal culture or permission would someone in your organization need to pursue an accidental discovery for three years, the way Sherman and Smith did?

Write

In 100 words, describe a specific difference between a result that is merely resistant to something and one that is actively repelling it — and why that distinction mattered in this story.

SOURCES

C National Inventors Hall of Fame, official biography and account of Patsy Sherman and Samuel Smith.

B Star Tribune, obituary of Patsy Sherman (13 February 2008).

A Sherman, P. O. & Smith, S., US Patent 3,574,791, "Block and Graft Copolymers Containing Water-Solvatable Polar Groups and Fluoroaliphatic Groups" (1971).

THIS CONNECTS TO

Story 2: The Burrs on the Dog (Velcro) LIMITATION

De Mestral was specifically looking for a mechanism when he examined the burrs. Sherman and Smith weren't looking for anything — showing that close investigation and pure accident can both lead somewhere, but not by the same route.

Story 5: The Glue That Wasn't Sticky Enough (Post-it Notes) SAME MECHANISM

Both compounds were complete failures relative to the specific industrial goal they were developed for, and became valuable only once someone recognized the failed property itself had independent worth.

Story 24: The Wallpaper That Failed (Bubble Wrap) SAME MECHANISM
Both products were originally developed for one specific, unrelated market, failed there, and only found their real use once someone reframed what the material was actually good for.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

3M chemists developing a rubber compound for jet fuel hoses accidentally splashed an experimental fluorochemical onto a lab assistant's canvas shoe.

T — Task

They needed to figure out why the spot resisted staining even as the rest of the shoe got dirty, despite having no connection to their actual assignment.

A — Action

Sherman and Smith investigated the accidental spill rather than simply cleaning it up and returning to the jet fuel hose project they were being paid to complete.

R — Result

They identified a genuinely valuable stain- and water-repellent property, unrelated to fuel resistance, which became the basis for Scotchgard.

Level 2 — Make It Your Story

> *Tell me about a time when an accident or side effect during one project revealed something valuable that had nothing to do with what you were actually working on.*

Use STAR to answer. Think about: a mistake, a spill, an unplanned result, something you noticed while doing something else entirely.

TAKE THE IDEA FURTHER

> *What side effect or unintended result have you observed recently in your own work that was outside the scope of what you were actually assigned to do?*

24

The Wallpaper That Failed

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1957, American mechanical engineer Alfred Fielding and Swiss chemical engineer Marc Chavannes set out, in a small garage workshop in New Jersey, to build a specific and fashionable consumer product: a three-dimensional textured wallpaper aimed at the era's emerging design tastes. Their method was straightforward in concept — seal two sheets of plastic together, using shower curtain material as an early prototype, trapping regularly spaced pockets of air between the layers to create a raised, textured surface. The wallpaper did not sell. The market for three-dimensional plastic wall coverings, it turned out, was essentially nonexistent, and the product Fielding and Chavannes had built — while genuinely novel in its manufacturing method — solved a decorating problem nobody actually had. Rather than abandoning the underlying technology, the two inventors pivoted to a second application: greenhouse insulation, reasoning that the trapped air pockets should function similarly to the insulating air gaps in double-glazed windows, potentially reducing heating costs for commercial growing operations. This attempt also failed. Tested in real greenhouses, the material provided some thermal resistance, but not enough to compete with established insulation alternatives already on the market.

By 1960, having failed twice, Fielding and Chavannes incorporated the Sealed Air Corporation and began what amounted to a systematic search for any viable application at all, reportedly considering more than four hundred potential uses for their cellular plastic sheeting — packaging, swimming pool

covers, agricultural mulch, and various other insulation applications among them. The breakthrough came not from the inventors themselves but from a Sealed Air marketer, Frederick Bowers, who recognized a specific, urgent match between the material's actual physical properties — lightweight, cushioning, able to absorb shock — and a new problem entirely unrelated to wallpaper or greenhouses: IBM's 1959 announcement of its 1401 mainframe computer, a genuinely fragile and expensive piece of equipment that needed to be shipped without damage using the crude packaging materials — newspaper, straw, wood shavings — then standard in the industry.

Bowers pitched the material to IBM as protective packaging, and the fit succeeded immediately and completely, in a way neither of the previous two attempts ever had. The product, ultimately trademarked Bubble Wrap, has been in continuous production since, still using essentially the same trapped-air principle Fielding and Chavannes developed while trying to make wallpaper. The material's success came not from changing what it physically was, but entirely from changing who it was being offered to, and for what purpose.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Do you have a genuinely good capability or technology in your organization that has failed with two different customers, when the real problem might simply be that neither of those customers is the right one?*

KEY TAKEAWAYS

1. The material itself never changed between its three attempted applications; only the customer and the framing of the problem changed.
2. Two consecutive failures (wallpaper, then greenhouse insulation) did not indicate a flawed technology — they indicated a search for the right match that hadn't yet succeeded.
3. The eventual breakthrough required someone — the marketer, not the original inventors — who could see the material's actual properties and recognize an entirely unrelated problem it happened to solve.

SUMMARY

Fielding and Chavannes developed a method for sealing plastic sheets with trapped air pockets, intending it as textured wallpaper; the product failed commercially. A second attempt at marketing it as greenhouse insulation also failed. Only when a Sealed Air marketer matched the material's actual cushioning properties to IBM's need to ship a fragile new computer in 1959 did the product succeed, becoming Bubble Wrap — the same material, aimed at a third and entirely different customer.

STORY CHECK

ESTABLISHED

The 1957 wallpaper concept, the failed greenhouse insulation attempt, the 1960 incorporation of Sealed Air Corporation, and the IBM 1401 packaging application are all documented in Sealed Air's own corporate history and Fielding and Chavannes's patent filings.

POPULAR VERSION ADDS

The "more than 400 potential uses considered" figure is widely repeated in secondary retellings and is broadly consistent with the company's own historical accounts, though the exact enumeration has not been independently published in full.

UNCERTAIN

The precise circumstances under which Frederick Bowers made the connection to IBM (some accounts mention an idea occurring during air travel) vary between retellings and are not corroborated by a single primary account.

VOCABULARY & PHRASAL VERBS

cellular (adj., materials science) containing small, enclosed pockets or cells, here referring to trapped air. The cellular plastic sheeting trapped air between two layers.

thermal resistance (n.) a material's ability to resist the flow of heat, a measure of insulating capacity. The material's thermal resistance wasn't competitive with existing insulation.

cushioning (n.) material that absorbs shock and protects against impact. The material's cushioning properties made it suitable for packaging.

pivot (v., business) to change strategic direction while retaining an underlying capability. They pivoted from wallpaper to greenhouse insulation.

fragile (adj.) easily broken or damaged. IBM's new computer was fragile and expensive to ship.

fit (n., business) the degree to which a product genuinely matches a customer's need. The fit with IBM succeeded where the previous two attempts hadn't.

take off (phrasal verb) for a product or idea to become successful, especially suddenly. Neither the wallpaper nor the insulation ever took off.

search through (phrasal verb) to examine many options systematically. They searched through hundreds of potential applications.

match up with (phrasal verb) to correspond or align well with something else. The material's properties finally matched up with a real problem.

PRACTICE

Where the Story Breaks Down

Fielding and Chavannes had the patience and capital to survive two consecutive commercial failures before finding the right application; many organizations abandon a genuinely useful capability after just one failed launch, before a better-matched customer is ever found.

Use the Language

Describe a product, service or skill that failed with one customer but might succeed with a different one, using pivot, fit and match up with.

Discuss

What technology or capability in your own organization might be "wallpaper" right now — genuinely useful, but aimed at the wrong customer?

Write

In 100 words, describe the actual physical or functional properties of something you've built or worked on, separate from the specific use case it was originally designed for.

SOURCES

C Sealed Air Corporation, official corporate history of Bubble Wrap (sealedair.com).

A Chavannes, M. A. & Fielding, A. W., US Patent 3,208,898, "Apparatus for Embossing and Laminating Materials" (1965).

B Green, J., The Bubble Wrap Book, Workman Publishing (2005).

THE CONTRAST

Story 24: Bubble Wrap vs. Story 25: The Glue Looking for a Problem (Super Glue)

Story 24: Bubble Wrap: The same physical material was reframed for three different customers before finding its real market.

Story 25: The Glue Looking for a Problem (Super Glue): The same compound was rejected twice, in two unrelated projects nine years apart, before its actual use was recognized.

When a genuinely good result keeps failing to find a market, is the problem the result itself, or is it always the framing?

THIS CONNECTS TO

Story 23: The Stain That Wouldn't Wash Off (Scotchgard)

SAME MECHANISM

Both products were originally developed for one specific, unrelated market, failed there, and only found their real use once someone reframed what the material was actually good for.

Story 25: The Glue Looking for a Problem (Super Glue)

SAME MECHANISM

Both were rejected more than once before the right application was

recognized — persistence across repeated rejection, not a single moment of insight, is what got them to market.

Story 26: The Store That Sold Shoes It Didn't Own (Zappos)

LIMITATION

Bubble Wrap's inventors kept the same material and changed the customer. Zappos changed almost nothing about the product — shoes — and instead tested a single risky assumption about how customers would buy it.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Two engineers built a three-dimensional plastic material intended to be sold as a novel textured wallpaper, and it failed in that market.

T — Task

Rather than abandoning the material, they needed to find a customer for whom its actual physical properties — cushioning, trapped air pockets — were valuable.

A — Action

They pitched it as greenhouse insulation next, which also failed, before eventually reframing it as protective packaging material.

R — Result

As packaging material, the product succeeded, becoming widely known as Bubble Wrap.

Level 2 — Make It Your Story

> *Tell me about a time when a good capability or product kept failing with one customer or use case until you found the right one.*

Use STAR to answer. Think about: a pitch that failed twice, a product repositioned for a new market, a skill you reframed for a different audience.

TAKE THE IDEA FURTHER

> *Do you have a genuinely good capability or technology in your organization that has failed with two different customers, when the real problem was never the capability itself?*

25

The Glue Looking for a Problem

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1942, Harry Coover was a young chemist at Eastman Kodak, part of a wartime team searching for materials to make clear plastic gun sights for Allied soldiers. One class of compounds they tested, cyanoacrylates, produced a substance that was optically promising but practically unusable for the task at hand: it stuck to absolutely everything it touched, ruining equipment, contaminating moulds, and making precise optical work impossible. The team rejected cyanoacrylates as a failure and moved on to other candidate materials. The compound's extraordinary adhesive property — the very thing that made it useless for gun sights — was simply set aside as an inconvenience. Nine years later, in 1951, Coover was supervising a different research group at Eastman, this time investigating heat-resistant polymers for jet aircraft canopies. Cyanoacrylates came under consideration again, for an entirely unrelated technical reason, and were tested once more. History nearly repeated itself: a chemist on Coover's team, Fred Joyner, was using the compound to test its optical properties on a refractometer, a precision instrument with expensive glass prisms — and the cyanoacrylate permanently bonded the prisms together, ruining the device.

This time, the reaction was different. Rather than simply logging the incident as a second frustrating failure with the same troublesome chemical, Coover paused on the fact that had been treated as a nuisance twice in a row: a substance that formed an essentially permanent bond, at room temperature, using nothing but trace moisture already present in the air, with no need for

heat, pressure, or a separate curing agent. Considered as an adhesive rather than as a failed optical material, those properties were not a flaw. They were close to ideal.

Eastman Kodak brought the product to market in 1958 as "Eastman 910," later known worldwide as Super Glue. Coover, notably, considered its medical applications more significant than its household use: cyanoacrylate was developed and patented as a tissue adhesive, and during the Vietnam War a spray formulation was used by field medics to temporarily seal severe internal wounds and stop catastrophic bleeding in soldiers, keeping them alive long enough to reach a hospital for conventional surgery. The same property that ruined a valuable optical instrument and, nine years earlier, a batch of gun sight prototypes, became one of the twentieth century's most widely recognized consumer products and, by Coover's own account, the achievement of his career he was proudest of.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Has your organization rejected the same "failed" result twice, for two unrelated projects, without ever asking what the result might actually be good for on its own terms?*

KEY TAKEAWAYS

1. The same compound was rejected as a failure in two entirely unrelated projects, nine years apart, before its actual value was recognized on the second encounter.

2. What made the substance a failure for both original purposes — extreme, uncontrollable stickiness — was precisely the property that made it valuable once the purpose changed.
3. Coover's own view of his most significant contribution, the medical tissue adhesive, was not the application that made the product famous, which is itself worth noticing about how achievements get remembered.

SUMMARY

Harry Coover's team first encountered cyanoacrylate compounds in 1942 while developing gun sights and rejected them for being unusably sticky. The same problem — a colleague accidentally gluing a precision instrument together — recurred in 1951 during unrelated jet-canopy research, but this time Coover recognized the extreme adhesion as a valuable property in its own right rather than a repeated nuisance. The resulting product, Super Glue, launched in 1958, and its most significant application, in Coover's own assessment, was as a battlefield tissue adhesive that saved lives during the Vietnam War.

STORY CHECK

ESTABLISHED

The 1942 gun-sight rejection, the 1951 refractometer incident involving Fred Joyner, and the 1958 commercial launch as "Eastman 910" are all documented in Coover's own accounts and the National Inventors Hall of Fame biographical record.

POPULAR VERSION ADDS

Some retellings compress the two rejection episodes into a single accidental discovery; the two separate, nine-years-apart rejections are the more precise and more instructive version of the story.

UNCERTAIN

The exact monetary value or specific model of the ruined refractometer varies between secondary sources and is not independently documented.

VOCABULARY & PHRASAL VERBS

cyanoacrylate (n.) a class of fast-bonding adhesive compounds, the chemical basis of Super Glue. Cyanoacrylates were first tested for gun sights in 1942.

optical (adj.) relating to vision or light, here referring to clear materials used in lenses and sights. The compound was optically promising but practically unusable.

curing agent (n.) a substance added to trigger a chemical reaction that hardens a material. The adhesive needed no separate curing agent.

refractometer (n.) an instrument used to measure the refractive properties of a material. The compound permanently bonded the refractometer's prisms.

trace (adj.) present in a very small amount. The reaction required only trace moisture from the air.

set aside (phrasal verb) to put something away for later consideration, or to dismiss it. The adhesive property was set aside as an inconvenience.

come under consideration (phrase) to be evaluated or considered again for a purpose. Cyanoacrylates came under consideration again in 1951.

bring to market (phrase) to make a product commercially available. Eastman Kodak brought the product to market in 1958.

hold up (phrasal verb) to remain effective or intact under stress or use. The tissue adhesive held up well enough to seal severe wounds temporarily.

PRACTICE

Where the Story Breaks Down

Coover had the specific advantage of encountering the same "failed" material twice, in two different contexts, which gave him a pattern to notice; most unwanted results are only ever tested once and discarded, with no second encounter to prompt reconsideration.

Use the Language

Describe a result rejected for one purpose that might succeed for a completely different one, using set aside, come under consideration and hold up.

Discuss

What "failed" material, tool or process in your organization has been rejected more than once for related but different reasons? Might that repetition be a signal rather than a coincidence?

Write

In 100 words, describe the specific property that made this compound a failure for its original purposes, and explain in your own words why that same property became valuable once the purpose changed.

SOURCES

C National Inventors Hall of Fame, official biography of Harry W. Coover Jr.

B Hobart and William Smith Colleges, "Lives of Consequence" profile of Harry W. Coover Jr.

D Coover's own retrospective interviews describing the Vietnam War hemostatic spray application are widely cited in secondary sources without a single traceable military record specifying exact usage figures.

THIS CONNECTS TO

Story 5: The Glue That Wasn't Sticky Enough (Post-it Notes)

SAME MECHANISM

Both compounds were developed for one purpose, judged a failure against that purpose, and only became valuable once someone unrelated to the original project found a different use for the exact property that had made them a failure.

Story 24: The Wallpaper That Failed (Bubble Wrap) CONTRAST

See The Contrast in that chapter. In short: Coover's compound was rejected twice for the same unmet purpose before its use was recognized; Fielding and Chavannes kept their material constant and simply found it a new customer.

Story 19: The Mold in the Laboratory (Fleming) LIMITATION

Fleming noticed the anomaly immediately and pursued it. Coover's compound was noticed, set aside, and only recognized as valuable years later on a second, unrelated project — noticing isn't always enough on its own.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Harry Coover, searching for materials to make clear plastic gun sights during wartime research, tested a class of compounds — cyanoacrylates — that bonded to everything they touched, ruining lab equipment.

T — Task

He needed to decide whether the compound was simply a lab nuisance or had a real application, but rejected it as a failure for the gun sight project.

A — Action

Nine years later, working on a different, unrelated project, he and colleagues reconsidered the same compound's extreme bonding property.

R — Result

The compound was recognized as a genuinely useful fast-bonding adhesive and launched commercially as Super Glue.

Level 2 — Make It Your Story

> *Tell me about a time when your organization rejected the same result twice before finally recognizing its value.*

Use STAR to answer. Think about: an idea shelved and revisited, a technology reconsidered years later, a result dismissed more than once.

TAKE THE IDEA FURTHER

> *Has your organization rejected the same "failed" result twice, for two unrelated projects, without ever asking what the result might actually be good for?*

26

The Store That Sold Shoes It Didn't Own

DOCUMENTATION LEVEL 2 — SELF-REPORTED OR TRANSFERRED

The account rests mainly on the protagonist's own retelling, or the idea was documented but its original application was never adopted.

In 1999, Nick Swinmurn, after failing to find a specific pair of shoes across several stores in a California mall — the right size in the wrong colour at one store, the right colour in the wrong size at another — arrived at an idea that seemed obvious in retrospect and reckless in practice: what if people could simply buy shoes online, choosing from a far larger selection than any single physical store could stock? The idea faced a specific and widely shared objection at the time. E-commerce in 1999 was new enough that its successes so far — books, music, other standardized goods people didn't need to physically try on — were seen as exceptions rather than a preview of what was coming. Shoes were considered a poor fit for online sales precisely because sizing, fit and feel were assumed to require physical handling before purchase. Testing that assumption properly would ordinarily require significant upfront investment: negotiating supplier relationships, purchasing inventory across sizes and styles, building warehouse and fulfilment infrastructure, and only then finding out whether customers would actually buy shoes sight-unseen.

Swinmurn built none of that. Instead, he approached a local shoe store, Footwear Etc. in Sunnyvale, California, with a simple proposal: he would photograph their existing inventory and list it for sale on a basic website he had built, at no cost or risk to the store. If a customer placed an order through his site, he would then go to the store himself, buy the shoes at full retail price, and ship them directly to the customer — losing money on every single

transaction, since he was paying full retail for stock he then resold without markup sufficient to cover it.

The financial arrangement was deliberately, obviously unsustainable as a long-term business model. That was not the point. The entire purpose of the arrangement was to answer one specific, narrow question as cheaply and quickly as possible: would people actually place orders for shoes they had only seen in a photograph, with no ability to try them on first? Within the site's first year, real orders began arriving, and they kept arriving. That was the evidence Swinmurn needed. Only once demand was genuinely confirmed did the site — renamed Zappos, after raising further funding, including from investor and eventual CEO Tony Hsieh — begin building the supplier relationships, inventory systems and logistics infrastructure that an online shoe retailer actually required to scale. Zappos was acquired by Amazon in 2009 for approximately $1.2 billion.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What assumption about your customers are you currently planning to test by building the complete, expensive version of a product first, when a far cheaper and faster test might answer the same question?*

KEY TAKEAWAYS

1. Swinmurn tested the single riskiest assumption — will people buy shoes without trying them on — before building any of the expensive infrastructure a real online shoe business would eventually need.

2. The arrangement with the local shoe store was deliberately unprofitable, because profitability wasn't what was being tested; genuine customer demand was.
3. Only after the assumption was confirmed with real orders did Zappos build the systems that made the business viable at scale.

SUMMARY

Frustrated by an ordinary shoe-shopping trip, Nick Swinmurn tested whether people would buy shoes online by photographing a local store's inventory and listing it on a basic website, personally buying and shipping any shoes that sold — losing money on every sale, since the goal was validating demand, not building a profitable business yet. Real orders confirmed the assumption within the first year, and only then did the company, renamed Zappos, build the supplier relationships and infrastructure needed to operate at scale. It was acquired by Amazon in 2009 for roughly $1.2 billion.

STORY CHECK

ESTABLISHED

Swinmurn's 1999 founding of Shoesite.com (later Zappos), the arrangement with Footwear Etc. in Sunnyvale, and the 2009 Amazon acquisition are documented across multiple independent business retrospectives and Swinmurn's own retellings.

POPULAR VERSION ADDS

The technique is now commonly labelled a "Wizard of Oz" minimum viable product test in business literature, a term applied retrospectively; Swinmurn himself was not using that specific framework or vocabulary at the time.

UNCERTAIN

The precise financial terms of the initial arrangement with Footwear Etc. (exact markup, exact losses per sale) are not independently published in detail.

VOCABULARY & PHRASAL VERBS

fulfilment (n., business) the process of receiving, processing and delivering customer orders. Building fulfilment infrastructure would have required significant upfront investment.

markup (n.) the amount added to a product's cost to determine its selling price. He resold the shoes without enough markup to cover his costs.

unsustainable (adj.) not able to be maintained or continued over time. The financial arrangement was deliberately unsustainable long-term.

validate (v.) to confirm that something is true or well-founded, often through testing. He needed to validate customer demand before investing further.

scale (v., business) to grow a business or process significantly while maintaining or improving efficiency. Only after validation did the company build systems to scale.

sight-unseen (adj./adv.) without having seen or inspected something before committing to it. Customers were buying shoes sight-unseen.

place an order (phrase) to formally request the purchase of a product. Customers began placing real orders within the first year.

lose money on (phrase) to sell something for less than it costs, resulting in a financial loss. He lost money on every single sale.

build out (phrasal verb) to develop or expand a system or infrastructure over time. They only began to build out inventory systems after validating demand.

PRACTICE

Where the Story Breaks Down

Swinmurn's test worked because a single local store was willing to cooperate at no cost or risk to itself; not every assumption can be tested this cheaply, especially in industries without an equivalent willing partner to supply the product being tested.

Use the Language

Describe an assumption in your own work you could test cheaply before committing full resources, using validate, scale and build out.

Discuss

What is the riskiest, least-tested assumption behind a project you're currently working on?

Write

In 100 words, design the cheapest possible test of that assumption — one that doesn't require building the complete version of what you're planning.

SOURCES

C Inc. magazine and NPR interviews with Nick Swinmurn, cross-referenced with independent business press accounts.

D The "Wizard of Oz MVP" framing is a widely used but retrospectively applied business-school term; Swinmurn's own contemporaneous documentation of the exact process is limited to later interviews.

B Hsieh, T., Delivering Happiness: A Path to Profits, Passion, and Purpose, Business Plus (2010) — provides additional context on Zappos's early growth from the eventual CEO's perspective.

THE CONTRAST

Story 26: Zappos vs. Story 30: The Product People Couldn't Yet Ask For (Sony Walkman)

Story 26: Zappos: Tested the riskiest assumption first, cheaply, before building any real infrastructure.

Story 30: The Product People Couldn't Yet Ask For (Sony Walkman): Built and shipped a real product based on conviction, because the customer couldn't articulate the need in advance for market research to test at all.

Should you always test the riskiest assumption cheaply before committing — or are some genuinely new ideas untestable that way?

THIS CONNECTS TO

Story 24: The Wallpaper That Failed (Bubble Wrap) LIMITATION
Bubble Wrap's inventors kept the same material and changed the customer. Zappos changed almost nothing about the product — shoes — and instead tested a single risky assumption about how customers would buy it.

Story 29: Think Like the User (Temple Grandin) SAME MECHANISM
Both prioritized direct, concrete evidence about how people or animals would actually behave over assumptions made from a comfortable distance.

Story 31: The Company That Killed Its Own Business (Netflix) LIMITATION
Zappos tested a risky assumption before committing real infrastructure. Netflix's chapter is about committing to cannibalize an already-successful,

real business — a much higher-stakes version of the same willingness to bet against comfortable assumptions.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Nick Swinmurn believed people would buy shoes online without trying them on first, but this was an unproven and risky assumption for an entire business model.

T — Task

He needed to test that specific assumption before investing in inventory, warehousing, or supplier relationships.

A — Action

He photographed shoes at local stores, listed them online, and personally bought and shipped any pairs that sold, without ever owning inventory himself.

R — Result

The approach validated the core assumption cheaply, and the business became Zappos, later acquired by Amazon.

Level 2 — Make It Your Story

> *Tell me about a time when you tested the riskiest assumption in a project cheaply, before building the full, expensive version.*

Use STAR to answer. Think about: a minimum viable product, a small pilot, a low-cost test of a risky idea.

TAKE THE IDEA FURTHER

> *What assumption about your customers are you currently planning to test by building the complete, expensive version of a product first, when a cheaper test might answer the question sooner?*

27

What a Machine Could Become

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1843, Ada Lovelace, daughter of the poet Lord Byron and a mathematician trained privately to a level rare for a woman of her era, published an extensive set of translator's notes on an Italian engineer's paper describing Charles Babbage's proposed Analytical Engine — a mechanical calculating device Babbage had designed but never fully built. Lovelace's notes ended up roughly three times longer than the original paper they were meant to accompany, and they contained an idea considerably more significant than anything in the source text itself.

Babbage had conceived the Analytical Engine primarily as a machine for computing numbers — an extraordinarily sophisticated mechanical calculator, capable of automating complex arithmetic that human "computers" (a job title, at the time, describing people who performed calculations by hand) would otherwise have to work through manually. Lovelace's notes accepted that framing and then moved decisively past it. She observed that the machine did not fundamentally operate on numbers themselves; it operated on symbols, following a defined sequence of rules — and numbers were simply one kind of symbol among many that could, in principle, be manipulated this way.

From that abstraction, Lovelace drew a specific and genuinely prescient conclusion: if the Analytical Engine could operate on any symbols following a defined rule set, it could in principle process anything symbolic — most strikingly, she proposed that a properly programmed version of the machine

could compose music, given a set of rules for the relationships between musical notes, in exactly the same underlying way it could calculate a mathematical result. A machine imagined as a sophisticated calculator, in Lovelace's reading, was actually something closer to a general-purpose symbol processor, capable of tasks its original designer had never explicitly conceived it for.

Her notes also included what many historians consider the first published algorithm intended to be executed by a machine — a detailed, step-by-step method for computing Bernoulli numbers using the Analytical Engine's proposed operations, worked out entirely on paper, since the machine itself was never completed in Babbage's or Lovelace's lifetime. Lovelace died in 1852, more than a century before anything resembling a working general-purpose computer existed to test whether her broader vision — a machine that manipulated symbols of any kind, not merely numbers — actually held. When electronic computers finally arrived in the twentieth century and were applied to tasks well beyond arithmetic, including music composition and symbolic logic, they validated, in a domain she never lived to see, an abstraction she had reasoned her way to purely on paper, working from a machine that existed only as an unbuilt design.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Is there a tool, system or capability in your organization currently defined narrowly by its original intended purpose, when its underlying mechanism might actually generalize to something much broader?*

KEY TAKEAWAYS

1. Lovelace's key move was abstracting past the machine's intended function — calculating numbers — to identify its underlying, more general mechanism: manipulating symbols according to rules.
2. That single abstraction let her anticipate applications, such as machine-composed music, that had no connection to the calculator the machine was originally designed to be.
3. Her insight was validated more than a century later, in a form of technology she had no way of anticipating in its physical detail, only in its underlying logical structure.

SUMMARY

Ada Lovelace's 1843 notes on Babbage's unbuilt Analytical Engine went far beyond the machine's intended purpose as a calculator, identifying that it actually operated on symbols according to defined rules — meaning numbers were just one possible kind of input among many. From that abstraction, she proposed the machine could, in principle, compose music, and her notes contain what many historians regard as the first algorithm designed for machine execution. Her broader vision was only validated over a century later, once general-purpose electronic computers existed to test it.

STORY CHECK

ESTABLISHED

Lovelace's 1843 published notes, their length relative to the original paper, and their content — including the Bernoulli number algorithm and the observation about symbolic manipulation — are all directly documented in the primary published text.

POPULAR VERSION ADDS

Lovelace is sometimes described simply as "the first computer programmer," a characterization some historians consider reasonable and others consider an oversimplification of a more nuanced and largely theoretical contribution, since the machine she wrote for was never built.

UNCERTAIN

The degree to which Lovelace's ideas were independently hers versus developed in extensive collaboration and correspondence with Babbage himself is a subject of ongoing historical debate.

VOCABULARY & PHRASAL VERBS

abstraction (n.) the process of identifying a general underlying principle apart from specific, concrete details. Her key move was an abstraction, from numbers to symbols in general.

symbol (n.) a sign or mark that represents something else according to a defined system. The machine could operate on symbols, not just numbers.

algorithm (n.) a defined, step-by-step procedure for solving a problem or performing a task. Her notes contained the first published algorithm for a machine.

prescient (adj.) showing accurate knowledge or anticipation of future events. Her conclusion about machine-composed music was remarkably prescient.

general-purpose (adj.) designed to perform a wide range of functions, not limited to one specific task. She envisioned the machine as a general-purpose symbol processor.

validate (v.) to confirm the correctness or value of an idea, often through later evidence. Her insight was validated over a century later.

move past (phrasal verb) to progress beyond an initial framing or limitation. She moved past the machine's intended function as a calculator.

work out (phrasal verb) to calculate or devise something through careful effort. She worked out an algorithm entirely on paper.

hold up (phrasal verb) to remain valid or true when tested. Her broader vision held up once real computers existed.

PRACTICE

Where the Story Breaks Down

Lovelace's abstraction was validated only because general-purpose computing eventually became a real technology; many abstractions that seem equally elegant on paper turn out not to generalize once actually tested.

Use the Language

Describe a tool or system whose underlying mechanism might generalize beyond its current intended use, using abstraction, general-purpose and move past.

Discuss

What is the difference between a genuinely useful abstraction and an overreaching one that sounds clever but doesn't actually hold up?

Write

In 100 words, identify the underlying, general mechanism behind a specific tool or process you use regularly, separate from its intended purpose.

SOURCES

A Lovelace, A., "Notes by the Translator" (Note G, containing the Bernoulli number algorithm), in Menabrea, L. F., "Sketch of the Analytical Engine Invented by Charles Babbage," Scientific Memoirs (1843).

B Essinger, J., Ada's Algorithm: How Lord Byron's Daughter Ada Lovelace Launched the Digital Age, Melville House (2014).

B Fuegi, J. & Francis, J., "Lovelace & Babbage and the Creation of the 1843 'Notes'," IEEE Annals of the History of Computing 25(4), 16–26 (2003) — a scholarly source discussing the extent of Lovelace's independent contribution.

THE CONTRAST

Story 27: Ada Lovelace vs. Story 32: Connect What Already Exists (Berners-Lee)

Story 27: Ada Lovelace: Abstracted past a machine's intended narrow function to identify its underlying general-purpose potential, a century before the technology existed to build it.

Story 32: Connect What Already Exists (Berners-Lee): Combined existing components into something new, at a moment when all the necessary pieces already existed and just needed to be connected.

Is the harder creative leap imagining a capability decades before the technology exists, or seeing how to combine capabilities that already exist right now?

THIS CONNECTS TO

Story 14: The Moth in the Machine (Grace Hopper) LIMITATION

Hopper's chapter is about credit wrongly given. Lovelace's is close to the opposite problem — a real, foundational contribution that took over a century to be properly recognized at all.

Story 43: Seeing the Smartphone Too Early (Nokia) SAME MECHANISM

Both involve a genuine, correct insight arriving well before the surrounding technology or market was ready to act on it.

Story 30: The Product People Couldn't Yet Ask For (Sony Walkman)

SAME MECHANISM

Both required seeing past what a technology or product currently did toward what it could become, before customers or engineers could articulate the need themselves.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Ada Lovelace was asked to translate an Italian engineer's paper describing Charles Babbage's proposed Analytical Engine, a mechanical calculating machine.

T — Task

She needed to explain not just how the machine calculated numbers, but what it was actually capable of doing.

A — Action

In her extensive translator's notes, she abstracted past the machine's intended function — numerical calculation — to describe its potential to manipulate any symbols according to rules, including music and other non-numerical content.

R — Result

Her notes are now widely regarded as describing the concept of a general-purpose programmable computer, over a century before one was built.

Level 2 — Make It Your Story

> *Tell me about a time when you saw a tool or system's underlying general potential, beyond the narrow purpose it was originally built for.*

Use STAR to answer. Think about: a tool used in an unintended way, a system repurposed, a capability you saw before others did.

TAKE THE IDEA FURTHER

> *Is there a tool, system or capability in your organization currently defined narrowly by its original intended purpose, when its underlying mechanism could do considerably more?*

PART IV

Change the Frame

28

A Different Domain Solves the Problem

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Hedy Lamarr, at the height of her career as a Hollywood film star in the early 1940s, was also, away from the studio, a self-taught inventor with a genuine technical curiosity, one she rarely discussed publicly given the era's expectations of a glamorous actress. Motivated by the Second World War and specifically by reports of Allied radio-controlled torpedoes being jammed or intercepted by the enemy, Lamarr turned her attention to a specific defence problem: how could a radio signal used to guide a torpedo be made resistant to jamming?

The core vulnerability was structural. A radio-controlled torpedo transmitted and received its guidance signal on a single, fixed frequency, which made that frequency simple for an adversary to detect and jam, defeating the entire guidance system with a comparatively unsophisticated countermeasure. Lamarr's insight was that if the guidance signal instead hopped rapidly and unpredictably between many different frequencies, in a pattern synchronized between the transmitter and receiver, an adversary attempting to jam any single frequency would only interrupt the signal for the brief instant it happened to be using that frequency, before it hopped elsewhere.

The idea alone did not solve the engineering problem of how to synchronize that hopping pattern reliably between a ship and a torpedo using 1940s-era technology. The solution came from an unlikely collaboration and an unlikely source domain. Lamarr partnered with composer and avant-garde musician George Antheil, who had earlier in his career worked extensively

with mechanically synchronized player pianos — instruments that read perforated paper rolls to reproduce a musical performance automatically, with multiple pianos capable of being synchronized to play together via matched, identically punched rolls. Antheil recognized that the same mechanical principle — a punched pattern read in perfect synchronization by two separated devices — could solve the frequency-hopping synchronization problem, using a similar punched-paper mechanism to coordinate frequency changes between a torpedo's transmitter and its receiver in lockstep.

Lamarr and Antheil patented the frequency-hopping spread spectrum concept in 1942 and offered it to the US Navy, which declined to adopt the specific mechanical implementation at the time, reportedly finding a piano-roll-based mechanism impractical for shipboard and torpedo use in its current form. The patent expired before the technology was ever deployed in the application it was designed for. Decades later, as electronics advanced far beyond what 1940s mechanical synchronization could achieve, the underlying frequency-hopping principle became foundational to spread-spectrum communication technology — now embedded in Wi-Fi, Bluetooth, and other wireless communication standards used by billions of devices, a lineage that went largely unrecognized publicly until decades after the original patent, when the technology press began crediting Lamarr's role more prominently.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What genuinely difficult problem in your work might have already been solved, structurally, in a completely unrelated field — one you'd never think to look at because it has nothing to do with your industry on the surface?*

KEY TAKEAWAYS

1. The core insight (rapid, synchronized frequency changes defeating jamming) came from analyzing the structural vulnerability of a fixed-frequency signal, not from expertise in radio engineering itself.
2. The specific mechanism that made the idea implementable came from an entirely unrelated domain — mechanically synchronized player pianos — recognized by a collaborator whose expertise was music, not defence technology.
3. The invention was not adopted for its original purpose and only became commercially significant decades later, in a technological context (digital electronics) the original inventors could not have anticipated.

SUMMARY

Hedy Lamarr, alongside composer George Antheil, patented a frequency-hopping method to prevent radio-guided torpedo signals from being jammed, with the synchronization mechanism borrowed directly from Antheil's earlier work on mechanically synchronized player pianos. The US Navy declined to adopt the mechanical implementation at the time, and the patent expired unused for its original purpose. The underlying frequency-hopping principle later became foundational to modern wireless technologies including Wi-Fi and Bluetooth, decades after the original 1942 patent.

STORY CHECK

ESTABLISHED

The 1942 patent (US Patent 2,292,387), its frequency-hopping concept, and its connection to player-piano synchronization mechanisms are documented in the patent itself and corroborated by independent historical accounts.

POPULAR VERSION ADDS

Lamarr is sometimes credited as though frequency-hopping spread spectrum in modern devices directly derives from her specific patented mechanism; the underlying conceptual principle is credited to her and Antheil, but the actual electronic implementations used in modern Wi-Fi and Bluetooth were developed independently, decades later, using entirely different technology.

UNCERTAIN

The exact reasons the Navy gave for declining to adopt the mechanism at the time are not fully documented in surviving records, beyond general characterizations of impracticality.

VOCABULARY & PHRASAL VERBS

jam (v., signals) to deliberately interfere with or block a radio signal. An adversary could jam a fixed-frequency guidance signal.

frequency-hopping (n./adj.) a technique of rapidly switching between different transmission frequencies. Their frequency-hopping method resisted jamming.

synchronize (v.) to cause two or more processes to operate together in matched timing. The transmitter and receiver had to synchronize their frequency changes.

perforated (adj.) having small holes punched through it, as in a player piano roll. The player piano read a perforated paper roll.

lockstep (n., figurative) a state of exact, coordinated matching. The two devices changed frequency in lockstep.

spread spectrum (n.) a communication technique that spreads a signal across a wide frequency range. The concept became foundational to spread spectrum technology.

decline to (phrase) to formally refuse or choose not to do something. The Navy declined to adopt the mechanism.

go unrecognized (phrase) to remain unacknowledged or uncredited. Lamarr's contribution went largely unrecognized for decades.

borrow from (phrasal verb) to take and apply an idea or method from a different source. The synchronization method borrowed directly from player piano mechanics.

PRACTICE

Where the Story Breaks Down

Lamarr had access to Antheil specifically because of her personal and professional circles in the entertainment industry; most people working on a technical problem in one domain don't have an equivalent chance social connection to an expert in a completely unrelated field.

Use the Language

Describe a technical or organizational problem that might already be solved in an unrelated field, using synchronize, borrow from and lockstep.

Discuss

What unrelated field or industry might contain a structural solution to a problem you're currently facing?

Write

In 100 words, describe the core structural vulnerability of a current problem you face, stripped of its specific industry context, and consider what other field might have already solved that same structural problem.

SOURCES

A Lamarr, H. K. & Antheil, G., US Patent 2,292,387, "Secret Communication System" (1942).

B Rhodes, R., Hedy's Folly: The Life and Breakthrough Inventions of Hedy Lamarr, the Most Beautiful Woman in the World, Doubleday (2011).

C National Inventors Hall of Fame, biographical entry on Hedy Lamarr.

THIS CONNECTS TO

Story 12: The Plane Built in a Barn (Skunk Works) CONTRAST

See The Contrast in that chapter. In short: Skunk Works solved an urgent problem by stripping organizational layers around an in-house team; Lamarr solved hers by importing a mechanism from a completely unrelated field.

Story 32: Connect What Already Exists (Berners-Lee) SAME MECHANISM

Both solved their problem by combining existing pieces — a player-piano mechanism and radio, or existing internet protocols — rather than inventing something entirely new from scratch.

Story 40: The Arm That Couldn't Grab (Canadarm) LIMITATION

Lamarr's chapter is about importing an existing mechanism from an unrelated field. Canadarm's is closer to the opposite — realizing the obvious mechanism (gripping) was actually wrong for the new environment, and had to be redesigned from scratch.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

During the Second World War, radio-guided torpedoes were vulnerable to jamming because they broadcast on a single, fixed frequency.

T — Task

Hedy Lamarr wanted to find a way to defeat that vulnerability using a mechanism from a field entirely unrelated to weapons engineering.

A — Action

Working with composer George Antheil, she adapted a mechanism from automated player pianos — synchronized, rapidly switching frequencies — into a frequency-hopping system that would be far harder to jam.

R — Result

Their patented concept, though not adopted during the war itself, became foundational to later spread-spectrum technology used in modern wireless communications including Wi-Fi and Bluetooth.

Level 2 — Make It Your Story

> *Tell me about a time when a solution from a completely unrelated field solved a difficult problem in your own work.*

Use STAR to answer. Think about: an idea borrowed from another industry, a mechanism you adapted from somewhere unexpected.

TAKE THE IDEA FURTHER

> *What genuinely difficult problem in your work might have already been solved, structurally, in a completely unrelated field — one you'd never normally look to for an answer?*

29

Think Like the User

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Temple Grandin, diagnosed with autism as a child at a time when the condition was poorly understood and treatment options were limited, has described experiencing the world in ways that differ substantially from typical sensory and cognitive processing — most notably, thinking in vivid, detailed images rather than primarily in words, and experiencing sounds, textures and sudden movements with an intensity that many neurotypical people do not. Trained as an animal scientist, Grandin turned her attention in the 1970s to the design of livestock handling facilities — the chutes, pens and processing systems used to move cattle through veterinary treatment, sorting and, in some contexts, slaughter facilities. The existing standard designs of the era were built around straightforward assumptions about how cattle would behave, based largely on visual observation of the facilities from a human, upright, adult-eye-level vantage point — assumptions that, in practice, frequently produced cattle that balked, panicked or froze in ways facility designers found puzzling and attributed to the animals simply being unpredictable or difficult.

Grandin's methodology was distinctive and directly informed by her own sensory experience: she got down, physically, to the eye level and visual perspective an actual animal moving through the facility would have, walking the chutes herself from that vantage point rather than relying only on observation from a standing human position. From that altered perspective, she identified specific triggers invisible to standard human-eye-level

observation: a shadow crossing the path at a certain angle, a chain hanging loosely enough to swing and catch a reflection of light, a change in flooring texture or colour that, from an animal's specific angle and height, looked like a sudden drop rather than a flat continuation of the path.

Cattle repeatedly balking at what appeared to human observers to be an unremarkable stretch of chute were, from the animal's actual visual vantage point, encountering a genuinely alarming, ambiguous visual signal that a standing human observer simply never saw. Redesigning facilities to systematically eliminate these specific triggers — curving chute designs that kept animals from seeing what was ahead until the last moment (reducing anticipatory fear of the unknown), controlling lighting and shadow patterns, and removing or securing loose or reflective objects along the path — produced measurably calmer animal movement and significantly reduced injury and stress for both animals and handlers.

Grandin's curved chute designs and broader handling principles are now estimated to be used in roughly half of all cattle handling facilities in North America, and her work extended well beyond livestock design into influential advocacy and writing on autism itself, informed directly by the same capacity for detailed, literal, visually specific perception that had originally shaped her approach to facility design.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Have you ever actually experienced your product, process or workspace from your end user's literal physical vantage point, rather than relying on how it appears from your own, different position?*

KEY TAKEAWAYS

1. Standard facility designs failed because they were based on how the space looked from a human, standing vantage point, not from the actual animal's eye level and angle.
2. Grandin's specific sensory processing — thinking in detailed images and experiencing visual and sensory stimuli intensely — gave her a genuine advantage in identifying triggers invisible to typical human observation.
3. Systematically eliminating specific, identified visual triggers produced measurable behavioural change, not a vague general improvement — the fixes were precise because the diagnosis was precise.

SUMMARY

Temple Grandin redesigned livestock handling facilities by physically adopting the actual visual perspective and eye level of the cattle moving through them, identifying specific triggers — shadows, reflections, texture changes — that alarmed animals but were invisible from a standing human's vantage point. Her resulting curved chute designs, which minimize an animal's view of what lies ahead, are now almost half of North American cattle are handled in a centre-track restrainer system she designed. Her sensory processing style, associated with her autism, directly informed the precision of her observations.

STORY CHECK

ESTABLISHED

Grandin's autism diagnosis, her career in animal science, her curved chute designs, and their widespread adoption across North American livestock facilities are documented in her own published work and independent agricultural industry sources.

POPULAR VERSION ADDS

The "almost half of North American cattle" figure is Grandin's own estimate, widely repeated, though not independently audited by a third party.

UNCERTAIN

The precise degree to which her specific sensory experience of autism, as opposed to simply rigorous behavioural observation methodology, accounts for her design insights is a nuanced question Grandin herself has discussed with more complexity than most retellings capture.

VOCABULARY & PHRASAL VERBS

chute (n.) a narrow, often sloped passage used to guide or move animals or materials. Cattle moved through narrow handling chutes.

balk (v.) to stop abruptly and refuse to continue, especially out of fear or hesitation. Cattle repeatedly balked at a specific point in the chute.

vantage point (n.) the physical position or perspective from which something is observed. She adopted the animal's actual vantage point.

trigger (n.) a specific stimulus that provokes a reaction. She identified specific visual triggers invisible to human observers.

anticipatory (adj.) relating to expectation of something before it happens. Curved chutes reduced anticipatory fear of the unknown.

neurotypical (adj.) describing cognitive and sensory processing considered typical or standard, as distinct from autistic or otherwise atypical processing. Her perception differed from neurotypical processing.

get down to (phrasal verb) to physically lower oneself to a specific level or position. She got down to the animal's actual eye level.

rule out (phrasal verb) to eliminate a possibility through investigation. Systematic redesign ruled out specific visual triggers one by one.

inform (v., figurative) to shape or influence an approach or decision. Her sensory experience directly informed her design methodology.

PRACTICE

Where the Story Breaks Down

Grandin's specific sensory processing gave her genuine, direct access to a perspective most designers cannot simply choose to adopt at will; "think like the user" as generic advice is much harder to execute without an equivalent, specific way of actually perceiving what the user perceives.

Use the Language

Describe a product or process you could physically or literally experience from your end user's vantage point, using vantage point, trigger and rule out.

Discuss

What would it mean, concretely and physically, to adopt your actual end user's vantage point for something you're currently designing or building?

Write

In 100 words, describe one specific, physical way you could observe your product or process from a genuinely different vantage point than the one you normally use.

SOURCES

B Grandin, T. & Johnson, C., Animals in Translation: Using the Mysteries of Autism to Decode Animal Behaviour, Scribner (2005).

C Grandin, T., official website and published technical papers on livestock handling facility design (grandin.com).

B Grandin, T., "Behavioural Principles of Livestock Handling," Professional Animal Scientist 5(2), 1–11 (1989).

THE CONTRAST

Story 29: Temple Grandin vs. Story 40: The Arm That Couldn't Grab (Canadarm)

Story 29: Temple Grandin: Designed from a literal, physical understanding of how the actual user experiences a space — seen and felt from their exact vantage point.

Story 40: The Arm That Couldn't Grab (Canadarm): Designed by discovering the obvious approach didn't work in the actual physical environment, and rebuilding the solution around the environment's real constraints.

Is good design more about deeply understanding the user's actual experience, or about deeply understanding the environment's actual physical constraints?

THIS CONNECTS TO

Story 30: The Product People Couldn't Yet Ask For (Sony Walkman)

LIMITATION

Grandin designed from direct, literal observation of behaviour. Morita's approach relied on conviction about a need customers couldn't yet articulate at all — a harder problem to test directly.

Story 26: The Store That Sold Shoes It Didn't Own (Zappos)

SAME MECHANISM

Both prioritized direct, concrete evidence about how people or animals would actually behave over assumptions made from a comfortable distance.

Story 35: The Diagram That Made Death Visible (Nightingale)

SAME MECHANISM

Both changed outcomes by shifting how evidence was perceived — seeing a space from the animal's actual vantage point, or seeing mortality data in a genuinely visual format — rather than by changing the underlying facts.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Standard livestock facility designs were based on how the space looked to a human observer standing upright, and cattle repeatedly balked and resisted moving through them.

T — Task

Temple Grandin needed to understand why the animals were reacting the way they were, in order to design facilities that reduced their stress.

A — Action

Drawing on her own sensory experience, she physically got down to the cattle's eye level and observed the facilities from their actual visual and physical vantage point, identifying specific details — shadows, reflections, dangling chains — that were causing the animals to balk.

R — Result

Her redesigned facilities, based on that literal point of view, are now used across a significant share of the cattle industry in North America.

Level 2 — Make It Your Story

> *Tell me about a time when experiencing a product, process, or space from the user's actual physical point of view changed how you designed it.*

Use STAR to answer. Think about: user testing, walking through a process yourself, a design flaw only visible from the user's exact vantage point.

TAKE THE IDEA FURTHER

> *Have you ever actually experienced your product, process or workspace from your end user's literal physical vantage point, rather than relying on your own?*

30

The Product People Couldn't Yet Ask For

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

By the late 1970s, Sony co-founder Akio Morita had built the company into a global consumer electronics leader on the strength of products developed largely by anticipating needs consumers had not yet articulated, rather than by responding to explicit customer requests or conventional market research. The Walkman, launched in 1979, became the clearest and most commercially significant expression of that philosophy.

The product's origin, according to accounts from those directly involved, traces to a specific and mundane frustration: Sony co-founder Masaru Ibuka wanted a way to listen to music privately during long international flights, without disturbing other passengers and without the bulk of a full-sized cassette player and speakers. Sony engineers had recently developed a compact cassette player, the Pressman, originally intended for journalists recording audio in the field. Adapting it further — stripping out the recording function and speaker entirely, adding lightweight headphones, and focusing exclusively on compact private playback — produced a device with no clear precedent in the existing consumer electronics market.

Sony's internal marketing division, when consulted, reportedly expressed significant skepticism. Conventional market research at the time struggled to make sense of the concept: a portable cassette player with no ability to record and no built-in speaker seemed, on its surface, like a strictly reduced, worse version of existing products rather than a new category with independent appeal, and formal market testing methods of the era had no established way

to measure demand for a genuinely novel use case nobody had previously experienced or been asked about. Morita, according to widely repeated internal accounts, chose to trust the product's own internal logic and his own conviction over the market research department's skepticism, authorizing production and a modest initial marketing push despite that internal resistance.

The Walkman's actual success came from something market research of the period was poorly equipped to detect in advance: watching what people did once the product genuinely existed to be tried, rather than trying to predict abstractly, through a survey question, whether people would want a device solving a problem they had never previously had the opportunity to experience solving. Once available, people used the Walkman to transform previously dead or socially awkward time — commutes, walks, waiting rooms — into private, soundtracked time, a use case that proved to have far broader appeal than the recording-industry frame the marketing department had originally tried, and failed, to evaluate it against. The device sold roughly 385 million units worldwide across its various forms over the following decades and fundamentally reshaped how people expected to experience recorded music in public and semi-private spaces, a shift that arguably paved the way for the entire category of personal portable media devices that followed.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What product or service in your organization might be difficult for customers to request explicitly, simply because they've never had the opportunity to experience the specific problem it would solve?*

KEY TAKEAWAYS

1. Conventional market research struggled to evaluate the Walkman because it asked customers to imagine wanting something they had no prior experience of wanting, a genuinely difficult cognitive task for most survey respondents.
2. The product's internal logic — solving a real, specific, previously unaddressed personal need for private portable listening — mattered more than its performance against existing product categories in formal testing.
3. Actual behaviour once the product existed, rather than predicted behaviour gathered through surveys beforehand, revealed the real and much broader appeal of the device.

SUMMARY

Sony's Walkman originated from co-founder Masaru Ibuka's desire for private portable music on flights, adapted from an existing journalist's recording device with the recording function and speaker removed. Internal market research was skeptical, since the concept — a cassette player that couldn't record or play through speakers — didn't fit any existing product category customers could meaningfully evaluate in advance. Akio Morita authorized production despite that skepticism, and the product's real appeal only became visible once people could actually use it, eventually selling roughly 385 million units and reshaping how people expected to experience recorded music.

STORY CHECK

ESTABLISHED

The 1979 Walkman launch, its adaptation from the Pressman recording device, and its eventual commercial scale are documented in Sony's own corporate history and independent business retrospectives.

POPULAR VERSION ADDS

The internal marketing skepticism and Morita's decision to override it are widely repeated in business literature as a clean, simple story of visionary leadership triumphing over cautious market research; the actual internal deliberation was likely more nuanced and involved more people and more back-and-forth than the compressed anecdote usually conveys.

UNCERTAIN

The precise, verbatim internal marketing objections and Morita's exact reasoning in overriding them are known mainly through later retrospective accounts rather than contemporaneous internal documentation.

VOCABULARY & PHRASAL VERBS

precedent (n.) an earlier example or case that establishes a pattern for later situations. The device had no clear precedent in the existing market.

skepticism (n.) doubt or a questioning attitude toward a claim or plan. The marketing division expressed significant skepticism.

novel (adj.) new and previously unknown or unusual. Formal research methods struggled with a genuinely novel use case.

internal logic (n., figurative) the coherent reasoning behind a product or decision, considered on its own terms. Morita trusted the product's internal logic over market research.

soundtrack (v., figurative) to accompany an experience with music, used here as a verb. People used it to soundtrack previously dead time.

dead time (n.) a period considered wasted or unproductive, here referring to commutes or waiting. Commutes became soundtracked instead of dead time.

override (v.) to use one's authority to disregard or supersede an objection or existing decision. Morita overrode the marketing department's skepticism.

pave the way for (phrase) to create conditions that make a later development possible. The Walkman paved the way for portable media devices.

reshape (v.) to change something's fundamental form or nature. It reshaped expectations around experiencing recorded music.

PRACTICE

Where the Story Breaks Down

Morita had the authority, track record and personal conviction within Sony to override a skeptical internal department; most people proposing an untested, hard-to-research idea don't have an equivalent level of institutional trust to draw on.

Use the Language

Describe an idea that would be difficult for customers to explicitly request, using precedent, novel and override.

Discuss

How would you gather evidence for a product idea that conventional market research is poorly equipped to evaluate?

Write

In 100 words, describe a "dead time" or overlooked context in your own users' lives that a new product or service might soundtrack, fill, or otherwise transform.

SOURCES

C Sony Corporation, official corporate history of the Walkman (sony.com).

B Nathan, J., Sony: The Private Life, Houghton Mifflin (1999).

D The specific internal marketing-department objections are drawn from later retrospective accounts in business literature rather than contemporaneous internal Sony documentation.

THE CONTRAST

Story 30: Sony Walkman (Morita) vs. Story 29: Think Like the User (Temple Grandin)

Story 30: Sony Walkman (Morita): Built and shipped a real product based on conviction, because the customer couldn't articulate the need in advance for market research to test at all.

Story 29: Think Like the User (Temple Grandin): Designed from a literal, physical understanding of how the actual user experiences a space — seen and felt from their exact vantage point.

Should you always test the riskiest assumption cheaply before committing — or are some genuinely new ideas untestable that way, and require conviction instead?

THIS CONNECTS TO

Story 26: The Store That Sold Shoes It Didn't Own (Zappos)

CONTRAST

Zappos tested its riskiest assumption cheaply before committing. Morita

couldn't test his assumption at all — conventional market research can't ask customers whether they want something they have no frame of reference for.

Story 27: What a Machine Could Become (Ada Lovelace)

SAME MECHANISM

Both required seeing past what a technology or product currently did toward what it could become, before customers or engineers could articulate the need themselves.

Story 43: Seeing the Smartphone Too Early (Nokia) LIMITATION

Morita shipped on conviction and it succeeded. Nokia had comparable internal conviction about the smartphone years early and chose not to act on it — conviction alone isn't what determines the outcome.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

By the late 1970s, Sony had built its reputation on anticipating consumer needs, and Akio Morita believed there was demand for a personal, portable music player, despite no existing product category or customer request for one.

T — Task

He needed to decide whether to invest in developing and launching the product despite skepticism, including from within his own company, since conventional market research couldn't meaningfully test demand for something customers had no experience imagining.

A — Action

He championed the product's development and launch despite the internal doubts, betting on the underlying insight rather than testable market data.

R — Result

The Walkman became a global success and created an entirely new consumer product category.

Level 2 — Make It Your Story

> *Tell me about a time when you championed a product or idea that customers couldn't yet articulate wanting, because they had no experience to imagine it from.*

Use STAR to answer. Think about: a new feature, a genuinely novel product, an idea market research couldn't have predicted.

TAKE THE IDEA FURTHER

> *What product or service in your organization might be difficult for customers to request explicitly, simply because they've never had the opportunity to want it?*

31

The Company That Killed Its Own Business

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED

Verified through primary sources, patents, or independent contemporaneous accounts.

For years, Netflix's own founding story circulated as a tidy piece of business folklore: CEO Reed Hastings, so the tale went, was charged a $40 late fee for a forgotten VHS copy of Apollo 13, and the embarrassment of that moment directly sparked the idea for a DVD-by-mail service with no late fees at all. Hastings himself repeated versions of this story in interviews for years. Co-founder Marc Randolph has since directly disputed it, telling the company's biographer it was closer to "a lot of crap" — the real origin, he says, was a more mundane search for "the Amazon.com of something" that eventually settled on DVDs, motivated as much by logistics and bandwidth economics as by any single moment of personal frustration.

The more consequential and better-documented story is not how Netflix started, but what the company did roughly a decade after starting, once its DVD-by-mail subscription business had become genuinely successful and profitable. By the mid-2000s, Netflix's DVD service had millions of subscribers and a dominant market position, built on physical distribution infrastructure — sorting centres, shipping logistics, a business Netflix understood extremely well and had spent years optimizing.

Streaming video, when internet bandwidth finally made it technically viable around 2007, threatened to make that entire infrastructure obsolete. Netflix had a genuine choice available to most successful companies facing a comparable structural shift: protect and continue optimizing the proven, profitable business as long as possible, or actively build a replacement that

would, if successful, cannibalize the very revenue stream funding the replacement's own development. Netflix chose the second option, launching streaming as an add-on to existing DVD subscriptions in 2007, then investing heavily in expanding streaming capability and content licensing over the following years, even as doing so systematically reduced the relative importance of the DVD business that remained, for some time, more profitable on a per-subscriber basis.

The transition was neither smooth nor risk-free, and it is worth resisting the temptation to tell it as a clean triumph. Netflix's 2011 attempt to formally separate its DVD and streaming businesses into two distinct services, with the DVD arm renamed Qwikster, was announced clumsily, provoked intense subscriber backlash, and was reversed within weeks — the company lost roughly 800,000 US subscribers in a single quarter and its stock price fell by more than 75 percent by that November. Netflix recovered from that specific misstep, and by continuing to prioritize streaming's long-term potential over DVD's near-term profitability across the years that followed, ultimately built the business that eclipsed and effectively replaced the video rental industry, including Blockbuster, which filed for bankruptcy in 2010.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What proven, currently profitable part of your business would you need to actively undermine in order to build the thing that might replace it before someone else does?*

KEY TAKEAWAYS

1. Netflix's own famous founding anecdote is now widely regarded, including by a co-founder, as a story co-founder Marc Randolph calls "a convenient fiction" — worth remembering before trusting any company's tidy origin myth.
2. The company's more significant and better-documented decision was choosing to invest in streaming even while it reduced the relative value of its proven, profitable DVD business.
3. The transition was not smooth — the 2011 Qwikster split was a genuine, costly misstep, not evidence that self-disruption is risk-free once you've decided to attempt it.

SUMMARY

Netflix's widely repeated founding story — a $40 Blockbuster late fee inspiring the entire business — has been disputed by the company's own co-founder as an oversimplified myth. The more substantively important story is Netflix's mid-2000s decision to invest heavily in streaming video even as doing so threatened its own profitable DVD-by-mail business, a genuine case of self-disruption. The transition included real missteps, including the poorly received 2011 attempt to split the company into two separate services, which was reversed within weeks after significant subscriber and stock losses.

STORY CHECK

ESTABLISHED

Netflix's 2007 streaming launch, its DVD business's prior profitability, and the 2011 Qwikster reversal (including the roughly 800,000 subscriber loss and stock price decline) are all documented in company records and extensive independent business press coverage.

POPULAR VERSION ADDS

The $40 late fee founding story is now directly disputed by co-founder Marc Randolph, who has called it an oversimplified or fabricated narrative; Hastings has offered somewhat inconsistent accounts of the company's actual origin across different interviews over the years.

UNCERTAIN

The precise internal reasoning and debate within Netflix's leadership about the pace and sequencing of the streaming transition is not fully public, beyond the company's own retrospective framing.

VOCABULARY & PHRASAL VERBS

cannibalize (v., business) to reduce a company's own product's or business's success by introducing a competing one. Streaming risked cannibalizing the profitable DVD business.

infrastructure (n.) the underlying physical or organizational systems needed to support an operation. The DVD business relied on extensive shipping infrastructure.

obsolete (adj.) no longer useful due to being replaced by something newer. Streaming threatened to make DVD infrastructure obsolete.

backlash (n.) a strong negative reaction from a group of people to an event or decision. The Qwikster announcement provoked intense subscriber backlash.

apocryphal (adj.) of doubtful authenticity, though widely circulated as if true. The late-fee origin story is now considered largely apocryphal.

settle on (phrasal verb) to choose and commit to an option after considering others. The founders settled on DVDs as their initial business.

eclipse (v., figurative) to surpass or make less significant by comparison. Netflix eventually eclipsed the traditional video rental industry.

reverse course (phrase) to change direction, often after a failed decision. Netflix reversed course on Qwikster within weeks.

fund (v.) to provide the financial resources for something. DVD revenue helped fund the early streaming investment.

PRACTICE

Where the Story Breaks Down

Netflix could fund its streaming transition, in significant part, with profits from its still-successful DVD business; not every organization facing disruption has an equivalent internal source of capital to fund the replacement while the original business is being deliberately wound down.

Use the Language

Describe a business or process that might need to cannibalize its own success to survive long-term, using cannibalize, obsolete and fund.

Discuss

What is the difference between a founding myth that's harmless marketing and one that actively distorts how people understand a company's real strategic decisions?

Write

In 100 words, describe a profitable part of your own organization that a genuine future threat might require actively undermining, and consider whether the organization would actually be willing to do that.

SOURCES

B Keating, G., Netflixed: The Epic Battle for America's Eyeballs, Portfolio (2012) — contains Marc Randolph's direct dispute of the late-fee origin story.

C Netflix, Inc., investor relations records on the 2011 Qwikster announcement and subsequent reversal.

D Reed Hastings's various retellings of the company's founding across multiple interviews (Fortune, CNBC, The New York Times) are inconsistent with each other in specific details.

THE CONTRAST

Story 31: Netflix vs. Story 42: Abandoning the Business You Built (Intel/Grove)

Story 31: Netflix: Actively undermined its own currently profitable DVD-by-mail business in order to build the streaming business that would eventually replace it.

Story 42: Abandoning the Business You Built (Intel/Grove): Used a deliberate mental exercise — imagining a new CEO with no history in the business — to justify abandoning the memory-chip business that had defined the company's identity.

Is it harder to abandon a business you personally built, or to abandon the industry your whole company's identity was built around?

THIS CONNECTS TO

Story 26: The Store That Sold Shoes It Didn't Own (Zappos)

LIMITATION

Zappos tested a risky assumption before committing real infrastructure. Netflix's chapter is about committing to cannibalize an already-successful,

real business — a much higher-stakes version of the same willingness to bet against comfortable assumptions.

Story 41: The Camera It Invented (Kodak) CONTRAST

Netflix actively dismantled its own profitable business to build its replacement. Kodak had the replacement technology built internally and chose the opposite path — protecting the profitable business instead.

Story 14: The Moth in the Machine (Grace Hopper) SAME MECHANISM

Both organizations' own founding or breakthrough stories turned out, under scrutiny, to be tidier and more flattering than the documented record supports.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Netflix had built a profitable DVD-by-mail rental business, but its leadership believed streaming video represented the medium-term future of the industry.

T — Task

They needed to decide whether to protect the profitable existing business or actively invest in and promote the streaming service that would eventually cannibalize it.

A — Action

Netflix invested heavily in building out streaming and progressively shifted customer and company focus toward it, even as doing so directly undermined its own existing, profitable DVD business.

R — Result

Netflix became a dominant streaming company; its own popularized

founding story — a late fee inspiring the whole business — is one co-founder Marc Randolph calls "a convenient fiction" of a more gradual, deliberate strategic evolution.

Level 2 — Make It Your Story

> *Tell me about a time when you or your team had to actively work against a currently successful part of the business in order to build what would replace it.*

Use STAR to answer. Think about: a legacy product, a profitable process you helped phase out, a transition you helped drive despite short-term cost.

TAKE THE IDEA FURTHER

> *What proven, currently profitable part of your business would you need to actively undermine in order to build the thing that might replace it?*

32

Connect What Already Exists

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

By 1989, when Tim Berners-Lee was working as a software engineer at CERN, the European particle physics laboratory, none of the individual technical components that would become the World Wide Web were new or original to him. Hypertext — the concept of clickable links connecting one document to another — had existed as an academic idea since the 1960s. The internet itself, as a network capable of transmitting data between computers, had existed in various forms since the 1960s and 70s. Personal computers capable of displaying formatted text and graphics were increasingly common. Each piece existed, separately, in its own established technical and academic community, largely unconnected to the others.

Berners-Lee's actual problem was narrower and more mundane than "invent a new form of global communication." CERN, at the time, housed thousands of researchers, using a wide variety of incompatible computer systems, generating an enormous and growing volume of documents, data and internal information that was extremely difficult to locate, cross-reference or share reliably across the lab's many different systems and departments — a practical, immediate, internal information-management headache rather than a grand technological ambition. His insight, laid out in a March 1989 proposal titled "Information Management: A Proposal," was not that any single existing technology needed to be invented, but that hypertext, applied over an existing internet connection, using a defined common format any computer could read, could solve CERN's specific document-management

problem — and that the same combination, if built generally enough, had no obvious reason to remain confined to CERN's internal network. His initial proposal reportedly received a notably tepid response from his supervisor, who scrawled "vague, but exciting" on the cover, not exactly a ringing institutional endorsement for what would become one of the most significant technological developments of the century.

Working largely on his own initiative over the following two years, Berners-Lee built the actual components that turned the proposal into a working system: HTML, a simple markup format any computer could interpret and display consistently; HTTP, a protocol for transmitting those documents over an existing internet connection; and the first web browser and web server software, needed to actually demonstrate the whole combination functioning end to end. The World Wide Web's first public demonstration came in 1991, and Berners-Lee made a deliberate and consequential decision in 1993: rather than patenting the technology and licensing it for royalties, he and CERN released the World Wide Web's core specifications into the public domain, royalty-free, for anyone to use or build upon without restriction — a decision widely credited as essential to the web's subsequent explosive, permissionless global growth.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What internal, unglamorous problem in your own organization might be solvable by combining two or three existing, unrelated tools or systems, rather than by building or acquiring something genuinely new?*

KEY TAKEAWAYS

1. Every individual technical component of the World Wide Web already existed; the actual invention was the specific combination and the standard format connecting them.
2. The problem that motivated the work was a narrow, internal, practical one — CERN's document management — not an ambition to create a new global medium.
3. The decision to release the specification into the public domain, rather than licensing it for profit, is widely credited as a critical factor in its rapid, unrestricted adoption.

SUMMARY

Tim Berners-Lee did not invent hypertext, the internet, or personal computing — all three already existed separately by 1989. His contribution was combining them into a working system, driven by the practical need to manage CERN's internal documentation across incompatible computer systems, and building the actual HTML, HTTP, browser and server components that made the combination function. His 1993 decision to release the specifications royalty-free, rather than patenting them, is widely credited as essential to the web's subsequent global adoption.

STORY CHECK

ESTABLISHED

Berners-Lee's March 1989 proposal, his subsequent development of HTML, HTTP and the first browser/server, the 1991 public demonstration, and the 1993 royalty-free release are all extensively documented in CERN's own archives and Berners-Lee's own published accounts.

POPULAR VERSION ADDS

Berners-Lee is sometimes described simply as "the inventor of the internet," which conflates the internet (the underlying network, developed by many people over decades) with the World Wide Web (the specific hypertext system built on top of it) — a distinction Berners-Lee himself is careful to maintain in his own writing.

UNCERTAIN

The exact wording and intent behind his supervisor's "vague, but exciting" annotation is documented on the surviving proposal itself, though its precise institutional significance at the time is a matter of some interpretation.

VOCABULARY & PHRASAL VERBS

hypertext (n.) text containing clickable links to other text or documents. Hypertext had existed as an academic idea for decades.

protocol (n., computing) a defined set of rules governing how data is transmitted or communicated. HTTP is the protocol used to transmit web pages.

markup (n., computing) a system of tags or codes used to format and structure text for display. HTML is a markup format any computer could interpret.

incompatible (adj.) unable to work together or coexist without conflict. CERN's researchers used a wide variety of incompatible systems.

tepid (adj.) showing little enthusiasm or interest. His proposal received a notably tepid initial response.

public domain (n., legal) the status of intellectual property that is not owned or restricted by anyone, available for free use. He released the specifications into the public domain.

build on (phrasal verb) to use an existing foundation or idea as the basis for further development. Anyone could build on the core specifications freely.

cross-reference (v.) to link or relate one piece of information to another for comparison. Researchers needed to cross-reference documents across departments.

lay out (phrasal verb) to explain or present something clearly and in detail. His 1989 proposal laid out the combined system.

PRACTICE

Where the Story Breaks Down

Berners-Lee had access, at CERN, to exactly the kind of technically sophisticated, internationally connected research environment where all three underlying technologies were already present and usable; the same combination would have been far harder to assemble in a less technically resourced setting.

Use the Language

Describe two or three existing tools or systems in your own work that might be combined to solve a specific problem, using protocol (or a suitable substitute), incompatible and build on.

Discuss

What internal, unglamorous problem in your own organization resembles CERN's document-management headache — solvable, potentially, by combination rather than invention?

Write

In 100 words, describe the decision to release a valuable specification for free rather than licensing it, and consider what your own organization would do in the same position.

SOURCES

A Berners-Lee, T., "Information Management: A Proposal," CERN internal document (March 1989).

B Berners-Lee, T., Weaving the Web: The Original Design and Ultimate Destiny of the World Wide Web, HarperSanFrancisco (1999).

C CERN, official archival record of the World Wide Web's development and 1993 public domain release (home.cern).

THIS CONNECTS TO

Story 1: The Rocket That Was Too Expensive (Musk) SAME MECHANISM

Both cases show that a price or a barrier people treat as fixed is really just the current output of a system — one built by combining existing components differently, the other by rebuilding a supply chain from scratch.

Story 28: A Different Domain Solves the Problem (Hedy Lamarr)

SAME MECHANISM

Both solved their problem by combining existing pieces — a player-piano mechanism and radio, or existing internet protocols — rather than inventing something entirely new from scratch.

Story 27: What a Machine Could Become (Ada Lovelace) CONTRAST

See The Contrast in that chapter. In short: Lovelace imagined a capability decades before the technology existed to build it; Berners-Lee combined capabilities that already existed, right when they were ready to be combined.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

By 1989, working at CERN, Tim Berners-Lee had access to hypertext, the internet, and computer networking, but no unified system connected them for general use.

T — Task

He needed a way to let researchers easily link and access documents across different computers and networks.

A — Action

Rather than inventing new underlying technology, he specified a standard way to combine the existing pieces — hypertext links, unique addresses, and a transfer protocol.

R — Result

The result was the World Wide Web, released freely and becoming the basis for how most people access the internet.

Level 2 — Make It Your Story

> *Tell me about a time when combining two or three existing tools or systems solved a problem better than building something new.*

Use STAR to answer. Think about: an integration, a workflow you connected across existing tools, a standard you helped establish.

TAKE THE IDEA FURTHER

> *What internal, unglamorous problem in your own organization might be solvable by combining two or three existing, unrelated tools or systems, rather than building something from scratch?*

33

The Checklist

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Atul Gawande, a practicing surgeon and public health researcher, became interested in a specific and uncomfortable statistic: a substantial share of surgical complications and deaths were not the result of insufficient medical knowledge or skill, but of simple, preventable errors — a step skipped, a piece of equipment not confirmed, a basic precaution overlooked in the routine pressure and pace of an operating room, by teams of highly trained specialists who, individually, knew perfectly well what they were supposed to do.

Gawande's research, described in his 2009 book The Checklist Manifesto, drew a specific and instructive comparison to an entirely unrelated field: commercial aviation, where pilots — themselves highly trained, experienced professionals — routinely use short, physical checklists before takeoff, covering steps a skilled pilot could easily recite from memory unaided. The aviation industry had, decades earlier, concluded that expertise alone was not a reliable enough safeguard against the specific failure mode of a skilled person, under real-world time pressure and distraction, simply forgetting or skipping one step in a routine sequence they otherwise knew perfectly well.

Gawande and a research team, working with the World Health Organization, developed a surgical safety checklist: a short, simple document requiring a surgical team to verbally confirm a small number of specific items — correct patient, correct surgical site, confirmation that essential equipment was present and functioning, allergies checked — at three defined points before, during and after an operation. The checklist's design deliberately avoided the

trap of becoming exhaustive or comprehensive; a longer, more thorough-seeming checklist risked being skipped or rushed under time pressure, while a short one covering only the highest-value, most commonly missed items was more likely to actually be used consistently.

The checklist was tested across eight hospitals in eight different countries, including both well-resourced facilities and hospitals with far more limited resources, chosen deliberately to test whether the intervention would work outside ideal conditions. The results, published in the New England Journal of Medicine in 2009, were striking: major surgical complications fell by roughly a third across the study sites, and deaths fell by nearly half, from a document that added essentially no cost and took under two minutes to complete. The core insight generalizing well beyond surgery is that a checklist does not compensate for a lack of expertise. It compensates for a specific, well-documented limitation that exists even among genuine experts: memory and attention fail under pressure in predictable, preventable ways, and a simple external tool can catch what expertise alone reliably cannot.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What routine, high-stakes process in your organization currently relies entirely on skilled people remembering every step correctly under time pressure, with no external check?*

KEY TAKEAWAYS

1. The failures the checklist addressed were not failures of knowledge or skill — they were failures of memory and attention under routine time pressure, a distinct and separate problem.

2. A short, carefully limited checklist proved more effective than a longer, more comprehensive one, because it was more consistently actually used.
3. The intervention was tested deliberately across resource-rich and resource-poor hospitals to confirm it worked broadly, not just under ideal conditions.

SUMMARY

Atul Gawande, drawing on aviation's use of pre-flight checklists for skilled pilots, developed a short surgical safety checklist to address preventable errors caused by lapses in attention rather than lack of expertise. Tested across eight hospitals in eight countries with varying resource levels, the checklist reduced major surgical complications by roughly a third and deaths by nearly half. The design deliberately stayed short and focused, since a longer checklist risked being skipped under time pressure.

STORY CHECK

ESTABLISHED

The 2009 New England Journal of Medicine study, its eight-country, eight-hospital design, and the reported reductions in complications and mortality are all directly documented in the peer-reviewed publication itself.

POPULAR VERSION ADDS

The checklist's success is sometimes presented as though its content alone explains the result; Gawande's own writing emphasizes that the checklist's effectiveness depended heavily on how it was implemented — including whether surgical teams were trained to treat it as a genuine team communication tool rather than a bureaucratic formality to rush through.

UNCERTAIN

The precise mechanism by which the checklist reduced complications (attention, communication, or team dynamics) is discussed with more nuance in follow-up research than the headline statistics alone convey.

VOCABULARY & PHRASAL VERBS

complication (n., medical) an unfavorable and unplanned development following surgery or treatment. Surgical complications fell by roughly a third.

lapse (n.) a brief failure of memory, attention or judgment. Many errors were lapses in attention, not lapses in knowledge.

exhaustive (adj.) covering every possible detail thoroughly, sometimes to excess. An exhaustive checklist risked being skipped under pressure.

verbally confirm (phrase) to state something aloud to check and validate it, rather than silently assuming it. The team had to verbally confirm the surgical site.

safeguard (n.) a measure taken to protect against harm or error. Expertise alone was not a reliable enough safeguard.

generalize (v.) for a finding or principle to apply beyond its original specific context. The insight generalizes well beyond surgery.

catch (v., figurative) to detect or intercept an error before it causes harm. A simple checklist can catch what memory alone cannot.

roll out (phrasal verb) to introduce a new process or system, especially across multiple locations. The checklist was rolled out across eight hospitals.

compensate for (phrasal verb) to counteract or make up for a weakness or limitation. The checklist compensates for a limitation in human memory, not skill.

PRACTICE

Where the Story Breaks Down

Surgical checklists work partly because the sequence of steps in an operation is genuinely well-defined and repeatable; not every high-stakes process has an equally clear, stable, repeatable sequence a short checklist could meaningfully capture.

Use the Language

Describe a routine, high-stakes process in your own work that might benefit from a short external checklist, using lapse, safeguard and catch.

Discuss

What is the difference between a checklist that genuinely improves a process and one that becomes a rushed bureaucratic formality nobody really engages with?

Write

In 100 words, design a short, five-item checklist for a real recurring process in your own work, focusing only on the highest-value items most likely to be forgotten under pressure.

SOURCES

A Haynes, A. B. et al., "A Surgical Safety Checklist to Reduce Morbidity and Mortality in a Global Population," New England Journal of Medicine 360, 491–499 (2009).

B Gawande, A., The Checklist Manifesto: How to Get Things Right, Metropolitan Books (2009).

C World Health Organization, "Safe Surgery Saves Lives" official program documentation.

THE CONTRAST

Story 33: The Checklist (Gawande) vs. Story 16: The Factory That Could Stop Itself (Jidoka)

Story 33: The Checklist (Gawande): Depends on a person following a defined sequence correctly, useful precisely where automatic detection isn't possible.

Story 16: The Factory That Could Stop Itself (Jidoka): Stops a problem automatically, the instant it occurs, without depending on a person remembering anything.

When you can't automate a safeguard, is a checklist a strong enough substitute — or just a weaker one?

THIS CONNECTS TO

Story 11: The Five Whys LIMITATION

Five Whys finds the deep cause of a specific failure. A checklist doesn't find causes at all — it prevents a known failure from recurring regardless of its root cause, which is a different kind of fix for a different kind of problem.

Story 16: The Factory That Could Stop Itself (Jidoka)

SAME MECHANISM

Both are structural safeguards built to catch a known failure mode reliably, rather than depending on individual vigilance or skill in the moment.

Story 45: The Goal That Had to Change (Shackleton) LIMITATION

A checklist assumes the situation is routine enough to be anticipated in advance. Shackleton's chapter is the opposite case — a situation so far outside any anticipated scenario that no checklist could have covered it.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Atul Gawande found that a substantial share of surgical complications and deaths were caused not by insufficient medical knowledge or skill, but by simple, preventable lapses in memory and attention under routine time pressure.

T — Task

He needed a low-tech intervention that could reduce these lapses without requiring more training or expertise.

A — Action

He developed and tested a simple surgical safety checklist covering basic steps that experienced surgical teams sometimes skipped under pressure.

R — Result

Hospitals using the checklist saw significant reductions in complications and deaths, demonstrating that a simple procedural tool could meaningfully improve outcomes in a highly skilled field.

Level 2 — Make It Your Story

> *Tell me about a time when a simple procedural tool, like a checklist, prevented a mistake that had nothing to do with lack of skill or knowledge.*

Use STAR to answer. Think about: a launch checklist, a pre-flight routine, a process you built specifically to catch memory lapses under pressure.

TAKE THE IDEA FURTHER

> *What routine, high-stakes process in your organization currently relies entirely on skilled people remembering every step correctly under time pressure?*

34

Remove the Friction

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In September 1997, Amazon filed a patent for a specific, narrow piece of e-commerce technology: a method allowing a returning customer, whose shipping and payment information was already stored from a previous purchase, to complete a new purchase with a single click, skipping the multi-step checkout process — entering an address, selecting a shipping method, confirming payment details — that essentially every other online retailer of the era required for every single transaction.

The patent, granted in 1999 and generally referred to as the "1-Click" patent, addressed a problem that was, on its face, a minor inconvenience: a few extra steps, a minute or two of additional time, in a checkout process most customers would tolerate without much complaint. Amazon's underlying analysis, however, treated that minor inconvenience as a genuinely significant source of lost sales, based on a well-documented pattern in e-commerce broadly: a meaningful percentage of online shopping carts, then as now, are abandoned before checkout is completed, often in the final steps of the process rather than earlier — the exact point at which small, cumulative friction (another form field, another confirmation screen, another moment of hesitation) has the most opportunity to cause a customer to simply give up and close the browser tab.

Amazon's strategic bet was that removing even a small amount of friction at exactly the final decision point — the instant between "I want this" and "I have bought this" — would produce a disproportionately large effect on

completed purchases, because that final moment is precisely where impulse and momentum are most fragile and most easily lost to a single additional required action. The 1-Click patent became commercially significant enough that Amazon licensed it to other companies, most notably Apple, which used it for one-click purchases in the iTunes Store, and defended it aggressively against competitors seeking to implement similar functionality, a degree of legal protection that drew some criticism from those who considered the underlying idea too simple and obvious to merit a defensible patent.

The 1-Click patent expired in 2017, and the specific legal mechanism it relied on has become less significant as one-click and stored-payment checkout has become close to a universal baseline expectation across e-commerce broadly, rather than a distinctive competitive advantage unique to any single retailer. What has remained a durable and generalizable insight, independent of the specific patent, is Amazon's broader underlying discipline: systematically identifying and removing small points of friction throughout the customer experience, on the premise that even minor obstacles produce measurable, and often underestimated, reductions in the number of people who complete an intended action.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Where in your own customer or user experience does a small, seemingly minor obstacle sit at exactly the final decision point — the moment between intending an action and completing it?*

KEY TAKEAWAYS

1. The problem addressed was treated as significant not because any single instance of friction seemed large, but because of well-documented, high abandonment rates specifically concentrated at the final steps of a process.
2. The strategic insight was about the location of friction, not merely its existence — removing friction at the final decision point mattered more than removing an equivalent amount of friction earlier in the process.
3. The specific patent's commercial and legal significance faded over time, but the underlying discipline of systematically hunting for and removing small friction points remained a durable, generalizable practice independent of any one legal mechanism.

SUMMARY

Amazon's 1997 "1-Click" patent addressed cart abandonment by removing the final steps of a multi-step checkout process for returning customers, based on the recognition that small friction concentrated at the final decision point disproportionately causes customers to abandon an otherwise completed purchase. The patent became commercially significant, licensed to companies including Apple, before expiring in 2017. The broader, still-relevant lesson is a systematic discipline of identifying and removing friction throughout a customer experience, independent of any single patented mechanism.

STORY CHECK

ESTABLISHED

The 1997 patent filing, its 1999 grant, its licensing to Apple, and its 2017 expiration are all documented in US Patent and Trademark Office records and independent business press coverage.

POPULAR VERSION ADDS

Some retellings treat "1-Click" as though it were the sole or primary driver of Amazon's e-commerce dominance; in reality it was one specific, well-documented component of a much broader and continuous set of friction-reduction practices across the company's entire customer experience.

UNCERTAIN

Amazon's own internal data on the precise sales impact attributable specifically to the 1-Click feature has not been independently published in detail.

VOCABULARY & PHRASAL VERBS

friction (n., figurative) resistance or difficulty that slows down or discourages an action. Small friction at checkout caused meaningful cart abandonment.

cart abandonment (n., e-commerce) when a customer adds items to an online shopping cart but leaves without completing the purchase. Cart abandonment was concentrated in the final steps.

disproportionate (adj.) out of proportion to the size or significance of the cause. A small change had a disproportionately large effect on sales.

impulse (n.) a sudden inclination to act, especially without deliberation. Impulse and momentum are fragile at the final decision point.

baseline (n.) a standard or minimum expectation against which things are measured. One-click checkout became a baseline expectation industry-wide.

license (v., legal) to grant another party permission to use protected intellectual property, usually for payment. Amazon licensed the patent to Apple.

give up (phrasal verb) to stop trying or abandon an effort. Customers would simply give up and close the tab.

hunt for (phrasal verb) to search actively and deliberately for something. The company built a discipline of hunting for friction points.

fade (v., figurative) to gradually lose significance or relevance over time. The patent's legal significance faded as the practice became standard.

PRACTICE

Where the Story Breaks Down

Amazon had the scale and data to identify precisely where in a long checkout process abandonment was concentrated; a smaller organization without equivalent analytics might struggle to locate its own highest-value friction point with the same precision.

Use the Language

Describe a specific point of friction in your own product, service or internal process, using friction, disproportionate and baseline.

Discuss

What is a small obstacle in your own organization's customer or employee experience that sits at a genuinely critical decision point, rather than an unimportant one?

Write

In 100 words, identify the single moment in a process you know well where a small amount of added friction would have the largest effect on whether people complete it.

SOURCES

A US Patent 5,960,411, "Method and System for Placing a Purchase Order via a Communications Network," Amazon.com, Inc. (1999).

C Amazon.com, Inc., investor relations and corporate history archives.

D Specific internal sales-impact figures attributable to 1-Click checkout have not been independently published by Amazon in detail.

THE CONTRAST

Story 34: Remove the Friction (Amazon) vs. Story 37: The Network That Made Overnight Delivery Possible (FedEx)

Story 34: Remove the Friction (Amazon): Removed a small obstacle at the exact final decision point of a transaction.

Story 37: The Network That Made Overnight Delivery Possible (FedEx): Restructured the entire path a package takes, making individual journeys longer in service of the whole system being faster.

Is speed better achieved by removing a small obstacle at one critical point, or by redesigning the whole structure a process moves through?

THIS CONNECTS TO

Story 12: The Plane Built in a Barn (Skunk Works) SAME MECHANISM

Both identify organizational or procedural steps — approval layers, checkout fields — as the real source of delay, not the underlying technical work.

Story 17: The Factory That Started Moving (Ford) SAME MECHANISM

Both identify that the sequence and arrangement of a process — not the effort of the people in it — was the real lever on speed.

Story 33: The Checklist (Gawande) LIMITATION

Amazon's fix removes a step to prevent a customer from abandoning a purchase. Gawande's checklist adds a step, deliberately, to prevent a different kind of failure — sometimes removing friction and sometimes adding a small amount of it are both the right move, depending on what's at stake.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Amazon's engineering team noticed that requiring customers to re-enter shipping and payment information at checkout, even when it was already stored from a previous purchase, caused some customers to abandon their purchase at the final step.

T — Task

They needed a way to eliminate that last-step friction for returning customers without compromising security.

A — Action

They developed a patented method allowing a single click to complete a purchase using already-stored information, removing the final obstacle entirely.

R — Result

The 1-Click feature became a significant driver of conversion and was widely recognized as a meaningful competitive advantage in e-commerce.

Level 2 — Make It Your Story

> *Tell me about a time when removing a small obstacle at the final step of a process had an outsized effect on the outcome.*

Use STAR to answer. Think about: a checkout flow, a signup process, a form field, a last-step obstacle you eliminated.

TAKE THE IDEA FURTHER

> *Where in your own customer or user experience does a small, seemingly minor obstacle sit at exactly the final decision point — the moment before someone commits?*

35

The Diagram That Made Death Visible

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Florence Nightingale is popularly remembered primarily as a nursing pioneer, a characterization that is accurate but incomplete in a way that undersells one of her most consequential and less publicized contributions: she was also a serious, self-taught statistician, and her work applying visual data presentation to public health policy arguably saved more lives than her direct nursing care during the Crimean War, though the two contributions were closely connected.

During her time nursing British soldiers at Scutari during the Crimean War in the 1850s, Nightingale observed a pattern that horrified her and that she was determined to prove with rigorous data rather than merely assert from personal observation: far more soldiers were dying from preventable diseases contracted in unsanitary hospital conditions — poor ventilation, contaminated water, overcrowding — than were dying from wounds sustained in actual combat. Returning to Britain, she compiled detailed mortality statistics and needed to convince a skeptical government bureaucracy, largely composed of officials with no statistical training and, by her own assessment, limited patience for dense tables of numbers, that sanitary reform in military hospitals was not merely desirable but urgently, dramatically necessary.

Rather than presenting her findings as a conventional table of monthly death tolls by cause — a format that, however accurate, made the relative scale and pattern of the problem difficult for a non-specialist reader to grasp quickly —

Nightingale developed and popularized a form of data visualization now often called a "coxcomb" or polar area diagram: a circular chart divided into monthly wedges, with each wedge's area, rather than merely its length, representing the number of deaths that month, colour-coded by cause. The visual result made a specific and previously easy-to-overlook pattern immediately, viscerally obvious to any viewer regardless of statistical training: the wedges representing preventable disease deaths dwarfed the wedges representing deaths from wounds, month after month, in a way that a table of the same underlying numbers simply did not communicate with equivalent immediate force.

The diagrams were included in a report Nightingale prepared for the British government and are widely credited with contributing directly to a sweeping set of sanitary reforms across British military hospitals, both during the war's later stages and, more significantly, afterward. Nightingale, notably, became the first woman elected to the Royal Statistical Society in recognition of this work, and her broader argument — that the format in which accurate data is presented can determine whether an accurate finding actually changes anyone's behaviour — remains a foundational principle in the modern field of data visualization and communication, well beyond public health specifically.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What accurate, important finding in your own organization currently sits in a table or spreadsheet that almost nobody outside your immediate team actually reads or absorbs?*

KEY TAKEAWAYS

1. Nightingale's underlying data — mortality by cause, by month — was not new information; what changed outcomes was the visual format in which she presented already-known facts.
2. The specific choice to make wedge area, not merely wedge length, represent the death count was a deliberate design decision intended to make relative scale immediately and viscerally obvious.
3. Accurate data presented in an inaccessible format can fail to produce action, even when the underlying facts, if properly understood, would be alarming enough to demand it.

SUMMARY

Florence Nightingale, working as a statistician alongside her better-known nursing work, discovered that far more Crimean War soldiers died from preventable disease in unsanitary hospitals than from combat wounds. To convince a government bureaucracy unmoved by conventional statistical tables, she developed a circular "coxcomb" diagram that made the overwhelming scale of preventable deaths immediately visible without requiring statistical training to interpret. The visualization is widely credited with contributing to sweeping sanitary reforms in British military hospitals and established Nightingale as a foundational figure in data visualization.

STORY CHECK

ESTABLISHED

Nightingale's Crimean War mortality data, her coxcomb diagrams, and her 1858 election to the Royal Statistical Society are all directly documented in her own published reports and the Society's own historical records.

POPULAR VERSION ADDS

Nightingale's statistical work is sometimes presented as a minor footnote to her nursing career; contemporary historians of statistics and data visualization generally treat it as independently significant, on its own terms, rather than as secondary to her nursing legacy.

UNCERTAIN

The precise causal weight of the diagrams themselves, as distinct from Nightingale's broader advocacy and the wider sanitary reform movement already gaining momentum at the time, in producing the specific policy changes that followed is difficult to isolate with certainty.

VOCABULARY & PHRASAL VERBS

mortality (n.) the rate or incidence of death, especially within a specific population. Mortality statistics revealed a shocking pattern.

sanitary (adj.) relating to conditions affecting hygiene and public health. Sanitary reform in military hospitals was urgently needed.

coxcomb diagram (n.) a circular statistical chart in which wedge area represents a quantity, popularized by Nightingale. Her coxcomb diagram made the pattern immediately visible.

viscerally (adv.) in a way that is felt immediately and emotionally, not merely understood intellectually. The diagram communicated the problem viscerally.

bureaucracy (n.) an administrative system, especially of government, often associated with procedural complexity. She needed to convince a skeptical government bureaucracy.

dwarf (v., figurative) to make something seem much smaller by comparison. The disease-death wedges dwarfed the combat-death wedges.

grasp (v., figurative) to understand or comprehend, especially something complex. A table made the pattern difficult to grasp quickly.

credited with (phrase) formally acknowledged as responsible for a result. The diagrams are credited with contributing to sanitary reforms.

absorb (v., figurative) to take in and fully understand information. Few people outside a team ever absorb a dense spreadsheet.

PRACTICE

Where the Story Breaks Down

Nightingale's underlying data was genuinely clear and dramatic once visualized; not every important finding has an equally stark, visually compelling pattern waiting to be revealed by better presentation alone.

Use the Language

Describe an important finding in your own work that might need a different presentation format to actually be absorbed, using grasp, dwarf and absorb.

Discuss

What is a specific example, from your own experience, of accurate data that failed to produce action because of how it was presented?

Write

In 100 words, describe how you would redesign the presentation of one specific finding from your own work to make its scale immediately, viscerally obvious to a non-specialist audience.

SOURCES

A Nightingale, F., Notes on Matters Affecting the Health, Efficiency, and Hospital Administration of the British Army, Harrison and Sons (1858), containing the original coxcomb diagrams.

C Royal Statistical Society, official historical record of Florence Nightingale's 1858 election.

B Small, H., "Florence Nightingale's Statistical Diagrams," presented at the Research Conference of the Florence Nightingale Museum (1998).

THIS CONNECTS TO

Story 4: The Map That Challenged a Theory (John Snow) LIMITATION
Snow's map won the argument in part through its visual clarity. Nightingale's chapter takes that one step further: sometimes the data existed all along, and only the presentation was ever the real barrier.

Story 29: Think Like the User (Temple Grandin) SAME MECHANISM
Both changed outcomes by shifting how evidence was perceived — seeing a space from the animal's actual vantage point, or seeing mortality data in a genuinely visual format — rather than by changing the underlying facts.

Story 33: The Checklist (Gawande) LIMITATION
Nightingale's diagram made an existing problem impossible to ignore. Gawande's checklist goes a step further, building a structural safeguard so the problem doesn't depend on anyone being persuaded at all.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Florence Nightingale had detailed mortality data showing that far more soldiers in the Crimean War were dying from preventable disease than from combat wounds, but the British government and military establishment were not acting on the pattern.

T — Task

She needed to make the scale of the problem immediately, viscerally clear to a skeptical government bureaucracy unlikely to absorb a dense spreadsheet of numbers.

A — Action

She designed original data visualizations, including her now-famous coxcomb diagrams, presenting the same underlying data in a format that made the pattern impossible to miss.

R — Result

The diagrams are credited with contributing to sanitary reforms in military hospitals and helped establish Nightingale's influence on public health statistics.

Level 2 — Make It Your Story

> *Tell me about a time when you used information or data to persuade someone, and the format of the presentation mattered as much as the content.*

Use STAR to answer. Think about: a chart, a report, a presentation redesigned to make a pattern impossible to miss.

TAKE THE IDEA FURTHER

> *What accurate, important finding in your own organization currently sits in a table or spreadsheet that almost nobody outside your immediate team ever actually absorbs?*

36

Verify the Calculation

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Katherine Johnson joined NASA's predecessor agency, the National Advisory Committee for Aeronautics, in 1953, working initially as one of a group of Black women mathematicians, known internally as "computers," performing complex trajectory and orbital mechanics calculations by hand — work that was, at the time, both racially and professionally segregated from much of the rest of the agency's engineering staff, in facilities and social spaces kept formally separate under Virginia's segregation laws of the era.

Johnson's specific and defining professional strength was rigorous, independent mathematical verification — checking, by an entirely separate method of calculation, results that had already been produced by other means, specifically including the electronic computers NASA was beginning to rely on more heavily by the early 1960s. This was not make-work or redundant bureaucratic process. Early electronic computers were genuinely new, unproven technology in an application — human spaceflight — where a calculation error could be fatal, and there was no institutional track record yet establishing that the machines could be trusted without a rigorous, independently verified check.

The moment most directly associated with Johnson's career, and the one dramatized in the 2016 film Hidden Figures, involves astronaut John Glenn's 1962 orbital mission, the first time an American astronaut would orbit the Earth. According to widely repeated accounts, Glenn specifically requested that Johnson personally verify, by her own independent hand calculation, the

trajectory figures that had been generated by the electronic computers before he would agree to fly the mission — reportedly telling engineers, in essence, that if "the girl" checked the numbers and confirmed them correct, he would trust the flight. Johnson performed the verification, confirmed the computer's calculations were accurate, and Glenn's mission proceeded successfully.

It is worth being precise about what made Johnson's verification valuable, since the popular version of the story sometimes reduces it to a simple morale-boosting anecdote about personal trust in an individual: the actual professional function she was serving was a genuine, structurally important check on a new and not-yet-fully-trusted technology, performed by someone whose independent mathematical method — working through the same orbital mechanics by a different calculational path — provided a real safeguard against the specific kind of undetected systematic error an electronic computer, malfunctioning or incorrectly programmed, could produce without any obvious external sign that something had gone wrong. Johnson's broader career included calculations for the 1969 Apollo 11 mission and decades of further orbital mechanics work, and she was awarded the Presidential Medal of Freedom in 2015.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What decision or calculation in your organization currently relies entirely on a single system or method, with no independent verification by a genuinely different approach?*

KEY TAKEAWAYS

1. Johnson's verification work addressed a real, structural problem — new, unproven computing technology in a context where an undetected error could be fatal — not merely a symbolic or ceremonial check.
2. The value of an independent verification depends on it using a genuinely different method, not simply repeating the same calculation a second time by the same means.
3. The specific 1962 Glenn mission anecdote, while dramatized in later retellings, reflects a real and structurally important professional practice Johnson performed throughout much of her career, not an isolated one-time event.

SUMMARY

Katherine Johnson, working as a mathematician at NASA's predecessor agency, provided independent verification of orbital trajectory calculations at a time when electronic computers were new, unproven technology in a context — human spaceflight — where an undetected error could be fatal. Astronaut John Glenn reportedly requested her personal verification of the computer-generated trajectory figures before his 1962 orbital mission, and her confirmation preceded the mission's successful flight. Her broader career included work on the 1969 Apollo 11 mission, and she received the Presidential Medal of Freedom in 2015.

STORY CHECK

ESTABLISHED

Johnson's career at NASA and its predecessor agency, her role in trajectory verification, and her contributions to the Apollo 11 mission are extensively

documented in NASA's own historical records and her 2015 Presidential Medal of Freedom citation.

POPULAR VERSION ADDS

The specific John Glenn quotation and the precise framing of the request are drawn primarily from later retellings, including the popularized 2016 film adaptation, and the exact verbatim wording of Glenn's request is not independently documented in a contemporaneous written record.

UNCERTAIN

The precise timeline and internal NASA documentation of exactly which calculations Johnson verified, and in what sequence, for the specific Glenn mission are less granularly documented than the broader, well-established fact of her involvement.

VOCABULARY & PHRASAL VERBS

trajectory (n.) the path followed by an object moving through space, especially under the influence of given forces. She verified the trajectory calculations for the orbital mission.

orbital mechanics (n.) the branch of physics governing the motion of objects in orbit. Her expertise was in orbital mechanics.

segregated (adj.) separated, especially by race, under discriminatory laws or policy. The facilities were formally segregated under Virginia law.

redundant (adj.) unnecessarily repeated or superfluous, though in engineering contexts can also mean a valuable duplicate safeguard. The verification was not redundant bureaucratic process, but a genuine safeguard.

track record (n.) a record of past performance used to establish reliability or trust. Electronic computers had no established track record yet.

undetected (adj.) not noticed or discovered, especially referring to an error or flaw. An undetected systematic error could have been catastrophic.

check (something) against (phrasal verb) to compare one result with another to confirm accuracy. She checked the computer's output against her own hand calculations.

rely on (phrasal verb) to depend on something or someone for support or accuracy. NASA was beginning to rely more heavily on electronic computers.

proceed (v.) to continue or go ahead with a planned action. The mission proceeded once the calculations were confirmed.

PRACTICE

Where the Story Breaks Down

Johnson's verification worked because she had a genuinely independent, equally rigorous calculational method available to her — hand-calculated orbital mechanics; many modern systems lack any equally rigorous independent method to check against, which is a harder problem than simply assigning someone to "double check."

Use the Language

Describe a decision or calculation in your own work that relies on a single unverified method, using track record, undetected and check against.

Discuss

What would a genuinely independent verification method — not just a second look by the same process — look like for a system you currently trust in your own work?

Write

In 100 words, describe a new tool or system in your own organization that doesn't yet have an established track record, and propose what an independent verification method for it might look like.

SOURCES

C NASA, official biography and historical record of Katherine Johnson (nasa.gov).

C The White House, Presidential Medal of Freedom citation for Katherine Johnson (2015).

D The specific John Glenn quotation is drawn from popularized retellings, including Hidden Figures (2016), rather than a verified contemporaneous transcript.

THIS CONNECTS TO

Story 9: The Planes That Didn't Return (Wald) SAME MECHANISM

Both trusted a rigorous, independent method over an existing account others were already relying on without question.

Story 3: The Bicycle Makers Who Wanted to Fly (Wright Brothers) SAME MECHANISM

Both built or used an independent alternative method — Johnson's hand calculation, the Wrights' wind tunnel — to check a result the rest of the field was accepting without direct verification.

Story 14: The Moth in the Machine (Grace Hopper) LIMITATION

Johnson's verification made an unproven technology trustworthy. Hopper's chapter shows the limit of that kind of protection: verification secures the math, but it doesn't stop a good story from later attaching itself to the wrong person's name.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

NASA was beginning to rely on new, unproven electronic computers for orbital trajectory calculations, in a context — human spaceflight — where an error could be fatal.

T — Task

Before John Glenn's 1962 orbital mission, the computer-generated trajectory figures needed independent verification.

A — Action

Katherine Johnson checked the calculations by hand, using a genuinely different method from the computer's.

R — Result

She confirmed the figures were accurate, the mission proceeded successfully, and independent verification became an established part of how NASA validated new technology.

Level 2 — Make It Your Story

> *Tell me about a time when you checked an important assumption or calculation.*

Use STAR to answer. Think about: a report you double-checked, a new tool or system you didn't fully trust yet, a number you verified before presenting it.

TAKE THE IDEA FURTHER

> *What in your work currently relies on a single method or system, with no independent way to check it?*

37

The Network That Made Overnight Delivery Possible

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Fred Smith founded Federal Express in 1971 with a specific structural insight about a problem the existing air freight industry had never actually tried to solve directly: shipping companies of the era generally attempted to fly packages along direct point-to-point routes between the specific cities where they originated and where they were destined, an arrangement that seemed intuitively efficient — a direct route is, after all, the shortest physical distance — but that became genuinely inefficient once a company needed to serve a large number of different city pairs simultaneously, since a direct-route network requires an enormous number of separate flight paths to connect every origin to every possible destination.

Smith's proposed alternative, reportedly first outlined in a Yale economics term paper as an undergraduate (a paper that, according to a persistent though disputed anecdote, received a mediocre grade from a professor unconvinced of its practicality), was structurally different: rather than direct point-to-point routes, every package, regardless of its actual origin or destination, would first be flown to a single central hub, sorted there according to its final destination, and then flown back out to its destination city on a second flight. The approach, now generally called a hub-and-spoke model, appeared, at first inspection, obviously inefficient for any individual package, which would travel a physically longer total distance — first toward a

central hub, often in a direction away from its ultimate destination, then back out again — than a hypothetical direct flight would have required.

The structural advantage became visible only when considering the full network rather than any single shipment in isolation. A hub-and-spoke system connecting a given number of cities requires dramatically fewer total flight routes than a fully connected direct point-to-point network serving the same cities, because every city needs only one route to the central hub rather than a separate direct route to every other city it might need to reach. Fewer total routes meant each individual route could be flown more frequently and with higher, more predictable aircraft utilization, which in turn made a genuinely reliable overnight delivery guarantee — a package picked up anywhere in the network would reliably reach any other point in the network by the next morning — actually achievable at a viable operating cost, something a direct point-to-point network serving the same number of cities could not economically sustain.

FedEx's first operational night in 1973, launched from a hub in Memphis, Tennessee, moved only a small number of packages — 186 packages to 25 cities — a genuinely inauspicious start for a company that would go on to process millions of packages nightly and become the standard model the broader package delivery, and later broader logistics and telecommunications routing, industries would eventually adopt as the structurally correct solution to serving a large, complex network efficiently.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What process in your organization is currently optimized for the efficiency of each individual transaction, when a structurally*

> *different network design might be less efficient for any single case but dramatically more efficient in aggregate?*

KEY TAKEAWAYS

1. The hub-and-spoke model made any single package's journey longer, not shorter — the efficiency gain only appeared when considering the full network of routes required to serve many cities simultaneously.
2. Fewer total routes, achieved through centralizing at a hub, allowed higher and more predictable utilization of each route, which is what actually made the overnight guarantee economically viable.
3. The company's first operational night moved a genuinely small number of packages, a reminder that a structurally correct model can still require years to reach the scale where its advantages become fully realized.

SUMMARY

Fred Smith founded FedEx around a hub-and-spoke network design, in which packages travel first to a central hub for sorting before flying to their final destination, rather than along direct point-to-point routes. Though this makes any individual package's journey physically longer, it dramatically reduces the total number of routes required to serve a large network of cities, enabling higher route utilization and making a reliable overnight delivery guarantee economically viable. The company's first operational night in 1973 moved only a small handful of packages before growing into an industry-defining logistics network.

STORY CHECK

ESTABLISHED

FedEx's 1973 launch from its Memphis hub and its hub-and-spoke network model are extensively documented in the company's own corporate history and independent business and logistics literature.

POPULAR VERSION ADDS

The story that Smith's original Yale term paper proposing the concept received a poor or mediocre grade is a widely repeated anecdote that Smith himself has referenced in interviews, though it is not independently verified by a surviving copy of the graded paper.

UNCERTAIN

The precise number of packages moved on the first operational night varies between six and eighteen depending on the specific historical source consulted, and no single authoritative count has been established.

VOCABULARY & PHRASAL VERBS

point-to-point (adj.) describing direct routes between two specific locations, without an intermediate stop. The old model used point-to-point routes between city pairs.

hub-and-spoke (n./adj.) a network design in which routes connect through a central hub rather than directly. FedEx pioneered the hub-and-spoke model in air freight.

utilization (n.) the degree to which a resource or capacity is actually used. Fewer total routes allowed higher utilization of each one.

in aggregate (phrase) considered as a whole, rather than as individual parts. The efficiency gain only appears in aggregate, across the full network.

inauspicious (adj.) not suggesting a favourable future outcome; an unpromising start. The first operational night was a genuinely inauspicious start.

viable (adj.) capable of working successfully; practically achievable. A reliable overnight guarantee became economically viable.

at first inspection (phrase) based on an initial, surface-level look, before deeper analysis. The model appeared inefficient at first inspection.

scale up (phrasal verb) to grow a process or system to a larger size while maintaining its function. The network scaled up from a handful of packages to millions.

sustain (v.) to maintain or keep something going over time, especially under strain. A direct-route network couldn't sustain the same guarantee economically.

PRACTICE

Where the Story Breaks Down

The hub-and-spoke advantage depends on serving a genuinely large number of origin-destination pairs simultaneously; for an operation serving only a small, fixed number of routes, a direct point-to-point model may remain more efficient, and centralizing through a hub would add unnecessary complexity.

Use the Language

Describe a process in your organization currently optimized transaction-by-transaction rather than as a network, using utilization, in aggregate and viable.

Discuss

What would a hub-and-spoke redesign look like for a process in your own work that currently connects many points directly to each other?

Write

In 100 words, describe a case where making one individual transaction slightly less efficient might make the overall system significantly more efficient.

SOURCES

C FedEx Corporation, official corporate history (fedex.com/en-us/about/history.html).

B Frock, R., Changing How the World Does Business: FedEx's Incredible Journey to Success, Berrett-Koehler (2006).

D The Yale term-paper anecdote and its grade are drawn from Smith's own retellings in interviews rather than an independently verified academic record.

THIS CONNECTS TO

Story 17: The Factory That Started Moving (Ford) LIMITATION
Ford's gain came from redesigning the physical flow of work within a single factory. Smith's came from redesigning the physical flow of an entire national shipping network — the same instinct applied at a very different scale.

Story 32: Connect What Already Exists (Berners-Lee) LIMITATION
Berners-Lee's innovation was combining pieces that already existed. Smith's was structural — redesigning how the pieces moved relative to each other — a different kind of systems insight.

Story 34: Remove the Friction (Amazon) CONTRAST
See The Contrast in that chapter. In short: Amazon removed a small obstacle at one critical point; FedEx restructured the entire path a package takes, making individual journeys longer so the whole system could be faster.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

The existing air freight industry generally shipped packages by whatever direct route was available between origin and destination, which was inefficient at scale and made reliable overnight delivery difficult to guarantee.

T — Task

Fred Smith wanted to build a shipping network that could reliably guarantee overnight delivery nationally.

A — Action

He designed a hub-and-spoke system, routing all packages through a central hub for sorting rather than shipping them directly, even though this made any individual package's journey longer.

R — Result

The counterintuitive design made the overall system faster and more reliable at scale, and became the foundational model for the modern express shipping industry.

Level 2 — Make It Your Story

> *Tell me about a time when a structurally different, less direct-looking approach turned out to be faster than the obvious direct one.*

Use STAR to answer. Think about: a routing decision, a workflow redesign, a process that looked slower on paper but was faster in practice.

TAKE THE IDEA FURTHER

> *What process in your organization is currently optimized for the efficiency of each individual transaction, when a structurally different network design might make the whole system faster?*

PART V

Build the System

38

The Farmer Who Did Less

DOCUMENTATION LEVEL 2 — SELF-REPORTED OR TRANSFERRED

The account rests mainly on the protagonist's own retelling, or the idea was documented but its original application was never adopted.

Masanobu Fukuoka trained as a microbiologist and worked as an agricultural scientist in Japan before a personal health crisis in his mid-twenties prompted him to reconsider the direction of his career and life entirely. He left his research position and eventually returned to his family's farm on the island of Shikoku, where he began, over subsequent decades, an unusual and initially controversial personal experiment: systematically removing, one at a time, agricultural interventions that conventional farming of the era treated as simply necessary — plowing the soil, applying synthetic fertilizer, using chemical pesticides, and even, eventually, flooding rice paddies in the standard way — to see which of these supposedly essential practices his land actually required, and which had simply become unquestioned convention.

Fukuoka's method, which he called "natural farming," was not agricultural neglect or a return to some romanticized pre-modern simplicity, though it was sometimes superficially mistaken for either. It required close, sustained, expert observation and highly specific, deliberate technique — precisely timed seeding, careful management of ground cover crops to naturally suppress weeds without tillage or herbicides, and the specific and now well-known practice of forming seeds into small clay pellets, protecting them from birds and insects until conditions were right for germination, then scattering them directly onto unplowed fields rather than planting them in conventionally prepared rows. Over years of refinement, Fukuoka reported achieving yields for rice and winter grain crops comparable to, and in some

years exceeding, conventional intensively managed farms in his region, achieved with dramatically less labour, no purchased chemical inputs, and soil that measurably improved in fertility and structure over time rather than gradually depleting, as intensively tilled and chemically fertilized soil often does.

The core insight Fukuoka drew from decades of this work, articulated most fully in his influential 1975 book The One-Straw Revolution, was that many agricultural interventions farmers had come to treat as simply necessary were, in fact, compensating for problems that earlier interventions had themselves created — plowing disrupted soil structure and microbial ecosystems in ways that then seemed to require fertilizer to compensate for lost fertility, and monoculture planting removed natural pest predators in ways that then seemed to require pesticides to compensate for lost natural pest control. Each new intervention was frequently addressing a problem the previous intervention had caused, rather than a problem that existed independently of human intervention in the first place. Fukuoka's approach and writing influenced the development of the broader permaculture movement internationally, and his central provocation — that removing an intervention can sometimes be a more sophisticated and effective act than adding one — has been applied by later writers and practitioners well beyond agriculture specifically.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What process, rule or intervention in your own organization exists primarily to compensate for a problem that an earlier, different process or rule originally created?*

KEY TAKEAWAYS

1. Fukuoka's method required close observation and precise technique — it was a sophisticated alternative practice, not an absence of practice or a simple return to doing nothing.
2. Several conventional interventions he removed were, on close examination, compensating for problems that other interventions had themselves created, rather than addressing an independent underlying need.
3. His approach achieved comparable or superior yields with dramatically reduced labour and no purchased chemical inputs, a result that took years of careful refinement, not an immediate transition.

SUMMARY

Masanobu Fukuoka, a trained microbiologist turned farmer, spent decades systematically removing conventional agricultural interventions — plowing, chemical fertilizer, pesticides — to determine which were genuinely necessary, developing a precise, low-intervention "natural farming" method involving techniques such as clay-pelleted seed broadcasting. He found that many interventions were compensating for problems that earlier interventions had caused, and achieved yields comparable to conventional farming with far less labour and no chemical inputs. His 1975 book The One-Straw Revolution influenced the international permaculture movement.

STORY CHECK

ESTABLISHED

Fukuoka's farming methods, his yield claims for his own farm, and the influence of The One-Straw Revolution on the permaculture movement are

documented in his own published writing and corroborated by agricultural researchers who visited and studied his farm.

POPULAR VERSION ADDS

"Natural farming" or "do-nothing farming" (a term sometimes applied to Fukuoka's approach) is occasionally misunderstood as advocating agricultural neglect; Fukuoka's own writing repeatedly emphasizes that his method required close observation and precise, deliberate technique, not an absence of active management.

UNCERTAIN

Independent, rigorously controlled agricultural studies directly comparing Fukuoka's specific yields against conventional regional farms under matched conditions are limited; most of the yield evidence comes from Fukuoka's own records and qualitative accounts by visiting researchers.

VOCABULARY & PHRASAL VERBS

tillage (n.) the preparation of soil for growing crops, typically by plowing or turning it. His method avoided conventional tillage entirely.

monoculture (n.) the agricultural practice of growing a single crop over a wide area. Monoculture planting removed natural pest predators.

deplete (v.) to gradually reduce the quantity or fertility of a resource. Chemically fertilized soil often gradually depletes over time.

germination (n.) the process by which a seed begins to grow. The clay pellets protected seeds until conditions favoured germination.

compensate for (phrasal verb) to counteract or make up for a deficiency or problem. Fertilizer was compensating for fertility that plowing had disrupted.

broadcast (v., agriculture) to scatter seed across a field rather than planting it in rows. He would broadcast the clay-pelleted seeds directly onto unplowed fields.

come to treat as (phrase) to gradually begin regarding something as true or necessary, often without questioning it. Farmers had come to treat plowing as simply necessary.

strip away (phrasal verb) to remove something layer by layer, especially something previously assumed necessary. He stripped away interventions one at a time to test their necessity.

hold up (phrasal verb) to remain valid or effective under scrutiny or testing. His yields held up against conventional farms over years of comparison.

PRACTICE

Where the Story Breaks Down

Fukuoka had decades of patient, sustained observation on a single piece of land to develop and refine his specific technique; an equivalent process of systematic removal in a fast-moving organizational context may not have the same luxury of time to properly test each removed intervention before consequences become clear.

Use the Language

Describe a rule or process in your organization that might be compensating for a problem an earlier process created, using compensate for, strip away and hold up.

Discuss

What is the difference between removing an intervention thoughtfully, after careful observation, and simply neglecting a process? How would you tell them apart in your own work?

Write

In 100 words, identify one rule, process or "necessary" step in your own organization, and consider what problem it might actually be a reaction to rather than a genuine independent requirement.

SOURCES

B Fukuoka, M., The One-Straw Revolution: An Introduction to Natural Farming, Rodale Press (1978; English translation).

C Permaculture research and advocacy organizations' documentation of Fukuoka's influence on the movement's founding principles.

D Independent, controlled agricultural yield comparisons specific to Fukuoka's farm are limited; most available evidence is qualitative or drawn from his own records.

THE CONTRAST

Story 38: Fukuoka vs. Story 44: Build Learning Into the System (Kaizen)

Story 38: Fukuoka: Removed interventions that were only compensating for problems earlier interventions had created in the first place.

Story 44: Build Learning Into the System (Kaizen): Adds a continuous, structured practice of small improvements on top of an existing system.

Is the better path to improvement removing accumulated, unnecessary intervention, or adding continuous, deliberate small improvements?

THIS CONNECTS TO

Story 2: The Burrs on the Dog (Velcro) LIMITATION

De Mestral's insight came from close investigation adding understanding. Fukuoka's came from decades of close observation leading him to subtract intervention rather than add it — both are patient, but pointed in opposite directions.

Story 45: The Goal That Had to Change (Shackleton) LIMITATION

Fukuoka's method was developed over years of patient, controlled observation. Shackleton's chapter is the opposite situation — a crisis with no time for that kind of careful, incremental study.

Story 52: Design for the World Coming (Buckminster Fuller)

SAME MECHANISM

Both pursued an unconventional, long-horizon design philosophy that looked impractical by the standards of their era, and required sustained conviction well before mainstream validation.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Masanobu Fukuoka, trained as a microbiologist, observed that conventional farming relied on an accumulating set of interventions — plowing, fertilizer, pesticides — many of which existed only to compensate for problems earlier interventions had caused.

T — Task

He wanted to develop a farming method that achieved comparable yields with dramatically less active intervention.

A — Action

Over decades of close observation on his own farm, he developed a natural farming method minimizing plowing, weeding, and chemical inputs, letting natural processes do work conventional farming did artificially.

R — Result

His method achieved yields comparable to conventional farming with far less labour and input, and his writing influenced sustainable and permaculture farming movements internationally.

Level 2 — Make It Your Story

> *Tell me about a time when removing an accumulated process or rule, rather than adding a new one, actually solved a problem.*

Use STAR to answer. Think about: a process simplified, a rule removed, a system that improved once something was taken away rather than added.

TAKE THE IDEA FURTHER

> *What process, rule or intervention in your own organization exists primarily to compensate for a problem that an earlier, different process created in the first place?*

39

Furniture Designed Around the Constraint

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1956, an employee at IKEA, the Swedish furniture company founded by Ingvar Kamprad, was attempting to fit a table into a car for delivery to a customer and, according to the company's own widely repeated internal account, found that the assembled table would not fit through the car's door. The practical solution — removing the table's legs so the pieces could be transported flat and reassembled at the destination — is credited internally as the specific moment that revealed a much larger structural opportunity than simply solving one delivery logistics problem. Once furniture was designed from the outset to be shipped disassembled, in flat, stackable packaging, rather than designed fully assembled and only disassembled as an afterthought when a delivery problem demanded it, a cascade of cost advantages became available that had nothing directly to do with the original delivery inconvenience that had prompted the discovery.

Flat-packed furniture could be shipped and warehoused far more densely than assembled furniture, since flat packages stack efficiently while assembled furniture, full of interior empty space, does not; this dramatically reduced the shipping and storage cost per unit across the company's entire supply chain. Flat packaging also meant customers, rather than the company, could transport purchased furniture themselves in an ordinary car, eliminating a significant delivery cost IKEA would otherwise have needed to absorb or pass on to customers as a separate charge.

IKEA restructured its entire product design process around this insight rather than treating flat-pack as an occasional convenient option for select items. Furniture was now designed, from the earliest stages of the design process, with disassembly, flat packaging density and self-assembly by an ordinary customer using simple tools as explicit, load-bearing design constraints — not an afterthought applied to an already-finished design, but a requirement shaping the design itself from the start, influencing decisions about joint types, material choices, and even overall furniture proportions to ensure pieces would pack efficiently and assemble reliably without professional installation. The resulting cost structure allowed IKEA to price furniture significantly below competitors offering functionally comparable assembled products, a competitive advantage that became central to the company's global expansion and its now-widely-recognized brand identity, extending even to the company's now-famous, occasionally frustrating, wordless pictorial assembly instructions — themselves a direct design consequence of needing instructions comprehensible to customers across dozens of countries and languages without translation. What began as an improvised fix for a single delivery problem became, once generalized and applied systematically, the structural foundation of the company's entire business model and cost advantage.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What logistical constraint or delivery problem in your own operation might, if treated as a design requirement from the start rather than an afterthought, reveal cost advantages well beyond the original problem it solved?*

KEY TAKEAWAYS

1. The original insight solved one specific, narrow delivery problem; its real value only emerged once the underlying principle was generalized and applied systematically across the entire product line and company structure.
2. Designing for disassembly and flat packaging from the earliest stages of product design, rather than as an afterthought, unlocked cost advantages in shipping, warehousing and delivery that had nothing directly to do with the original problem.
3. Treating a logistical constraint as a genuine, load-bearing design requirement — shaping joints, materials and proportions — rather than as an inconvenience to work around after the fact, produced a durable structural cost advantage.

SUMMARY

An IKEA employee's 1956 discovery that removing a table's legs solved a specific delivery-vehicle problem became, once generalized, the foundation of the company's entire flat-pack furniture design philosophy. Designing furniture from the outset for disassembly, dense flat packaging and customer self-assembly dramatically reduced shipping and warehousing costs across IKEA's supply chain, allowing prices significantly below assembled-furniture competitors. The insight originated as a narrow fix for one delivery problem but became the company's central structural cost advantage once treated as a systematic design requirement.

STORY CHECK

ESTABLISHED

IKEA's flat-pack design philosophy and its role in the company's cost structure and global expansion are extensively documented in the company's own corporate history and independent business case studies.

POPULAR VERSION ADDS

The specific 1956 table-and-car-door anecdote is IKEA's own official company history account; like many such founding anecdotes, its precise details (the specific employee involved, the exact circumstances) are difficult to independently verify beyond the company's own retelling.

UNCERTAIN

The exact timeline of how quickly the insight was generalized from one product to the company's entire design process, versus a more gradual evolution over subsequent years, is not documented in precise detail.

VOCABULARY & PHRASAL VERBS

flat-packed (adj.) describing furniture or products shipped disassembled in flat packaging. Flat-packed furniture stacks far more densely than assembled pieces.

cascade (n., figurative) a series of connected consequences following from an initial event. A cascade of cost advantages followed from the original insight.

load-bearing (adj., figurative) structurally essential, not merely decorative or optional. Flat-pack constraints became a load-bearing part of the design process.

afterthought (n.) something considered or added only after the main process is complete. Disassembly was no longer an afterthought but a starting requirement.

absorb (v., figurative) to bear a cost internally rather than passing it on to someone else. IKEA no longer had to absorb the cost of delivery.

pass on (phrasal verb) to transfer a cost or responsibility to another party. The delivery cost no longer needed to be passed on to customers.

structural advantage (n.) a competitive benefit built into the fundamental design of a system, not easily copied superficially. Flat-pack design became a genuine structural advantage.

generalize (v.) to extend a specific insight or principle to a broader range of applications. The insight was generalized across the entire product line.

comprehensible (adj.) able to be understood. Wordless instructions kept the design comprehensible across languages.

PRACTICE

Where the Story Breaks Down

IKEA's flat-pack advantage depends on customers being willing and able to perform assembly themselves; not every product category has customers with the same tolerance for self-assembly, so the same structural solution wouldn't transfer to every industry equally well.

Use the Language

Describe a logistical constraint in your own operation that might become a genuine design advantage if treated systematically, using cascade, afterthought and structural advantage.

Discuss

What narrow, specific problem has your organization solved that might contain a much larger, generalizable insight nobody has yet applied systematically?

Write

In 100 words, describe one delivery, distribution or logistical constraint in your own work, and consider what would change if it were treated as a starting design requirement rather than an afterthought.

SOURCES

C IKEA, official corporate history of flat-pack design (ikea.com/global/en/our-business/).

B Torekull, B., Leading by Design: The IKEA Story, HarperBusiness (1999) — the authorized company history, including the table-and-car-door account.

D The precise timeline of the insight's generalization across IKEA's full product line is not documented in granular detail beyond the company's own retrospective account.

THE CONTRAST

Story 39: IKEA vs. Story 18: The Square Peg and the Round Hole (Apollo 13)

Story 39: IKEA: Started from a hard physical constraint — what fits in a car — and let that constraint drive an entire design philosophy going forward.

Story 18: The Square Peg and the Round Hole (Apollo 13): Started from a hard physical constraint — what's on board — and solved one acute, specific problem within it.

Does a hard constraint work best as a one-time problem to solve, or as an ongoing design principle to build a whole strategy around?

THIS CONNECTS TO

Story 18: The Square Peg and the Round Hole (Apollo 13)

SAME MECHANISM

Both treat a hard physical constraint as the design requirement itself, rather than an obstacle to work around.

Story 15: 5,127 Prototypes (Dyson) SAME MECHANISM

Both treat a large number of small, specific attempts — prototypes, delivery problems — as informative data points that narrow the design space, rather than as simple repeated failure.

Story 1: The Rocket That Was Too Expensive (Musk) LIMITATION

Musk's reframe started by breaking a price down into its raw materials. IKEA's started from a physical constraint — what fits in a car — and let that constraint drive the whole design, a different route to the same kind of structural cost advantage.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

An IKEA employee, trying to fit a table into a car for delivery, struggled with the assembled piece's bulky shape.

T — Task

The company needed a way to make furniture easier and cheaper to transport and deliver.

A — Action

According to the company's widely repeated internal account, the solution was to remove the legs and pack the table flat.

R — Result

Flat-pack design, built around the transportation constraint, became a defining feature of IKEA's entire product line and business model.

Level 2 — Make It Your Story

> *Tell me about a time when a hard physical or logistical constraint ended up driving a better design than an unconstrained approach would have.*

Use STAR to answer. Think about: a shipping constraint, a budget limit, a space restriction that shaped a better solution.

TAKE THE IDEA FURTHER

> *What logistical constraint or delivery problem in your own operation might, if treated as a design requirement from the start rather than an afterthought, actually improve the product?*

40

The Arm That Couldn't Grab

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

When NASA invited Canada to contribute a component to the Space Shuttle program in the early 1970s, the National Research Council of Canada and the Toronto-area firm Spar Aerospace took on what looked like a straightforward extension of industrial robotics: a remotely operated arm that could lift a satellite out of the orbiter's cargo bay and place it in orbit, or retrieve one. Agreements between the NRC and NASA were signed in 1975 and 1976. The obvious way to frame the problem was the way anyone frames a lifting problem — how strong must the arm be, and how firmly must its hand grip?

Both halves of that framing were wrong. Strength was almost irrelevant: in orbit a payload weighs nothing, though it keeps every kilogram of its mass and all of its inertia. The finished arm weighed 410 kilograms, could manoeuvre payloads of up to 266,000 kilograms in weightlessness, and could not support its own weight in Earth's gravity — so its full range of motion could never be tested physically before launch, and engineers built a simulation facility, SIMFAC, to verify its behaviour and train the astronauts who would operate it. Gripping was the deeper problem. A claw closing on a free-floating object pushes against it, and the object, with nothing to brace against, drifts away or begins to tumble. An arm designed to grab things in space is an arm that shoves them away from itself.

The reframing is credited to a Spar engineer named Frank Mee, and its essential move was to stop treating the arm as the only part of the system that

could be designed. Instead of a cleverer hand able to seize arbitrary objects, the team built a mechanism that captures one deliberately standardized shape: a grapple fixture — a pin topped with a sphere, on a plate with guide ramps — bolted onto the payload itself. Inside the arm's cylindrical end effector, three wires form an open triangle that shrinks toward a point as it rotates, snaring the pin and drawing the payload into alignment rather than clamping down on it. The gripping problem had not been solved. It had been dissolved.

The first arm was accepted by NASA in early 1981, and the NRC's president, Larkin Kerwin, gave it the name that stuck: Canadarm. On 13 November 1981, on the second Shuttle mission, pilot Richard Truly deployed it from Columbia's cargo bay and reported that it moved more freely than it had in training simulations. NASA bought four more. Over three decades the arm and its successors flew ninety missions, deploying satellites, capturing Hubble for servicing, carrying space-walking astronauts and inspecting the orbiter's own heat shield — all of it resting on the decision to standardize the thing being grabbed rather than perfect the thing doing the grabbing.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What problem in your organization are you trying to solve by improving one component, when the real solution may require redesigning the relationship between the components?*

KEY TAKEAWAYS

1. The apparent problem — build a hand strong and dextrous enough to grab objects in space — was the wrong problem; in weightlessness, gripping force pushes the target away rather than securing it.
2. The solution required redesigning both sides of the interface: a standardized grapple fixture mounted on every payload, and a three-wire snare in the arm that captures that one shape with almost no applied force.
3. Because the arm could not support its own weight on Earth, its full range of motion could never be tested physically before flight, and confidence in the design had to be built through simulation instead.

SUMMARY

Asked to build a robotic arm for the Space Shuttle, Canada's National Research Council and Spar Aerospace found the obvious framing — a stronger arm with a better grip — failed in orbit, where a closing claw simply pushes a free-floating payload away. The team instead designed a standardized grapple fixture to be mounted on the payloads themselves, captured by three rotating wires inside the arm's end effector. The 410-kilogram arm, which could not support its own weight on the ground but could manoeuvre 266,000 kilograms in weightlessness, first flew on STS-2 on 13 November 1981 and served across ninety missions before retiring in 2011.

STORY CHECK

ESTABLISHED

The NRC–NASA agreements of 1975 and 1976, Spar Aerospace's role, the snare end effector and standardized grapple fixture, the mass and payload figures, the arm's inability to support its own weight in Earth gravity, and its

first deployment on STS-2 on 13 November 1981 are documented in Canadian Space Agency, Parks Canada and Smithsonian records.

POPULAR VERSION ADDS

The Canadarm is usually retold as a story about national pride and mechanical capability — a large Canadian arm on an American spacecraft — with the actual design insight, that the payloads had to be redesigned as much as the arm did, rarely appearing in the popular version at all.

UNCERTAIN

The frequently repeated account of Frank Mee first demonstrating the snare concept with a model built at home from cardboard and string rests largely on later recollection rather than contemporaneous documentation, as does the exact sequence in which competing end-effector concepts were evaluated and rejected.

VOCABULARY & PHRASAL VERBS

payload (n.) the cargo a vehicle carries, as distinct from the vehicle itself. The arm lifted payloads out of the orbiter's cargo bay.

end effector (n., engineering) the working device at the end of a robotic arm — the equivalent of a hand. Three wires inside the end effector closed around the pin.

grapple (v./n.) to seize and hold something firmly; in spaceflight, the fixture designed to be seized. Every payload carried a standard grapple fixture.

degrees of freedom (n., engineering) the number of independent ways a mechanism can move. The arm's six degrees of freedom allowed precise positioning.

interface (n.) the shared boundary at which two separate systems meet and interact. They redesigned both sides of the interface at once.

remotely operated (adj.) controlled by an operator at a distance rather than on site. It was a remotely operated arm, worked from inside the cabin.

manoeuvre (v.) to move something carefully into a required position. In weightlessness it could manoeuvre 266,000 kilograms.

latch onto (phrasal verb) to attach firmly to something and hold it. The snare latched onto the pin without pushing it away.

design around (phrase) to shape a design so that it accommodates something treated as fixed. Rather than design around the problem, they changed both components.

PRACTICE

Where the Story Breaks Down

Redesigning both sides of an interface is only available when you control both sides. Spar could specify the grapple fixture because NASA would require every Shuttle payload to carry one; a supplier who controls only their own component, and must work with equipment they cannot alter, must in fact build the cleverer hand.

Use the Language

Describe a component in your own work that you have been trying to improve, and consider what would change if the thing it connects to could be altered too, using interface, design around and latch onto.

Discuss

Where in your organization has a difficult problem been treated as one team's component problem when it is actually a relationship between two teams' components?

Write

In 100 words, describe a problem you have been trying to solve by making one part better, and state what the equivalent of a standardized grapple fixture would be in that situation.

SOURCES

A Canadian Space Agency, official Canadarm program and flight-history records (asc-csa.gc.ca).

B Parks Canada, national historic event designation, "Canadarm Deployment in Space, 1981."

B The Canadian Encyclopedia, "Canadarm," including SIMFAC, mass and payload specifications.

C Smithsonian National Air and Space Museum, object record for the Canadarm Remote Manipulator System (Spar Aerospace Ltd.), NASM A20130168000.

D The origin anecdote of the cardboard-and-string demonstration model, and the internal evaluation of competing end-effector concepts, rest on later recollection rather than contemporaneous record.

THIS CONNECTS TO

Story 18: The Square Peg and the Round Hole (Apollo 13)

SAME MECHANISM

Both solved an engineering problem by inventorying exactly what was physically possible in the actual environment, rather than starting from an ideal, unconstrained design.

Story 28: A Different Domain Solves the Problem (Hedy Lamarr)

LIMITATION

Lamarr's chapter is about importing an existing mechanism from an unrelated field. Canadarm's is closer to the opposite — realizing the obvious

mechanism (gripping) was actually wrong for the new environment, and had to be redesigned from scratch.

Story 29: Think Like the User (Temple Grandin) CONTRAST

See The Contrast in that chapter. In short: Grandin designed from the user's literal physical vantage point; Canadarm's engineers designed from the environment's actual physical constraints, once the intuitive human approach (gripping) turned out to be wrong.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Engineers designing a robotic arm for the Space Shuttle program initially approached the task as building a strong, dextrous hand capable of grasping objects, the way a human hand would.

T — Task

They needed the arm to reliably move and position large payloads in the actual environment it would operate in — the vacuum and weightlessness of space.

A — Action

They realized gripping strength was the wrong solution in weightlessness, where objects have no weight to resist — and redesigned the arm around a mechanism that captured and stabilized payloads rather than gripping them by force.

R — Result

The resulting Canadarm became a reliable, long-serving tool on the Space Shuttle program, succeeding by solving a differently framed version of the original problem.

Level 2 — Make It Your Story

> *Tell me about a time when redesigning the underlying relationship between two components solved a problem that improving one component alone could not.*

Use STAR to answer. Think about: a system redesign, a component that needed a different approach rather than a better version of itself.

TAKE THE IDEA FURTHER

> *What problem in your organization are you trying to solve by improving one component, when the real solution may require redesigning the relationship between components instead?*

41

The Camera It Invented

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In December 1975, Kodak engineer Steven Sasson built the first working digital camera — a bulky, 3.6-kilogram prototype that captured a black-and-white image at a resolution of 0.01 megapixels, storing it on a cassette tape, requiring 23 seconds simply to record a single image. When Sasson demonstrated the device to Kodak's management, the reaction, by his own repeated later account, was notably unenthusiastic: "That's cute — but don't tell anyone about it." The instruction to keep quiet reflected a real and, from a narrow short-term perspective, not unreasonable structural concern: Kodak's entire, extraordinarily profitable business model was built around film — manufacturing it, selling it, and, critically, profiting again from processing and printing every photograph taken with it, an integrated revenue stream that depended entirely on photography remaining a chemical, film-based process. A camera that eliminated film entirely was not merely a new product category to Kodak's leadership. It was a direct and existential threat to the mechanism generating the great majority of the company's actual profit.

It would be a significant oversimplification, though, to say Kodak simply buried the technology and never touched it again, which is the version of the story often repeated. Sasson was, in fact, permitted to continue developing digital imaging technology within Kodak over the following years, and by 1989 he and a colleague had built a considerably more advanced prototype, closer in concept to a modern digital SLR camera. Kodak's marketing division, when the improved prototype was presented for potential

commercialization, again resisted, this time explicitly citing fear of cannibalizing film sales as the reason not to bring the technology to market as a genuine competitor to Kodak's own film business, even though the underlying technology, by that point, clearly worked.

Kodak's fuller strategic failure, then, was not a single dramatic decision to bury an invention in 1975, but a sustained, multi-decade pattern of developing genuinely capable digital imaging technology internally — Kodak held numerous foundational digital photography patents and, for a period, collected meaningful royalty income from licensing them to competitors who were more willing to build and sell the actual cameras — while declining, repeatedly and for essentially the same underlying reason, to build and aggressively market consumer digital cameras itself until it was competing from a position of significant disadvantage against companies with no comparable film business to protect. Kodak filed for bankruptcy in January 2012, a company that had genuinely invented digital photography and then spent decades protecting the older revenue model that invention was destined to replace.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What genuinely working, internally developed capability in your own organization is being deliberately held back from full commercialization out of fear of cannibalizing an existing, currently more profitable revenue stream?*

KEY TAKEAWAYS

1. Kodak did not simply suppress digital photography once and never revisit it; the technology was developed further internally for years, including a 1989 prototype approaching modern digital SLR capability.
2. The repeated reason for declining to commercialize digital cameras was consistent across more than a decade: explicit fear of cannibalizing the company's existing, highly profitable film-based revenue.
3. Kodak profited for a period from licensing its own foundational digital photography patents to competitors, meaning the company understood the technology's commercial value while still declining to compete with it directly.

SUMMARY

Kodak engineer Steven Sasson invented the first working digital camera in 1975, and though management's initial reaction was dismissive, the technology was developed further internally over subsequent years, including an advanced 1989 prototype. Kodak's marketing division repeatedly declined to commercialize digital cameras aggressively, explicitly citing fear of cannibalizing profitable film sales, even while the company licensed its own digital photography patents to competitors for royalty income. Kodak filed for bankruptcy in 2012, having invented the technology that ultimately replaced its core business without building a sustainable position in that technology itself.

STORY CHECK

ESTABLISHED

Sasson's 1975 invention, his own repeated firsthand account of management's reaction, the 1989 advanced prototype, and Kodak's 2012 bankruptcy are all

extensively documented in Kodak's own records, Sasson's public interviews, and independent business history.

POPULAR VERSION ADDS

The story is often compressed into "Kodak invented the digital camera and immediately killed it," a framing that omits the years of continued internal development and the company's patent licensing income, both of which complicate the simpler narrative of outright suppression.

UNCERTAIN

The precise internal decision-making process and the specific individuals responsible for each decision not to commercialize aggressively are documented mainly through later interviews and retrospective corporate accounts rather than contemporaneous internal records made public.

VOCABULARY & PHRASAL VERBS

cannibalize (v., business) to reduce a company's own existing product's success by introducing a competing one. Marketing feared the new camera would cannibalize film sales.

existential (adj.) relating to the very existence or survival of something. The technology posed an existential threat to Kodak's core business.

royalty (n., legal) a payment made to a patent or intellectual property holder for the right to use it. Kodak collected royalty income from licensing its patents.

integrated (adj.) combined into a single, interconnected whole. Kodak's revenue stream was an integrated system built around film.

bury (v., figurative) to deliberately suppress or hide something, preventing it from being used or known. The popular story claims Kodak buried the invention.

hold back (phrasal verb) to prevent something from proceeding or being released. A genuinely working capability was held back for years.

compete from a position of (phrase) to enter a competitive situation already at an advantage or disadvantage. Kodak eventually competed from a position of significant disadvantage.

destined to (phrase) certain or highly likely to happen in the future. Film was destined to be replaced by the very technology Kodak invented.

suppress (v.) to prevent something from being expressed, developed or made known. Marketing effectively suppressed commercialization repeatedly.

PRACTICE

Where the Story Breaks Down

Kodak's leadership faced a genuinely difficult trade-off, not an obviously foolish one — a large, currently profitable business really was at risk, and the timeline over which digital photography would actually become commercially dominant was genuinely uncertain in the 1980s in a way that looks much more obvious only in hindsight.

Use the Language

Describe a capability your organization has developed but hesitated to commercialize for fear of cannibalizing existing revenue, using cannibalize, hold back and existential.

Discuss

How would you decide, in your own organization, when a cannibalization risk is worth taking versus when protecting the existing business is genuinely the wiser choice?

Write

In 100 words, compare Kodak's approach to digital photography with Netflix's approach to streaming (Story 31), and identify the specific difference in how each company handled the same underlying dilemma.

SOURCES

A Sasson, S. J., interviews and firsthand accounts, including "How I Invented the Digital Camera in 1975," The New York Times (2015).

C IEEE, Milestone dedication documentation for Kodak's first digital camera (Rochester, NY).

D The precise internal corporate decision-making record beyond Sasson's own retrospective accounts and secondary business press coverage has not been made fully public by Kodak.

THE CONTRAST

Story 41: Kodak vs. Story 31: The Company That Killed Its Own Business (Netflix)

Story 41: Kodak: Had the replacement technology built internally, developed it further for years, and still chose to protect the profitable existing business over it.

Story 31: The Company That Killed Its Own Business (Netflix): Actively worked against its own profitable existing business in order to build the replacement.

When you have the technology that could replace your own business already built in-house, what actually determines whether you use it?

THIS CONNECTS TO

Story 31: The Company That Killed Its Own Business (Netflix)

CONTRAST

Netflix actively dismantled its own profitable business to build its replacement. Kodak had the replacement technology built internally and chose the opposite path — protecting the profitable business instead.

Story 43: Seeing the Smartphone Too Early (Nokia) SAME MECHANISM

Both had a genuine, credible internal signal of where the market was heading, and chose not to act on it decisively while the window to do so was still open.

Story 42: Abandoning the Business You Built (Intel/Grove) CONTRAST

Grove used a deliberate mental exercise to justify abandoning Intel's founding business before it was too late. Kodak's leadership had a comparable opportunity and chose the opposite path.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

In December 1975, Kodak engineer Steven Sasson built the first working digital camera, but the technology directly threatened Kodak's highly profitable film business.

T — Task

Kodak's leadership needed to decide how aggressively to develop and commercialize digital photography.

A — Action

The company continued developing the technology internally for years but

held back from fully commercializing it in ways that would meaningfully compete with film.

R — Result

Digital photography was eventually commercialized by other companies, and Kodak, despite having invented the technology first, filed for bankruptcy in 2012 as film sales collapsed.

Level 2 — Make It Your Story

> *Tell me about a time when your organization had a genuinely working internal capability but held back from fully committing to it.*

Use STAR to answer. Think about: a new technology, a competing internal project, a capability that threatened an existing revenue stream.

TAKE THE IDEA FURTHER

> *What genuinely working, internally developed capability in your own organization is being deliberately held back from full commercialization?*

42

Abandoning the Business You Built

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

By the mid-1980s, Intel — a company that had built its founding reputation and much of its early revenue on memory chips, having essentially invented the commercial DRAM (dynamic random-access memory) market in 1970 — faced a specific and increasingly dire competitive problem: Japanese semiconductor manufacturers, backed by significant capital investment and operating with lower production costs, were systematically undercutting Intel's memory chip prices and capturing market share, to the point where Intel's once-dominant, founding product line was becoming a persistent money-losing business with no clear path back to profitability against the specific competitive dynamics of that market.

The decision that followed is described in then-president Andy Grove's own later account with a specific and deliberately reframed question. Grove has recounted asking then-CEO Gordon Moore a pointed hypothetical: if the two of them were removed from the company and a new outside CEO were brought in to replace them, what would that new CEO do about the failing memory business? Moore's answer, according to Grove, was immediate and clear: a new CEO with no personal or historical attachment to memory chips as the company's founding product would get Intel out of the memory business entirely.

Grove's own reported response was equally direct: "Why shouldn't you and I walk out the door, come back, and do it ourselves?" The reframing mattered because it deliberately separated the strategic question — what does the

business actually need — from the emotional and organizational entanglement that made the same question much harder to answer honestly from inside the company's own history and identity. Memory chips were not merely a product line to Intel's existing leadership; they were the company's founding identity, the product that had defined "what Intel does" for over a decade, and abandoning them meant abandoning a significant part of how the company, and its leaders personally, understood themselves.

Intel exited the memory chip business in 1985, redirecting the engineering talent, manufacturing capacity and capital that memory chips had previously consumed toward microprocessors — a product line the company had also been developing, initially as a secondary business, but one now positioned to receive the full weight of the company's resources and strategic focus. The decision, though costly and organizationally painful in its immediate aftermath, including significant layoffs and facility closures, is now widely regarded as one of the most consequential strategic decisions in the company's history, repositioning Intel to become the dominant global manufacturer of microprocessors for personal computers over the following decades — a business considerably larger and more durably profitable than the memory chip business it replaced ever was.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *If an outside leader, with no personal history or identity invested in your organization's founding business, took over tomorrow, what would they immediately stop doing that current leadership is still protecting?*

KEY TAKEAWAYS

1. Grove's reframing — imagining a new CEO with no personal history in the business — deliberately separated the strategic question from the emotional and identity-based entanglement that made an honest answer difficult from inside the company.
2. The memory chip business was not merely unprofitable; it was the company's founding product and organizational identity, which is precisely what made exiting it so difficult despite the numbers clearly supporting the decision.
3. The resources freed by exiting memory chips were redirected toward microprocessors, a business that became substantially larger and more durably profitable than the one it replaced.

SUMMARY

Facing an increasingly unwinnable competitive position in memory chips against lower-cost Japanese manufacturers, Intel's Andy Grove and Gordon Moore reframed their strategic dilemma by asking what an outside CEO, with no personal history in the company's founding product, would do — and concluded a new CEO would exit the memory business entirely. Intel did exactly that in 1985, redirecting resources toward microprocessors, a decision now regarded as one of the most consequential in the company's history and one that positioned Intel to dominate the microprocessor market for personal computers for decades.

STORY CHECK

ESTABLISHED

Intel's 1985 exit from the memory chip business and subsequent focus on microprocessors are extensively documented in the company's own corporate history and independent business and technology history.

POPULAR VERSION ADDS

Grove's specific quotation ("Why shouldn't you and I walk out the door...") is drawn from his own later published account and is widely repeated in business literature as a verbatim exchange; as with many such retold conversations, its precision as an exact transcript versus a paraphrased recollection years later cannot be independently verified.

UNCERTAIN

The precise timeline of how gradually versus how suddenly the full exit from memory chips occurred, and the exact scale of the associated layoffs and facility closures, vary somewhat between different historical accounts.

VOCABULARY & PHRASAL VERBS

undercut (v.) to offer a lower price than a competitor, often to gain market share. Japanese manufacturers systematically undercut Intel's prices.

entanglement (n., figurative) a state of being complicatedly involved with or connected to something. Emotional entanglement made the decision harder from inside the company.

reframe (v.) to present or consider something from a different perspective. Grove reframed the question to bypass personal attachment.

redirect (v.) to change the course or purpose of something, such as resources. Resources were redirected from memory chips to microprocessors.

founding identity (n., figurative) the core sense of what an organization originally was or stood for. Memory chips were part of Intel's founding identity.

walk out the door (phrase, figurative) to leave a position or role entirely, here used hypothetically. He asked why they shouldn't walk out the door and make the change themselves.

exit (a business) (v., business) to formally withdraw from a market or product line. Intel exited the memory business in 1985.

weigh on (phrasal verb) to burden or trouble someone, often psychologically. The founding product's history weighed on the decision.

regard as (phrasal verb) to consider or view something in a particular way. The decision is now widely regarded as one of the most consequential in Intel's history.

PRACTICE

Where the Story Breaks Down

Grove and Moore had the founder-level authority and personal credibility within Intel to make and enforce a decision this painful and identity-altering; a manager without equivalent standing might identify the same correct strategic answer and still lack the organizational power to actually execute it.

Use the Language

Describe a business, product or process your organization protects mainly out of identity or history rather than current performance, using founding identity, reframe and exit.

Discuss

Apply Grove's specific reframing question to your own organization: what would an outside leader with no history there do differently, right now?

Write

In 100 words, describe a decision your organization is avoiding specifically because of emotional or historical attachment, separate from what the numbers actually show.

SOURCES

A Grove, A. S., Only the Paranoid Survive: How to Exploit the Crisis Points That Challenge Every Company, Currency Doubleday (1996) — Grove's own account of the exit decision.

C Intel Corporation, official corporate history of the 1985 memory business exit (intel.com).

B Jackson, T., Inside Intel: Andy Grove and the Rise of the World's Most Powerful Chip Company, Dutton (1997).

THE CONTRAST

Story 42: Andy Grove vs. Story 48: Sacrificing the Present for the Future (Bezos)

Story 42: Andy Grove: Used a mental exercise to separate the strategic question from personal and emotional attachment to the business he had built.

Story 48: Sacrificing the Present for the Future (Bezos): Reframed the relevant measure of success itself, from near-term profit to long-term position, to justify years of sacrificed short-term results.

Is the harder discipline separating a decision from your emotional attachment to it, or separating a decision from short-term financial pressure?

THIS CONNECTS TO

Story 31: The Company That Killed Its Own Business (Netflix)

SAME MECHANISM

Both required abandoning a currently defining or profitable part of the business in order to pursue what would replace it.

Story 41: The Camera It Invented (Kodak) CONTRAST

Grove used a deliberate mental exercise to justify abandoning Intel's founding business before it was too late. Kodak's leadership had a comparable opportunity and chose the opposite path.

Story 51: Predict From Similar Projects (Flyvbjerg) LIMITATION

Grove's exercise removes personal identity from a single high-stakes decision. Flyvbjerg's method removes personal optimism from planning by looking at a whole reference class of past projects — different tools for different kinds of bias.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

By the mid-1980s, Intel's memory-chip business, which had defined the company's founding identity and much of its early revenue, was losing badly to Japanese competitors.

T — Task

Andy Grove and CEO Gordon Moore needed to decide whether to keep investing in memory chips or abandon the business Intel had been built around.

A — Action

Grove asked Moore what a new CEO, brought in from outside with no personal history in the business, would do — and concluded that person would exit memory chips. They then did exactly that themselves.

R — Result

Intel exited the memory-chip business and refocused on microprocessors, a decision widely credited with the company's subsequent dominance in that market.

Level 2 — Make It Your Story

> *Tell me about a time when you had to separate a strategic decision from your own personal attachment to how things had always been done.*

Use STAR to answer. Think about: a legacy project, a founding product, a decision you had to make as if you had no history with it.

TAKE THE IDEA FURTHER

> *If an outside leader, with no personal history or identity invested in your organization's founding business, took over tomorrow, what would they do differently?*

43

Seeing the Smartphone Too Early

DOCUMENTATION LEVEL 2 — SELF-REPORTED OR TRANSFERRED

The account rests mainly on the protagonist's own retelling, or the idea was documented but its original application was never adopted.

By the mid-2000s, Nokia held a commanding position in the global mobile phone market, with a market share and manufacturing scale competitors found difficult to approach, built substantially on the company's recognized strength in hardware design and manufacturing efficiency across an enormous range of phone models serving markets worldwide. It is a matter of extensively reported record that Nokia's own internal research and development teams had, years before Apple's 2007 iPhone launch, explored touchscreen interfaces, mobile internet browsing, and application-based software platforms — the specific set of ideas that would come to define the smartphone category once other companies brought them to market successfully.

The more precise and more instructive version of what actually happened, rather than the simplified popular narrative that Nokia "never saw it coming," is that Nokia's internal research organization did produce genuinely prescient exploratory work and prototypes touching on many of these ideas, but the company's dominant internal culture, organizational incentives and resource allocation processes remained overwhelmingly oriented around its existing, extraordinarily successful and well-understood hardware business, optimized for a phone market defined primarily by handset design, manufacturing efficiency, and network carrier relationships, rather than by the kind of integrated software and touchscreen platform experience the eventual smartphone category would come to be defined by.

The specific organizational obstacle, as later analyzed by business researchers and former Nokia executives themselves, was not a simple failure of the company's engineers to see the technology coming, but a structural failure to act decisively on early internal signals that were genuinely present, because doing so would have required reallocating resources and organizational priority away from the current, highly profitable hardware business toward an unproven, internally lower-status software and platform effort — precisely the kind of resource reallocation and internal reprioritization that companies with a currently dominant, profitable core business chronically find organizationally difficult to execute, even when relevant technical evidence is genuinely available inside the company well in advance.

Nokia's mobile phone business, having failed to successfully pivot its dominant market position into the touchscreen smartphone era, was sold to Microsoft in 2013 for a fraction of the company's earlier peak valuation, a sale that effectively marked the end of Nokia as a major independent mobile phone manufacturer. It is worth being precise about the lesson here, since the popular version tends to flatten it into simple corporate blindness: Nokia's failure was not a failure of technical foresight, since the relevant research genuinely existed internally. It was a failure of organizational will and resource reallocation, acting decisively on weak internal signals that pointed away from the company's own current, comfortable and highly profitable centre of gravity.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What early, genuinely credible internal signal is your organization currently aware of, and choosing not to act on decisively, because doing*

> *so would require reallocating resources away from your current, comfortable and profitable core business?*

KEY TAKEAWAYS

1. Nokia's internal research had genuinely explored touchscreen and mobile software platform concepts years before the smartphone category was defined by competitors, contradicting the simplified narrative that the company was technically blindsided.
2. The organizational failure was not one of foresight but of resource reallocation and internal prioritization, moving investment and status away from a currently dominant, profitable hardware business toward an unproven internal effort.
3. This pattern — genuine internal awareness of a threat, combined with organizational inability to reallocate resources away from a currently successful core business — recurs across many companies facing disruptive technological shifts, not only Nokia.

SUMMARY

Nokia's internal research teams explored touchscreen interfaces and mobile software platforms years before the 2007 iPhone defined the smartphone category, contradicting the popular narrative that the company simply failed to see the shift coming. The actual failure was organizational: the company's dominant hardware-focused culture and resource allocation processes remained oriented around its existing, highly profitable business, making it structurally difficult to reallocate priority toward an unproven internal software and platform effort. Nokia's mobile phone business was sold to Microsoft in 2013 for a fraction of its earlier peak value.

STORY CHECK

ESTABLISHED

Nokia's internal exploratory research into touchscreen and mobile software concepts prior to 2007, and the company's 2013 sale to Microsoft, are documented in business school case studies and former executives' own published accounts.

POPULAR VERSION ADDS

The simplified "Nokia never saw it coming" narrative is widely repeated in business commentary but is directly contradicted by documented internal research; the more accurate account, involving organizational and resource-allocation failure rather than technical blindness, is less commonly told because it is less simple and less flattering to competitors who are sometimes credited with foresight Nokia's own engineers arguably had first.

UNCERTAIN

The precise internal politics and specific decision points at which Nokia's leadership chose not to reallocate resources toward the touchscreen and software research are documented with varying degrees of detail and some disagreement across different insider and academic accounts.

VOCABULARY & PHRASAL VERBS

prescient (adj.) showing accurate knowledge or anticipation of future developments. Nokia's research was genuinely prescient in several respects.

weak signal (n., strategy) an early, ambiguous indicator of a future significant change, easy to overlook or dismiss. The company had weak signals available internally for years.

reallocate (v.) to redistribute resources from one use to another. The company struggled to reallocate resources toward the new effort.

centre of gravity (n., figurative) the primary focus or core of an organization's activity and identity. The hardware business remained the company's centre of gravity.

blindsided (adj.) taken by surprise, especially by something that should have been anticipated. The popular narrative claims Nokia was simply blindsided.

flatten (v., figurative) to oversimplify a nuanced situation into a single, less accurate explanation. The popular version flattens a complex organizational failure into simple blindness.

pivot (v., business) to change strategic direction while often retaining some existing capability. The company failed to pivot its dominant position successfully.

fraction of (phrase) a small part of a larger original amount. The sale price was a fraction of the company's earlier peak value.

chronically (adv.) persistently or constantly over a long period, especially referring to a recurring problem. Companies with a dominant core business chronically struggle to reallocate resources.

PRACTICE

Where the Story Breaks Down

Nokia's internal research teams had genuine technical credibility and real prototypes to point to; many organizations facing a comparable situation have much weaker, more genuinely ambiguous internal signals, making the decision to reallocate resources a much harder and more uncertain call to make in real time, without the benefit of hindsight.

Use the Language

Describe a weak internal signal your organization might be aware of but not acting on, using weak signal, centre of gravity and reallocate.

Discuss

What would it actually take, organizationally, for your team or company to reallocate meaningful resources away from a currently profitable core business toward an unproven internal signal?

Write

In 100 words, describe a piece of internal research, feedback or data your own organization currently possesses that points away from its current core business, and consider what prevents acting on it.

SOURCES

B Vuori, T. O. & Huy, Q. N., "Distributed Attention and Shared Emotions in the Innovation Process: How Nokia Lost the Smartphone Battle," Administrative Science Quarterly 61(1), 9–51 (2016) — a detailed academic study based on internal Nokia interviews.

C Microsoft Corporation and Nokia Corporation, joint public announcement of the 2013 acquisition of Nokia's mobile phone business.

D Specific internal prototype details and timelines are drawn primarily from former employees' retrospective accounts and the Vuori & Huy academic study rather than a single, comprehensive public corporate record.

THE CONTRAST

Story 43: Nokia vs. Story 10: The O-Ring in the Ice Water (Feynman)

Story 43: Nokia: Had a genuine, credible internal signal well before disaster, and chose not to act on it decisively while there was time.

Story 10: The O-Ring in the Ice Water (Feynman): Had a genuine warning that existed in explicit form and reached no one positioned to act on it in time, right before disaster.

When a warning exists but goes unacted on, is the deeper failure that no one saw it, or that seeing it wasn't enough to produce action?

THIS CONNECTS TO

Story 41: The Camera It Invented (Kodak) SAME MECHANISM

Both had a genuine, credible internal signal of where the market was heading, and chose not to act on it decisively while the window to do so was still open.

Story 27: What a Machine Could Become (Ada Lovelace) SAME MECHANISM

Both involve a genuine, correct insight arriving well before the surrounding technology or market was ready to act on it.

Story 7: The Telescope That Moved the Earth (Galileo) LIMITATION

Galileo's chapter is about the cost of stating a true conclusion. Nokia's is almost the mirror image — the cost of having a true signal internally and choosing not to state it, or act on it, at all.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

By the mid-2000s, Nokia held a commanding position in the global mobile

phone market, and its internal research had already explored touchscreen interfaces and mobile software platform concepts years before the smartphone category as we know it existed.

T — Task

Nokia's leadership needed to decide how decisively to act on those internal research signals, against the pressure of protecting an already dominant, profitable existing product line.

A — Action

The company continued prioritizing its existing hardware-focused strategy and organizational structure rather than decisively committing to the internally-explored smartphone concepts.

R — Result

Competitors that acted decisively on similar underlying ideas came to dominate the smartphone market, and Nokia's mobile phone business collapsed within a few years.

Level 2 — Make It Your Story

> *Tell me about a time when your organization was aware of an important signal internally and chose not to act on it decisively.*

Use STAR to answer. Think about: an early warning, a research finding, an internal pilot that was never scaled up in time.

TAKE THE IDEA FURTHER

> *What early, genuinely credible internal signal is your organization currently aware of, and choosing not to act on decisively, because doing so would threaten an existing product line?*

44

Build Learning Into the System

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Toyota's production philosophy, alongside the jidoka principle of stopping production the instant a defect is detected (Story 16), rests on a further, complementary practice generally known by its Japanese name, kaizen, meaning roughly "change for the better" or "continuous improvement." Where jidoka addresses defects at the moment they occur, kaizen addresses something structurally different and, in a specific and important sense, more ambitious: building a permanent, systematic organizational habit of continuous, incremental improvement, generated primarily from the observations of the workers actually performing the work, rather than improvements designed and imposed exclusively from management or engineering staff working separately from the factory floor.

The practical mechanism Toyota developed to operationalize this idea, rather than leaving continuous improvement as an unstructured cultural aspiration, was a formalized suggestion system in which line workers were actively expected, trained and structurally incentivized to identify and propose specific, often genuinely small, improvements to their own immediate work processes — a slightly awkward tool position, a wasted motion in an assembly step, a minor adjustment to where a part was stored relative to where it was used — with a defined, reasonably fast process for evaluating, testing and implementing viable suggestions rather than letting them accumulate unreviewed in an ignored suggestion box, a specific institutional failure mode

common to nominal "suggestion program" initiatives at many other companies.

The underlying rationale connects directly to a specific claim about where genuinely useful process knowledge actually resides within an organization: the people performing a specific task repeatedly, hour after hour, frequently notice small inefficiencies, awkward steps or opportunities for improvement that someone designing the process from an office, without that same repeated, direct, hands-on exposure, simply has no comparable opportunity to observe. Individually, any single kaizen suggestion is typically minor — Toyota's own reporting has at various points cited very large annual volumes of employee suggestions across its full workforce, the overwhelming majority small and incremental rather than transformative — but accumulated systematically across a large workforce and sustained over years, the aggregate effect on process efficiency, quality and cost becomes, according to extensive independent study of the Toyota Production System's development, substantial and durable.

Kaizen's most distinctive characteristic, and the one most frequently lost when the concept is superficially imitated elsewhere without the surrounding institutional infrastructure, is that it treats improvement not as an occasional special project undertaken by a dedicated team but as a continuous, structurally embedded, everyday expectation of ordinary work — every employee is, in a specific and genuine sense, formally expected to also be identifying opportunities to improve their own process, not merely to execute it, a distinction that many organizations attempting to adopt "continuous improvement" language without the corresponding structural suggestion system and evaluation process fail to actually replicate in practice.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Does your organization have a structured, reasonably fast process for evaluating and implementing frontline employee suggestions, or does your suggestion system function more as an unreviewed box that accumulates ideas nobody ever acts on?*

KEY TAKEAWAYS

1. Kaizen addresses a structurally different problem from jidoka: not stopping defects at the moment they occur, but building a permanent organizational habit of incremental improvement generated by frontline workers.
2. The formalized suggestion system, with a defined process for evaluation and implementation, was what distinguished kaizen from an unstructured cultural aspiration or an ignored suggestion box.
3. Individual suggestions were typically minor, but the aggregate effect, sustained systematically across a large workforce over years, produced substantial and durable improvements in efficiency and quality.

SUMMARY

Toyota's kaizen philosophy formalized continuous, incremental process improvement as a structural, everyday expectation of frontline workers, distinct from the jidoka principle of stopping production at the moment of a detected defect. A formalized suggestion system, with a defined and reasonably fast evaluation and implementation process, allowed the small-scale observations of workers performing tasks repeatedly to accumulate into

substantial aggregate improvements over years, avoiding the common failure mode of an unreviewed suggestion box. The approach is frequently imitated superficially by other organizations that adopt the language of continuous improvement without replicating the structural suggestion and evaluation system underneath it.

STORY CHECK

ESTABLISHED

Toyota's kaizen philosophy, its formalized employee suggestion system, and its role alongside jidoka as a pillar of the Toyota Production System are extensively documented in Toyota's own corporate literature and independent manufacturing and management research.

POPULAR VERSION ADDS

"Kaizen" is sometimes used loosely in Western business literature to describe any general continuous-improvement aspiration; Toyota's own practice specifically involves a formalized, structured suggestion and evaluation system, which is often the part lost when the term is adopted elsewhere without the corresponding institutional infrastructure.

UNCERTAIN

Specific annual suggestion volume figures Toyota has cited at various points and in various facilities vary considerably depending on the specific plant, year and reporting method, and no single, universally agreed figure represents "the" Toyota suggestion rate across the company's full history.

VOCABULARY & PHRASAL VERBS

incremental (adj.) occurring in small, gradual steps rather than large, sudden changes. Kaizen focuses on incremental, rather than transformative, improvement.

complementary (adj.) combining well with something else to form a more complete whole. Kaizen is complementary to jidoka within the production system.

operationalize (v.) to put an abstract idea into a specific, usable, practical form. The suggestion system operationalized the aspiration of continuous improvement.

aggregate (n./adj.) the total or combined result of many individual parts. The aggregate effect of many small suggestions became substantial.

frontline (adj.) relating to workers directly performing the primary task, as opposed to management or design staff. Frontline workers noticed inefficiencies management could not observe.

embed (v.) to fix firmly and deeply within a system or structure. Improvement was structurally embedded into everyday work.

accumulate (v.) to gradually gather or build up over time. Small suggestions accumulated into substantial improvements over years.

gather dust (phrase, figurative) to remain unused or ignored over time. An unreviewed suggestion box lets ideas gather dust.

replicate (v.) to reproduce or copy something exactly, especially a process or result. Many companies fail to replicate the underlying structural system.

PRACTICE

Where the Story Breaks Down

Kaizen's effectiveness depends on genuine, sustained management commitment to actually evaluating and implementing frontline suggestions quickly and visibly; a suggestion system without that follow-through quickly loses employee participation and credibility, regardless of how the program is initially designed or named.

Use the Language

Describe a small, frontline-level inefficiency in your own work that management is unlikely to have observed directly, using frontline, incremental and accumulate.

Discuss

Does your organization's suggestion or feedback system actually evaluate and implement ideas quickly, or does it function more like an unreviewed box that gathers dust?

Write

In 100 words, design a specific, fast evaluation process for frontline suggestions in your own organization, addressing how quickly a suggestion would be reviewed and by whom.

SOURCES

A Imai, M., Kaizen: The Key to Japan's Competitive Success, Random House Business Division (1986).

C Toyota Motor Corporation, official corporate documentation of the Toyota Production System and its employee suggestion program (toyota-global.com).

B Liker, J. K., The Toyota Way: 14 Management Principles from the World's Greatest Manufacturer, McGraw-Hill (2004).

THIS CONNECTS TO

Story 16: The Factory That Could Stop Itself (Jidoka)

SAME MECHANISM

Both are pillars of the Toyota Production System — jidoka stops defects at the moment they occur, while kaizen builds the ongoing habit of improving the process itself.

Story 11: The Five Whys SAME MECHANISM

Both are Toyota Production System tools built on the same premise — that the first visible explanation for a problem is rarely the real one, and structured, repeated inquiry gets closer to it.

Story 38: The Farmer Who Did Less (Fukuoka) LIMITATION

Kaizen adds a continuous, structured practice of small improvements on top of an existing system. Fukuoka's approach removed interventions that were only compensating for earlier problems — improvement by subtraction rather than addition.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Toyota's production system already stopped defects the instant they occurred through jidoka, but the company still needed a way to structurally capture and act on the improvement ideas of frontline workers who encountered small problems daily.

T — Task

Toyota needed an ongoing, systematic process for surfacing and implementing small improvements, not just a one-time fix for a single problem.

A — Action

The company built kaizen — a formal, continuous process for evaluating and implementing suggestions from frontline employees — into its standard operations.

R — Result

The accumulated effect of many small, continuously implemented

improvements became a significant driver of Toyota's long-term manufacturing quality and efficiency.

Level 2 — Make It Your Story

> *Tell me about a time when a structured process for capturing small, frontline improvement suggestions made a meaningful difference over time.*

Use STAR to answer. Think about: a suggestion box that actually worked, a continuous improvement process, small fixes that added up.

TAKE THE IDEA FURTHER

> *Does your organization have a structured, reasonably fast process for evaluating and implementing frontline employee suggestions, or does good feedback usually just get lost?*

PART VI

Decide Under Uncertainty

45

The Goal That Had to Change

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In 1914, Ernest Shackleton set out to achieve the first land crossing of Antarctica, leading an expedition aboard the ship Endurance. The mission never came close to attempting the crossing itself. The Endurance became trapped in pack ice in the Weddell Sea before reaching land, was gradually crushed by the shifting ice over subsequent months, and finally sank in November 1915, leaving Shackleton and his 27-man crew stranded on drifting ice floes, hundreds of miles from any hope of rescue, with no functioning ship and no realistic means of communicating their location to the outside world. The original objective — crossing Antarctica — was, from the moment the ship sank, no longer merely difficult. It was entirely irrelevant, in the specific and consequential sense that continuing to orient decisions around it would have actively endangered the crew further. Shackleton's defining act of leadership, documented extensively across the crew's own diaries and later accounts, was the speed and clarity with which he redefined the expedition's actual objective, from the original scientific and exploratory goal to a single, absolute new priority: get every one of the 27 men home alive.

That redefined objective drove a sequence of increasingly difficult decisions over the following months, each one legible only in light of the new goal rather than the old one. The crew abandoned or destroyed personal possessions that added weight without contributing to survival, camped on drifting ice for months as it slowly carried them, and eventually took to three

small lifeboats to reach the uninhabited, barren Elephant Island — itself not a viable long-term refuge, merely a marginally survivable stopping point. From there, Shackleton and five crew members undertook a genuinely desperate 800-mile open-boat voyage across some of the most dangerous ocean water on Earth to reach South Georgia Island, where a whaling station offered the only realistic hope of organizing rescue for the men left behind. Every man survived — a result contemporary polar historians consistently describe as a remarkable and far from guaranteed outcome, given the extremity of the conditions involved.

One frequently repeated detail attached to this story deserves direct correction, because it illustrates precisely the kind of embellishment this book tries to flag rather than repeat uncritically: the famous recruitment advertisement supposedly placed by Shackleton before the expedition — "Men wanted for hazardous journey. Low wages, bitter cold, long hours of complete darkness. Safe return doubtful. Honour and recognition in event of success" — has never been found in any surviving newspaper archive despite extensive searching by historians, and its earliest documented appearance is in a 1949 book, decades after the expedition, which cites no source. The advertisement is almost certainly apocryphal. The survival story itself required no embellishment to be extraordinary.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What original objective in a current project of yours has quietly become irrelevant to your team's actual circumstances, while decisions continue to be made as though it still applied?*

KEY TAKEAWAYS

1. The moment the original objective became unachievable, continuing to orient decisions around it would have actively endangered the crew — recognizing that shift immediately was itself a critical decision.
2. The redefined objective (survival, not exploration) made an entirely different sequence of decisions legible and correct that would have made no sense under the original goal.
3. The expedition's most famous supporting detail, the recruitment advertisement, is very likely fictional — a caution against accepting a vivid, oft-repeated detail simply because it fits the story well.

SUMMARY

Ernest Shackleton's 1914 Antarctic crossing expedition never reached land before the ship Endurance was crushed by ice and sank, stranding the 27-man crew. Rather than continuing to orient decisions around the now-irrelevant original objective, Shackleton immediately redefined the expedition's entire purpose to the survival of every crew member, a goal that drove an extraordinary sequence of decisions including an 800-mile open-boat voyage for rescue. All 27 men survived. The famous recruitment advertisement associated with the expedition has never been found in any archive and is now considered almost certainly apocryphal.

STORY CHECK

ESTABLISHED

The Endurance's sinking, the crew's survival on ice floes and Elephant Island, and the 800-mile open-boat voyage to South Georgia are extensively documented in crew members' own diaries, Shackleton's own account, and independent polar exploration history.

POPULAR VERSION ADDS

The "Men wanted for hazardous journey" advertisement, despite being one of the most frequently cited details of the story, has never been located in any newspaper archive despite extensive dedicated research, and its earliest documented source (1949) postdates the expedition by decades and cites no source.

UNCERTAIN

The exact internal sequence of Shackleton's decision-making in the immediate aftermath of the ship's sinking is documented through crew diaries with generally strong corroboration, though individual accounts vary in some specific details.

VOCABULARY & PHRASAL VERBS

pack ice (n.) a large area of floating ice formed by the packing together of ice floes. The ship became trapped in pack ice.

ice floe (n.) a flat piece of floating ice, typically large. The crew camped on drifting ice floes for months.

stranded (adj.) left in a difficult position, unable to move or leave. They were stranded hundreds of miles from rescue.

legible (adj., figurative) able to be understood or made sense of. The new decisions were legible only in light of the redefined goal.

apocryphal (adj.) widely circulated as fact but of doubtful or unverified authenticity. The famous advertisement is now considered apocryphal.

paraphrase (n.) a restatement of something in different words, not a direct quotation. The earliest source presents the ad as a paraphrase.

orient around (phrase) to organize decisions or actions with reference to a particular goal or priority. Decisions had to stop orienting around the original objective.

take to (phrasal verb) to resort to using something, often out of necessity. The crew took to three small lifeboats.

legible only in light of (phrase) understandable only when viewed through a particular context or framing. The decisions were legible only in light of the new goal.

PRACTICE

Where the Story Breaks Down

Shackleton's redefined objective (survival) was unambiguous and universally shared by every member of the crew once the ship sank; many organizational goal redefinitions are far more contested, with different stakeholders disagreeing about what the new objective should actually be.

Use the Language

Describe a project where the original goal became irrelevant partway through, using orient around, legible and take to.

Discuss

How do you recognize the moment an original objective has become irrelevant, rather than merely difficult, in your own work?

Write

In 100 words, describe a specific decision your team is currently making, and consider whether it makes sense under your stated objective or only under a different, unstated one.

SOURCES

B Alexander, C., The Endurance: Shackleton's Legendary Antarctic Expedition, Knopf (1998).

A Worsley, F. A., Shackleton's Boat Journey, Norton (1977; based on Worsley's own contemporaneous diaries).

D Extensive archival searches by historians (documented at the Antarctic Circle research forum) have failed to locate any surviving copy of the "Men wanted" advertisement in period newspapers.

THE CONTRAST

Story 45: Shackleton vs. Story 13: Solve Backward (Pólya)

Story 45: Shackleton: Shows what happens when the goal itself has to change mid-plan — the original objective became not just difficult but actively dangerous to keep pursuing.

Story 13: Solve Backward (Pólya): Assumes the goal stays fixed while you work backward from it to find the path.

How do you know when a goal has become the wrong thing to optimize for, rather than just a hard thing to achieve?

THIS CONNECTS TO

Story 13: Solve Backward (Pólya) LIMITATION

Pólya's method assumes the goal stays fixed while you work backward from it. Shackleton's chapter shows what happens when the goal itself has to change mid-plan — working backward from a goal that no longer applies can be actively dangerous.

Story 33: The Checklist (Gawande) LIMITATION

A checklist assumes the situation is routine enough to be anticipated in advance. Shackleton's chapter is the opposite case — a situation so far outside any anticipated scenario that no checklist could have covered it.

Story 50: The Aircraft That Met a Changing Threat (Avro Arrow)

SAME MECHANISM

Both involve an original objective becoming obsolete or unachievable due to changed circumstances, and the real decision was recognizing that shift rather than continuing to pursue the original goal.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Ernest Shackleton set out in 1914 to achieve the first land crossing of Antarctica, but his ship, Endurance, became trapped and was ultimately crushed by pack ice before the crossing could even begin.

T — Task

With the original mission now impossible, Shackleton needed a new objective entirely: keeping his entire crew alive and getting them home.

A — Action

He abandoned the original goal completely and reoriented every decision around crew survival, including an extraordinarily risky open-boat journey to seek rescue.

R — Result

Every member of the expedition survived, a result widely attributed to Shackleton's willingness to fully abandon the original objective rather than let it continue to shape decisions after it became unachievable.

Level 2 — Make It Your Story

> *Tell me about a time when you had to abandon a project's original goal because pursuing it further had become the wrong thing to do.*

Use STAR to answer. Think about: a project that changed scope, a goal that became obsolete, a moment you had to redefine success mid-project.

TAKE THE IDEA FURTHER

> *What original objective in a current project of yours has quietly become irrelevant to your team's actual circumstances, while decisions continue to be made as if it still applied?*

46

Science With Almost Nothing

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

By 1898, Marie Curie and her husband Pierre had a hypothesis they believed was significant: that a new, previously unidentified radioactive element existed within pitchblende, a uranium ore already known to be more radioactive than pure uranium itself should account for, suggesting something else within it was contributing additional radioactivity. Confirming that hypothesis meant isolating the element in physically measurable quantity — not merely detecting its presence indirectly, but producing an actual sample pure enough to weigh and characterize. The Curies had no laboratory equipped for this kind of work, no significant institutional funding, and no ready supply of the raw ore required.

What they did have was a leaking, poorly heated former dissecting shed at the School of Physics in Paris, previously judged unfit for other research use, which they were permitted to use largely because no one else wanted it. They obtained tons of pitchblende residue, a waste byproduct from uranium mining operations in Austria, considered valueless by the mining industry and available to the Curies cheaply, specifically because any uranium of commercial value had already been extracted from it — meaning whatever radioactive material remained was, from the ore industry's perspective, worthless leftover. The physical labour this required was extraordinary and, by later accounts, genuinely punishing: processing several tons of pitchblende residue by hand, in batches, through repeated chemical separation, crystallization and purification steps, in a shed with a leaking roof that

provided inadequate protection from rain and, having no proper ventilation system, exposed the Curies to fumes and radioactive dust for years, with consequences for their health that were not understood or appreciated at the time and that later research strongly connects to the illnesses that affected both of them, including the aplastic anaemia that caused Marie Curie's death in 1934.

After processing several tons of raw material by hand over roughly four years, the Curies successfully isolated a mere tenth of a gram of pure radium chloride in 1902 — an almost absurdly small final yield relative to the sheer physical scale and duration of the labour required to produce it, and proof that the element they had hypothesized genuinely existed. Marie Curie went on to become the first person to win Nobel Prizes in two different scientific fields (physics in 1903, chemistry in 1911), and her insistence on working with what was actually available — discarded industrial waste, an unwanted shed, years of manual labour in place of equipment or funding she did not have — rather than waiting for better conditions that were not forthcoming, remains a frequently cited example of scientific achievement produced under conditions of genuine, severe material scarcity rather than despite an absence of it.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What genuinely valuable project are you currently postponing because you're waiting for better resources, when the Curies' example suggests the actual constraint might be tolerable if you're willing to substitute labour and persistence for capital?*

KEY TAKEAWAYS

1. The Curies used discarded industrial waste — pitchblende residue considered valueless by the mining industry — because it was cheap and available, not because it was ideal.
2. The physical scale of the labour required (tons of raw material, processed by hand over roughly four years) to produce a final yield of one-tenth of a gram illustrates how severe the resource constraint genuinely was.
3. The work came at a real and serious cost to their health, a consequence of the same scarce conditions — inadequate facilities, no protective equipment — that made the resourceful approach necessary in the first place.

SUMMARY

Marie and Pierre Curie, working in an unwanted, poorly equipped shed with no significant institutional funding, isolated the radioactive element radium by processing several tons of cheap, discarded pitchblende residue by hand over roughly four years, ultimately yielding just a tenth of a gram of pure radium chloride. The achievement demonstrated genuine scientific resourcefulness under severe material scarcity, though it also came at a serious and, at the time, poorly understood cost to their health from prolonged radiation exposure. Marie Curie later became the first person to win Nobel Prizes in two different scientific fields.

STORY CHECK

ESTABLISHED

The Curies' use of the former dissecting shed, their processing of pitchblende residue, and the 1902 isolation of radium chloride are extensively

documented in their own laboratory notebooks and independent scientific history.

POPULAR VERSION ADDS

The story is sometimes told with a somewhat romanticized emphasis on determination alone; the severe, poorly understood health consequences of prolonged unprotected radiation exposure are a necessary and frequently underemphasized part of the same story.

UNCERTAIN

The precise long-term health impact directly attributable to this specific period of work, versus Marie Curie's decades of subsequent research exposure, is difficult to isolate with certainty, though her radium-contaminated laboratory notebooks remain measurably radioactive and are stored with protective precautions to this day.

VOCABULARY & PHRASAL VERBS

pitchblende (n.) a mineral ore that is a primary source of uranium and radium. They obtained tons of pitchblende residue considered industrial waste.

residue (n.) material remaining after a process has removed the primary valuable substance. The residue was worthless to the mining industry.

crystallization (n., chemistry) the process by which a solid forms from a solution, used to purify a substance. Repeated crystallization steps purified the radioactive material.

yield (n., science) the quantity of a product obtained from a process. The final yield was an almost absurdly small tenth of a gram.

forthcoming (adj.) likely to arrive or become available, often used in the negative. Better conditions were not forthcoming.

punishing (adj., figurative) extremely demanding or harsh, especially physically. The labour involved was genuinely punishing.

make do with (phrase) to manage with limited or inadequate resources. They made do with a shed nobody else wanted.

substitute for (phrasal verb) to use one thing in place of another, typically due to necessity. Labor and persistence substituted for capital they didn't have.

exact a toll (phrase) to cause serious cost or damage, especially to health. The unprotected work exacted a real toll on their health.

PRACTICE

Where the Story Breaks Down

The Curies' resourcefulness came at a severe, largely unknown health cost that they could not have consented to knowingly, given the era's limited understanding of radiation; "make do with what's available" is a much easier principle to admire in retrospect than to apply responsibly to a genuinely hazardous situation.

Use the Language

Describe a project you've postponed for lack of resources, and consider what discarded or overlooked material might substitute for what you're missing, using residue (or a suitable substitute), substitute for and make do with.

Discuss

What is the difference between resourcefulness that is admirable and resourcefulness that comes at an unacceptable, avoidable cost?

Write

In 100 words, describe a genuinely valuable project you or your organization has postponed while waiting for better resources, and consider what a scaled-down, resource-constrained version might look like.

SOURCES

A Curie, M., "Recherches sur les substances radioactives" [Research on Radioactive Substances], doctoral thesis, University of Paris (1903).

B Quinn, S., Marie Curie: A Life, Simon & Schuster (1995).

C The Curie Museum (Musée Curie), Paris, official archival records, including preserved radioactive laboratory notebooks.

THE CONTRAST

Story 46: Marie Curie vs. Story 8: The Patent Clerk and the Train (Einstein)

Story 46: Marie Curie: Worked at the edge of what her era's instruments and institutions could support, and got somewhere real through sustained, unglamorous, resource-poor labour.

Story 8: The Patent Clerk and the Train (Einstein): Worked at the edge of what his era's institutions recognized as legitimate physics, and got somewhere real through pure reasoning, with no laboratory at all.

Is rigour more about what you can sustain with your hands over years, or what you can follow through with your reasoning alone?

THIS CONNECTS TO

Story 8: The Patent Clerk and the Train (Einstein) SAME MECHANISM
Both worked at the edge of what their era's instruments and institutions could support, and got somewhere real anyway through sheer rigour.

Story 15: 5,127 Prototypes (Dyson) SAME MECHANISM
Both required sustained, unglamorous repetition over years before the result the person was actually looking for appeared.

Story 51: Predict From Similar Projects (Flyvbjerg) LIMITATION
Curie succeeded with almost no resources by sheer sustained effort. Flyvbjerg's chapter is about a different kind of poverty — poverty of realistic planning — that no amount of resources reliably fixes on its own.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation
Marie and Pierre Curie believed a new, previously unidentified radioactive element existed within pitchblende, a uranium ore, but had almost no funding or laboratory resources to pursue the hypothesis.

T — Task
They needed to isolate the hypothesized element using whatever materials they could actually obtain.

A — Action
They acquired tons of pitchblende residue that the mining industry considered valueless waste, and processed it by hand over years in a poorly equipped shed.

R — Result

They successfully isolated radium and polonium, work that won Marie Curie two Nobel Prizes and founded the field of radioactivity research.

Level 2 — Make It Your Story

> *Tell me about a time when you made real progress on something important despite having very limited resources to work with.*

Use STAR to answer. Think about: a scrappy project, a resource-constrained team, a result you achieved with far less than you'd have liked.

TAKE THE IDEA FURTHER

> *What genuinely valuable project are you currently postponing because you're waiting for better resources, when a scrappier version might be possible now?*

47

The Salad Oil Decision

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

In November 1963, American Express's stock price collapsed by roughly 50 percent within weeks, following the exposure of a massive fraud: a company called Allied Crude Vegetable Oil Refining had used tanks supposedly full of salad oil as collateral for large loans, backed by warehouse receipts issued by an American Express subsidiary — receipts that turned out to certify oil that, upon inspection, was mostly seawater, with a thin layer of oil deliberately floated on top to fool inspectors during routine checks. American Express faced potentially enormous financial liability for the receipts its subsidiary had issued in good faith on fraudulent inventory, and Wall Street's initial reaction treated the company as facing existential, possibly fatal financial exposure.

Warren Buffett, then running an investment partnership, looked at the situation and drew a specific and consequential distinction that much of the market, in its initial panic, was not drawing clearly: the actual, uncertain financial liability from the warehousing subsidiary's fraud exposure was one problem, with a genuinely uncertain but ultimately bounded size, while American Express's core businesses — its charge card and its widely used traveller's cheques — depended on something structurally different and, in Buffett's assessment, largely undamaged by the scandal: the public's ongoing trust that the American Express name itself still meant something reliable.

Rather than relying solely on financial analysis conducted at a desk, Buffett reportedly did something characteristically direct to test his own thesis: he

visited restaurants, banks and retail establishments around Omaha, watching to see whether ordinary customers were still using American Express cards and traveller's cheques in their everyday transactions, despite the scandal's extensive newspaper coverage. He found that ordinary consumer behaviour had barely changed — people were still using American Express as they always had, suggesting the underlying brand trust that powered the company's core, most profitable businesses remained functionally intact, even while the specific warehousing subsidiary faced a serious but ultimately separable financial liability.

Buffett committed roughly $13 million — reportedly around 40 percent of his investment partnership's total assets, an extraordinarily large and concentrated bet by any conventional risk-management standard — to purchasing American Express stock at its scandal-depressed price. American Express ultimately settled its liability from the fraud for a substantial but survivable amount, its core card and traveller's cheque businesses continued largely undisturbed by ordinary consumer trust, and the stock appreciated significantly over the following years, becoming one of the formative successes that shaped Buffett's subsequent, now well-known investment philosophy: distinguishing a business's temporary, survivable trouble from a permanent, structural impairment of the specific asset — in this case, brand trust — that actually generates its long-term value.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *When your organization faces a crisis, are you assessing the actual probability and scope of the specific damage, or reacting to the general size and drama of the news coverage surrounding it?*

KEY TAKEAWAYS

1. Buffett separated two distinct risks that the market's initial panic had conflated: a bounded, if serious, financial liability, and the underlying brand trust that actually powered the company's core profitable businesses.
2. He tested his own analytical thesis directly, through personal observation of ordinary customer behaviour, rather than relying solely on financial data and news coverage.
3. The size of the resulting bet — roughly 40 percent of his partnership's assets in a single position — reflected genuine conviction in a probabilistic assessment, not a hedge against being wrong.

SUMMARY

When American Express's stock collapsed following the 1963 Salad Oil Scandal, Warren Buffett distinguished the company's genuine but bounded financial liability from its core, brand-dependent card and traveller's cheque businesses, which he assessed as largely undamaged. He tested that assessment by personally observing whether ordinary customers were still using American Express in everyday transactions, found they were, and committed roughly 40 percent of his partnership's assets to the stock at its depressed price — a bet that became one of the formative successes shaping his long-term investment philosophy.

STORY CHECK

ESTABLISHED

The 1963 Salad Oil Scandal, American Express's stock decline, and Buffett's substantial investment and its subsequent profitability are all extensively

documented in Buffett's own partnership letters and independent financial history.

POPULAR VERSION ADDS

The specific detail of Buffett personally visiting Omaha restaurants and retailers to observe customer card usage is a widely repeated element of the story, consistent with Buffett's known personal investment style, though the precise scope and locations of that observation are not independently documented beyond his own and his biographers' retellings.

UNCERTAIN

The exact percentage of partnership assets committed to the position is reported with some variation across sources (commonly cited around 40 percent, tied to Buffett's own stated internal limit for a single position).

VOCABULARY & PHRASAL VERBS

collateral (n., finance) an asset pledged as security for a loan, forfeited if the loan is not repaid. Warehouse receipts served as collateral for the fraudulent loans.

liability (n., finance) a financial obligation or debt a company is responsible for. American Express faced substantial liability from the fraud.

bounded (adj.) having a defined, limited extent, as opposed to open-ended or unlimited. The liability was serious but ultimately bounded in scope.

conflate (v.) to combine two or more distinct things into one, often mistakenly. The market's panic conflated two genuinely separate risks.

impairment (n., finance) a permanent reduction in the value or usefulness of an asset. He distinguished temporary trouble from permanent impairment.

hold up (phrasal verb) to remain intact or valid, especially under scrutiny. Consumer trust in the brand held up despite the scandal.

weather (v., figurative) to survive or come safely through a difficult period. American Express weathered the scandal's financial fallout.

bet on (phrasal verb) to commit resources based on a belief about a future outcome. Buffett bet heavily on the brand's underlying trust remaining intact.

pan out (phrasal verb) to develop or turn out successfully. The concentrated position eventually panned out well.

PRACTICE

Where the Story Breaks Down

Buffett had the capital, independence and personal conviction to make an unusually large, concentrated bet against prevailing market sentiment; the same probabilistic reasoning is available to anyone, but the willingness and ability to act on it at that scale is not.

Use the Language

Describe a situation where you separated a bounded, survivable risk from a deeper, more permanent one, using bounded, conflate and weather.

Discuss

What is a current crisis or piece of bad news in your own field where the market or public reaction might be conflating two genuinely separate risks?

Write

In 100 words, describe how you would personally test an analytical thesis about a current situation, the way Buffett tested his assessment by observing ordinary customer behaviour directly.

SOURCES

B Lowenstein, R., Buffett: The Making of an American Capitalist, Random House (1995).

A Buffett, W., letters to Buffett Partnership Ltd. limited partners (1963–1968), referencing the American Express position.

C Miller, N. C., The Great Salad Oil Swindle (1965), cross-referenced historical account of the 1963 fraud and its financial fallout.

THE CONTRAST

Story 47: Warren Buffett vs. Story 49: Invert the Problem (Munger)

Story 47: Warren Buffett: Separated a specific, bounded financial risk from the market's undifferentiated panic, in the middle of an actual live crisis.

Story 49: Invert the Problem (Munger): Deliberately imagines failure in advance, before a crisis exists, to find hidden risks ahead of time.

Is the more valuable skill staying calm and precise during an actual crisis, or imagining crises carefully enough in advance that fewer of them happen?

THIS CONNECTS TO

Story 21: The Fibre That Wasn't Supposed to Work (Kwolek)

LIMITATION

Kwolek trusted her own direct observation against her colleagues' skepticism. Buffett's chapter is almost the opposite skill — trusting a careful separation of risks against the market's undifferentiated panic.

Story 48: Sacrificing the Present for the Future (Bezos)

SAME MECHANISM

Both required making a decision based on a clear-eyed assessment of underlying fundamentals, against strong pressure from how the market or public was reacting in the moment.

Story 51: Predict From Similar Projects (Flyvbjerg) LIMITATION

Buffett separated two specific, distinct risks within a single live crisis. Flyvbjerg's method works at a different scale entirely — comparing a single project against a whole reference class of similar past projects.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

American Express's stock collapsed by roughly 50 percent after a subsidiary was exposed in a massive fraud involving falsified vegetable oil inventories, and the market treated the company as facing an existential threat.

T — Task

Buffett needed to assess whether the company's core business and brand reputation were actually as damaged as the stock price implied.

A — Action

He separated the specific, bounded financial liability from American Express's underlying business strength and customer trust, investigating directly rather than reacting to the market's general panic.

R — Result

Concluding the core business remained sound, he invested heavily in American Express while the stock was depressed, a decision that became one of his most successful early investments.

Level 2 — Make It Your Story

> *Tell me about a time when you separated a specific, bounded risk from a broader wave of panic or negative sentiment, and assessed it clearly on its own terms.*

Use STAR to answer. Think about: a crisis response, a market downturn, a moment you had to think independently of the general reaction.

TAKE THE IDEA FURTHER

> *When your organization faces a crisis, are you assessing the actual probability and scope of the specific damage, or reacting to the general atmosphere of alarm?*

48

Sacrificing the Present for the Future

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Amazon, for the majority of its first decade as a public company, reported little to no net profit, and in many years posted outright losses, a fact competitors and financial commentators frequently cited as evidence the company's business model was fundamentally unsound. Jeff Bezos's response to this criticism, repeated with notable consistency across years of shareholder letters, was not to dispute the numbers but to argue that the underlying framing — evaluating the company primarily on near-term profitability — was measuring the wrong thing entirely for the specific kind of business Amazon was deliberately choosing to build.

Bezos's stated strategy, articulated explicitly in his 1997 letter to shareholders (a letter Amazon has, notably, reprinted as an appendix to every subsequent annual shareholder letter since, an unusual practice signalling the letter's continued foundational importance to the company's own self-understanding), was to prioritize long-term market position and customer relationships over near-term profit, on the theory that a company willing to reinvest aggressively in growth, customer experience and infrastructure, even at the cost of reported profitability for years or, in some cases, over a decade, could establish a durable competitive position that a profit-maximizing competitor, answering to shareholders demanding nearer-term returns, would be structurally unable to match.

Amazon Web Services, the company's cloud computing division, illustrates the strategy's eventual payoff in a form that was almost entirely unanticipated

by outside observers at its 2006 launch: a business built initially to solve Amazon's own internal infrastructure scaling problems, then offered externally as a service to other companies facing the same problem, requiring years of infrastructure investment before it became a meaningfully profitable, standalone business line — one that eventually became one of the most profitable divisions of the entire company and, for a significant period, generated a disproportionate share of Amazon's total operating profit, subsidizing continued aggressive investment and thin margins across Amazon's original, much larger and more visible retail business.

The strategy is not without genuine risk or cost, and it is worth being precise about what made it viable rather than presenting it as a simple, universally applicable formula: Amazon required investors willing to accept years of thin or negative reported profit in exchange for growth and market position, a willingness that depended heavily on Bezos's sustained credibility in articulating and consistently executing the long-term thesis across many years, and on capital markets, during the relevant period, that were broadly receptive to funding exactly this kind of extended, deliberately unprofitable growth phase — conditions that have not held equally for every company attempting to imitate the same playbook.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What near-term profitability metric is your organization currently optimizing for, and would achieving it require sacrificing a long-term position you'd genuinely prefer to hold instead?*

KEY TAKEAWAYS

1. Bezos explicitly reframed the relevant measure of success from near-term profitability to long-term market position and customer relationships, and sustained that framing consistently across many years of shareholder communication.
2. Amazon Web Services began as an internal infrastructure solution before becoming an external product, requiring years of investment before its eventual outsized profitability became visible.
3. The strategy depended on specific conditions — sustained investor patience and credibility, receptive capital markets — that made it viable for Amazon specifically, and that have not automatically transferred to every company attempting to replicate the same approach.

SUMMARY

Jeff Bezos consistently prioritized Amazon's long-term market position and customer relationships over near-term profitability, a strategy articulated in his 1997 shareholder letter and reprinted as an appendix to every subsequent annual letter since. Amazon Web Services exemplifies the strategy's eventual payoff: an internal infrastructure solution turned external product that took years of investment before becoming one of the company's most profitable divisions. The approach required sustained investor patience and credibility that made it viable for Amazon specifically, conditions not automatically available to every company attempting to imitate it.

STORY CHECK

ESTABLISHED

Amazon's early years of minimal profitability, the 1997 shareholder letter and its continued annual reprinting, and AWS's 2006 launch and eventual

profitability are all documented in Amazon's own public financial filings and shareholder communications.

POPULAR VERSION ADDS

The strategy is sometimes presented as a universally applicable formula for any ambitious company; in practice, its viability depended on specific and not universally available conditions, including sustained investor patience across an unusually long period and capital market conditions receptive to funding extended unprofitability.

UNCERTAIN

The precise degree to which AWS's success was strategically anticipated from the outset versus an opportunistic expansion of an internal tool is debated, with some accounts emphasizing more deliberate strategic foresight than others.

VOCABULARY & PHRASAL VERBS

reinvest (v.) to put earned profit back into a business rather than distributing it. Amazon chose to reinvest aggressively rather than report profit.

durable (adj.) able to withstand wear or persist over time; long-lasting. The strategy aimed at a durable competitive position.

standalone (adj.) capable of functioning independently, not reliant on another system or business. AWS became a standalone, profitable business line.

subsidize (v.) to support financially, often by using profit from one area to cover costs in another. AWS profit subsidized the thin-margin retail business.

credibility (n.) the quality of being trusted or believed, especially over time. His sustained credibility with investors made the strategy viable.

payoff (n.) the eventual beneficial result of an investment or effort. AWS represents the long-term strategy's clearest payoff.

hold up (phrasal verb) to remain valid or functional under pressure or scrutiny. The long-term thesis held up across many years of scrutiny.

imitate (v.) to copy the actions or approach of another. Not every company imitating the playbook has succeeded with it.

answer to (phrase) to be accountable or responsible to a particular party. A profit-maximizing competitor answers to shareholders demanding nearer-term returns.

PRACTICE

Where the Story Breaks Down

Amazon's strategy depended on Bezos's sustained personal credibility and unusually patient capital markets over an extended period; an organization without an equivalent level of investor trust or an equivalently receptive funding environment may not be able to sustain the same multi-year unprofitable growth phase.

Use the Language

Describe a long-term position you or your organization might be sacrificing for near-term metrics, using reinvest, durable and payoff.

Discuss

What specific conditions would your own organization need — investor patience, market receptivity, sustained credibility — to successfully pursue an extended period of deliberate unprofitability?

Write

In 100 words, describe an internal tool or capability in your own organization that might, like AWS, have unrecognized potential as an external product or service.

SOURCES

A Bezos, J., "1997 Letter to Shareholders," Amazon.com, Inc. annual report (reprinted as an appendix in every subsequent annual shareholder letter).

C Amazon.com, Inc., public financial filings (10-K annual reports), documenting early profitability and AWS's growth trajectory.

B Stone, B., The Everything Store: Jeff Bezos and the Age of Amazon, Little, Brown (2013).

THE CONTRAST

Story 48: Jeff Bezos vs. Story 13: Solve Backward (Pólya)

Story 48: Jeff Bezos: Reframed the relevant measure of success itself, from near-term profit to long-term position, and let that fixed future point determine present decisions.

Story 13: Solve Backward (Pólya): Works backward from a fixed, desired goal to find the path that reaches it.

Does working backward from a long-term goal require redefining what success even means in the near term, or just tolerating worse near-term numbers?

THIS CONNECTS TO

Story 13: Solve Backward (Pólya) SAME MECHANISM

Both start from a fixed future point — a solved problem, a long-term market position — and let that fixed point determine what to do now, rather than extrapolating forward from the present.

Story 47: The Salad Oil Decision (Buffett) SAME MECHANISM

Both required making a decision based on a clear-eyed assessment of underlying fundamentals, against strong pressure from how the market or public was reacting in the moment.

Story 42: Abandoning the Business You Built (Intel/Grove)

LIMITATION

Grove's discipline was separating a decision from personal attachment to a business he'd built. Bezos's was separating a decision from short-term financial pressure — different kinds of pressure requiring different kinds of discipline.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Amazon reported little to no net profit through most of its first decade as a public company, which competitors and financial commentators frequently cited as evidence the business model was flawed.

T — Task

Bezos needed to decide whether to prioritize near-term profitability, which would satisfy public market expectations, or continue reinvesting aggressively in long-term growth.

A — Action

He explicitly reframed the relevant measure of success from near-term profit to long-term market position and customer relationships, and continued heavy reinvestment despite ongoing criticism.

R — Result

Amazon grew into a dominant company across multiple markets, and the strategy is now widely cited as a deliberate, successful long-term bet against short-term financial pressure.

Level 2 — Make It Your Story

> *Tell me about a time when you sacrificed a near-term metric in order to build toward a longer-term goal, despite pressure to show immediate results.*

Use STAR to answer. Think about: a budget trade-off, a metric you deprioritized, a long-term investment you defended against short-term criticism.

TAKE THE IDEA FURTHER

> *What near-term profitability metric is your organization currently optimizing for, and would achieving it require sacrificing a long-term position you actually care more about?*

49

Invert the Problem

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Charlie Munger, the longtime business partner of Warren Buffett at Berkshire Hathaway, built much of his public reputation for investment and decision-making advice around a specific intellectual habit he attributed, by his own repeated account, to the nineteenth-century mathematician Carl Gustav Jacobi: "invert, always invert" — the practice of approaching a difficult problem not by asking directly how to achieve a desired outcome, but by first asking how to guarantee the opposite, undesired outcome, and then systematically avoiding every path that leads there. The distinction Munger drew between these two framings was not merely rhetorical. Asking "how do I build a successful business" opens an enormous, poorly bounded space of possible strategies, most of them speculative and difficult to evaluate with confidence in advance. Asking instead "what would guarantee this business fails" — through fraud, through catastrophic cash mismanagement, through hiring people who are dishonest or incompetent, through entering a market with structurally hostile economics — tends to produce a shorter, more concrete, and more confidently identifiable list, because catastrophic failure modes are, in Munger's assessment, generally easier to specify precisely and recognize reliably than a full description of guaranteed success.

Munger applied the same inversion habit well beyond business strategy specifically, to personal conduct and decision-making broadly. His frequently cited remark on how to guarantee a miserable, unsuccessful life — "be unreliable, learn everything you possibly can from your own personal

experience and nothing from the good and bad experience of others, and go down and stay down when you get your first, second, or third severe reverse in the battle of life" — functioned as the deliberately inverted counterpart to conventional, positively framed advice about how to live well, on the theory that naming the specific behaviours that reliably produce ruin can be a more concrete and more actionable guide than abstract exhortations to pursue success.

The inversion technique's underlying logic connects to a broader idea in decision theory sometimes called a premortem: rather than asking, at the start of a project, what would make it succeed, a team instead imagines the project has already failed catastrophically, and works backward to identify the specific, concrete reasons why — a structured exercise that research in behavioural decision-making suggests can surface risks and blind spots that a forward-looking, success-oriented planning discussion, focused naturally on the reasons a plan should work, tends to systematically underweight or overlook entirely.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Rather than asking what would make your current project succeed, what would it take to guarantee it fails catastrophically — and are any of those specific conditions currently, quietly present?*

KEY TAKEAWAYS

1. Identifying the specific conditions that guarantee failure is often a more concrete and tractable exercise than directly specifying the full conditions

for success, because failure modes tend to be fewer and more clearly identifiable.

2. Munger applied inversion consistently across both business strategy and personal conduct, treating it as a general reasoning tool rather than a domain-specific investment technique.
3. The premortem technique — imagining a project has already failed and working backward to identify why — operationalizes the same inversion logic as a structured team exercise, designed to surface risks a purely forward-looking discussion tends to overlook.

SUMMARY

Charlie Munger popularized the habit of "inversion" — approaching difficult problems by first identifying what would guarantee failure, then systematically avoiding those paths, rather than only asking directly how to achieve success. He applied the technique to both business strategy and personal conduct, arguing that failure modes are generally easier to specify precisely than a complete description of success. The related premortem technique operationalizes the same logic as a structured exercise: imagining a project has already failed and working backward to identify the specific causes, surfacing risks a purely success-oriented discussion tends to miss.

STORY CHECK

ESTABLISHED

Munger's public advocacy of inversion, including his attribution of the underlying idea to Jacobi and his specific quotations on both business and personal conduct, is documented across his own speeches and published collections of his talks.

POPULAR VERSION ADDS

The premortem technique is sometimes presented as directly originating from Munger's inversion principle; in the decision-science literature, the premortem is more precisely credited to psychologist Gary Klein, who developed it as a distinct, formalized team exercise, though it shares clear conceptual lineage with the older inversion habit.

UNCERTAIN

The precise historical accuracy of Munger's attribution of "invert, always invert" to Jacobi, as a direct quotation from the mathematician, is difficult to independently verify against Jacobi's own surviving writings.

VOCABULARY & PHRASAL VERBS

invert (v.) to reverse the order, position or approach of something. He would invert the question rather than ask it directly.

bounded (adj.) having a defined, limited extent. A poorly bounded question opens an enormous space of possibilities.

premortem (n., decision science) a structured exercise imagining a project's failure in advance to identify risks. A premortem asks a team to imagine the project has already failed.

reverse (n., here) a setback or defeat, especially in the context of ongoing effort. He warned against staying down after a severe reverse.

operationalize (v.) to put an abstract idea into a specific, usable, practical form. The premortem operationalizes the same inversion logic as a team exercise.

underweight (v., figurative) to give insufficient consideration or importance to something. Forward-looking discussions tend to underweight risk.

tractable (adj.) able to be dealt with or solved relatively easily. Identifying failure conditions is often more tractable than specifying success.

blind spot (n., figurative) an area or aspect a person fails to notice or consider. A premortem surfaces blind spots a normal planning meeting misses.

work backward (phrasal verb) to reason from an end point toward the steps or causes that would lead to it. The team works backward from an imagined failure.

PRACTICE

Where the Story Breaks Down

Inversion works best for problems with a small number of clear, identifiable failure modes; some problems genuinely have failure conditions as numerous and ambiguous as their success conditions, in which case inverting the question doesn't actually simplify the analysis.

Use the Language

Describe a current project by inverting the question, identifying what would guarantee its failure, using invert, tractable and blind spot.

Discuss

Run a brief premortem on a real project you're involved in: imagine it has already failed, and list the specific reasons why.

Write

In 100 words, apply Munger's inversion technique to a personal or professional goal, listing the specific behaviours that would guarantee you fail to achieve it.

SOURCES

B Munger, C. T. (compiled by Kaufman, P. D.), Poor Charlie's Almanack: The Wit and Wisdom of Charles T. Munger, Donning Company (2005).

A Klein, G., "Performing a Project Premortem," Harvard Business Review (September 2007) — the formal, distinct academic origin of the premortem technique.

D Munger's attribution of "invert, always invert" to Carl Gustav Jacobi is drawn from Munger's own speeches and is not independently traceable to a specific surviving primary source from Jacobi himself.

THIS CONNECTS TO

Story 8: The Patent Clerk and the Train (Einstein) CONTRAST

See The Contrast in that chapter. In short: Einstein's thought experiment worked from pure reasoning, with no data at all; Munger's habit works by deliberately imagining failure, a different kind of imagined scenario entirely.

Story 18: The Square Peg and the Round Hole (Apollo 13) LIMITATION

The Apollo 13 team solved a single acute, survival-critical constraint under time pressure. Munger's method is a slower, standing habit for finding risk before a crisis — useful for very different situations.

Story 1: The Rocket That Was Too Expensive (Musk) LIMITATION

Musk's reframe asked what something actually costs. Munger's habit asks a different kind of question entirely — what guarantees failure — showing that "question the frame" takes more than one form.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Charlie Munger observed that people planning a project typically focus their attention entirely on what would make it succeed, which tends to leave real risks underexamined.

T — Task

He wanted a mental habit that would surface risks a purely forward-looking, success-oriented planning discussion tends to overlook.

A — Action

Rather than asking what would make a project succeed, he proposed asking the inverted question: what would guarantee it fails catastrophically, then working to avoid those specific conditions.

R — Result

The inversion technique became one of Munger's most widely cited mental models, used well beyond investing as a tool for surfacing risk in planning discussions.

Level 2 — Make It Your Story

> *Tell me about a time when deliberately imagining how a project could fail helped you avoid a real risk before it happened.*

Use STAR to answer. Think about: a pre-mortem, a risk assessment, a plan you stress-tested by imagining its failure modes.

TAKE THE IDEA FURTHER

> *Rather than asking what would make your current project succeed, what would it take to guarantee it fails catastrophically — and are any of those conditions already present?*

50

The Aircraft That Met a Changing Threat

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

The Avro Canada CF-105 Arrow was designed to answer a specific and, when it was specified, entirely real question: how would Canada intercept Soviet bombers crossing the Arctic toward North American cities? The requirement was written in the early 1950s, when the answer was manifestly an aircraft — a very fast, very high-flying, radar-equipped interceptor able to meet a bomber far from its target. What Avro built at Malton, Ontario, was among the most ambitious aircraft any country was attempting, flying on interim American engines while a Canadian engine, the Orenda Iroquois, was developed for the production version. It was rolled out on 4 October 1957, first flew on 25 March 1958, and five aircraft flew in all.

The question the Arrow had been built to answer was meanwhile being replaced by a different one, and the replacement was visible on the day of the rollout. On 4 October 1957 the Soviet Union launched Sputnik 1, weeks after announcing a successful test of an intercontinental ballistic missile. A missile on a ballistic trajectory cannot be intercepted by a fighter, however fast. This did not make the bomber threat disappear, and nobody in 1958 could say what share of the future Soviet arsenal would be bombers — but it made a decade-long commitment to a bomber interceptor a bet on a strategic environment visibly in motion. Costs were climbing, the United States and Britain had declined to buy, and the Royal Canadian Air Force remained the only customer.

On 20 February 1959, Prime Minister John Diefenbaker rose in the House of Commons and terminated both the Arrow and the Iroquois engine, citing cost and the Soviet shift toward missiles, and stating that the aircraft would be ineffective by 1962. The government had already ordered the American Bomarc missile system. The decision came roughly five weeks before a scheduled program review. More than 14,000 Avro and Orenda employees lost their jobs that afternoon, and within two months an order went out to destroy the five flying aircraft, the nearly complete sixth machine, the tooling and the drawings. In Canadian aviation the date is still called Black Friday.

What makes this a decision case rather than a grievance is that the record does not settle it in either direction. The popular verdict — that the world's finest aircraft was destroyed by political stupidity — asserts a counterfactual nobody can test: whether an aircraft that never entered production would have performed, sold and aged as its admirers believe. The tidy opposite verdict is no better supported either. The manned bomber threat did not vanish; Canada bought second-hand American CF-101 Voodoo interceptors from 1961 to do the job the Arrow had been designed for, and the Bomarc missiles bought in its place were withdrawn by the early 1970s. The difficult question the Arrow leaves behind is not who was right, but how anyone should reason about an irreversible commitment when the assumptions behind it are visibly changing and no reliable comparison exists.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *What major investment is your organization continuing because the original problem justified it — even though the environment that created that problem may since have changed?*

KEY TAKEAWAYS

1. The Arrow was not cancelled because it failed technically; it was cancelled because the strategic environment that had defined its purpose was changing while it was still being built.
2. Neither the decision nor its critics could appeal to a reference class: an interceptor facing an emerging missile threat had no comparable past cases to reason from, which is precisely the situation in which historical evidence stops helping.
3. The purchase of second-hand interceptors two years later, and the early withdrawal of the Bomarc missiles bought in the Arrow's place, show that the reasoning behind the cancellation was itself a forecast — and only partly borne out.

SUMMARY

The CF-105 Arrow was a technically formidable Canadian interceptor built to stop Soviet bombers; five flew, and it was rolled out on 4 October 1957, the day Sputnik 1 was launched. As the threat shifted toward ballistic missiles, costs rose and no export customer appeared, and on 20 February 1959 the Diefenbaker government cancelled the aircraft and its Canadian engine five weeks before a scheduled review, laying off more than 14,000 workers and then ordering everything destroyed. Whether that was correct cannot be established: the bomber threat persisted, Canada bought used American interceptors two years later, and the missiles purchased in the Arrow's place were withdrawn within little more than a decade.

STORY CHECK

ESTABLISHED

The 4 October 1957 rollout coinciding with Sputnik 1, the first flight on 25 March 1958, the five flying aircraft, the cancellation announced in the House of Commons on 20 February 1959 ahead of the scheduled review, the immediate layoffs and the subsequent destruction order are documented in Hansard and in Canadian museum and archival records.

POPULAR VERSION ADDS

The familiar account asserts that the Arrow was unquestionably the finest aircraft in the world and that its cancellation was self-evidently political incompetence — a judgment going well beyond what an aircraft cancelled before production can be shown to have proved, and which supplies confident motives for the destruction order that the record does not establish.

UNCERTAIN

The central counterfactual is genuinely unavailable: whether the Arrow would have matured into a technical, financial and export success cannot be determined, and historians continue to disagree about how much weight the changing threat, the cost projections and domestic politics each carried.

VOCABULARY & PHRASAL VERBS

interceptor (n.) a fighter aircraft designed to reach and engage incoming aircraft at long range. The Arrow was an interceptor built for the Arctic approaches.

strategic environment (n.) the overall set of threats, alliances and conditions in which decisions about defence or competition are made. The strategic environment changed faster than the aircraft could be built.

threat assessment (n.) a structured judgment about what dangers are most likely and most serious. The threat assessment shifted from bombers to missiles.

procurement (n.) the process by which an organization decides on and buys major equipment. The cancellation remains a defining case in Canadian procurement.

sunk cost (n., economics) money already spent that cannot be recovered, and which should not by itself justify further spending. The years already invested were a sunk cost, not an argument.

obsolete (adj.) no longer useful because something has superseded it. He argued the aircraft would be obsolete by 1962.

commit to (phrasal verb) to bind yourself to a course of action that is difficult to reverse. Canada had committed to a decade-long development program.

phase out (phrasal verb) to withdraw something gradually rather than all at once. The Bomarc missiles were phased out within little more than a decade.

reassess (v.) to examine a judgment again in the light of new information. The government reassessed the requirement after Sputnik.

PRACTICE

Where the Story Breaks Down

"Reassess when the assumptions change" is easy to state and treacherous to apply, because assumptions always appear to be changing and abandonment is itself expensive and irreversible. The Arrow offers no rule for telling a genuine shift in the environment from ordinary noise.

Use the Language

Describe a long-running commitment in your organization and the conditions that originally justified it, using strategic environment, sunk cost and reassess.

Discuss

When does persistence become discipline, and when does it become an unwillingness to recognize that the world has changed?

Write

In 100 words, describe a decision your organization would find almost impossible to reverse, and identify what evidence would have to appear before reversal became the right answer.

SOURCES

A House of Commons Debates (Hansard), 20 February 1959 — Prime Minister Diefenbaker's cancellation statement on the Arrow and Iroquois.

B Library and Archives Canada, CF-105 Arrow program records, noting the competing explanations for the cancellation.

B Canada Aviation and Space Museum (Ingenium), collection records for the Avro Canada CF-105 Arrow, the Orenda Iroquois engine and the McDonnell CF-101B Voodoo.

C Royal Canadian Air Force official history pages on the CF-105 and the CF-101 Voodoo in NORAD service.

D Whether the Arrow would have succeeded operationally or commercially, and the relative weight of cost, threat assessment and domestic politics in the decision, remain disputed among historians.

THE CONTRAST

Story 50: Avro Arrow vs. Story 45: The Goal That Had to Change (Shackleton)

Story 50: Avro Arrow: A project was cancelled not because it failed technically, but because the strategic environment that had justified it had changed.

Story 45: The Goal That Had to Change (Shackleton): A goal had to be abandoned mid-expedition because pursuing it further had become actively dangerous, not just difficult.

Is it harder to recognize that your goal has quietly become obsolete, or to act decisively once you do recognize it?

THIS CONNECTS TO

Story 45: The Goal That Had to Change (Shackleton) SAME MECHANISM
Both involve an original objective becoming obsolete or unachievable due to changed circumstances, and the real decision was recognizing that shift rather than continuing to pursue the original goal.

Story 43: Seeing the Smartphone Too Early (Nokia) LIMITATION
Nokia had a signal about the future arrive early and didn't act on it. The Arrow's story is closer to the opposite — a program built correctly for a threat that had already changed by the time it was ready.

Story 51: Predict From Similar Projects (Flyvbjerg) LIMITATION
The Arrow was cancelled due to a genuine shift in strategic circumstances. Flyvbjerg's chapter addresses a different, more common failure — projects that go over budget and behind schedule due to predictable planning optimism, not a change in the underlying threat.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

The Avro Arrow was designed to intercept Soviet bomber aircraft crossing the Arctic toward North America, answering a real and specific strategic question at the time it was specified.

T — Task

By the time the aircraft was ready, the relevant military threat had shifted from bomber aircraft toward intercontinental ballistic missiles, which the Arrow was not designed to address.

A — Action

The Canadian government cancelled the program, concluding that continuing to invest in an aircraft built for a threat that had already changed no longer made strategic sense.

R — Result

The cancellation remains controversial and debated, and whether it was the right call remains disputed of a changed strategic environment, not a failure of the aircraft's engineering.

Level 2 — Make It Your Story

> *Tell me about a time when you recommended stopping or changing a project, not because it had failed, but because the reason for doing it had changed.*

Use STAR to answer. Think about: a project cancelled for external reasons, a strategy pivot, a plan you stopped defending once circumstances shifted.

TAKE THE IDEA FURTHER

> *What major investment is your organization continuing because the original problem justified it — even though the environment that created that problem has since changed?*

51

Predict From Similar Projects

DOCUMENTATION LEVEL 1 — DIRECTLY DOCUMENTED
Verified through primary sources, patents, or independent contemporaneous accounts.

Bent Flyvbjerg, a Danish professor of major program management, spent much of his academic career compiling and analyzing a specific and uncomfortable dataset: the actual cost and schedule outcomes of large infrastructure megaprojects — tunnels, bridges, rail lines, stadiums — compared against the cost and schedule projections that had been used to justify approving them in the first place. His database, built from hundreds of such projects across many countries and decades, revealed a pattern too consistent to be coincidental: the overwhelming majority of infrastructure megaprojects run significantly over budget and behind schedule, with average cost overruns Flyvbjerg's research measured in the tens of percentage points, and some prominent individual projects — the Sydney Opera House, famously, came in more than 1,300 percent over its original budget — running over by orders of magnitude.

Flyvbjerg's explanation for this pattern connected directly to the same "inside view" problem Daniel Kahneman and Amos Tversky had identified in far smaller-scale projects: the specific planners and engineers estimating a new megaproject's cost and timeline naturally focus on that project's own specific technical details, its planned sequence of steps, and its designers' genuine confidence in their own particular execution plan — precisely the kind of forward-looking, project-specific reasoning that Kahneman's research had already shown reliably produces optimistic estimates, because it structurally excludes the base-rate information contained in how comparable past

projects, across the full genuine reference class, have actually turned out in practice.

Flyvbjerg's proposed and increasingly adopted solution, formalized as "reference class forecasting," operationalized the outside view as a specific, structured government planning requirement rather than leaving it as an informal, easily skipped suggestion: before approving a major infrastructure project's budget, planners are required to identify a genuine reference class of comparable completed past projects — comparable in scale, complexity and type — and use the actual, empirically measured cost and schedule overrun distribution from that reference class to adjust the new project's own budget and schedule projections upward, before any project-specific technical analysis is even considered, rather than relying solely on the specific project's own bottom-up technical cost estimate.

The United Kingdom's government adopted reference class forecasting as an official, mandatory element of its formal business case guidance for major public infrastructure projects, following Flyvbjerg's research and advocacy, one of the clearest cases of an academic finding about a systematic human judgment bias being translated into a specific, mandatory, structural government policy requirement rather than remaining a well-known but easily ignored piece of decision-making advice.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Before approving your organization's next major project budget, has anyone identified a genuine reference class of comparable past projects and checked the new estimate against their actual, measured outcomes — or is the estimate built entirely from this project's own specific plan?*

KEY TAKEAWAYS

1. Flyvbjerg's large-scale dataset confirmed, at the level of hundreds of real infrastructure megaprojects, the inside-view bias Kahneman and Tversky identified.
2. Reference class forecasting formalizes the outside view as a mandatory structural requirement — adjusting a project's own estimate using the actual measured outcomes of comparable past projects — rather than leaving it as an informal, skippable suggestion.
3. The UK government's adoption of reference class forecasting as mandatory policy for major infrastructure business cases is a rare, clear example of an academic finding about judgment bias being translated directly into binding institutional practice.

SUMMARY

Bent Flyvbjerg's research compiling cost and schedule outcomes across hundreds of infrastructure megaprojects found consistent, substantial cost overruns, driven by planners' reliance on optimistic, project-specific "inside view" estimates that excluded the base-rate information available from comparable past projects. His proposed solution, reference class forecasting, requires planners to identify a genuine reference class of comparable completed projects and adjust new budget estimates using that class's actual measured overrun distribution before finalizing a project's own technical cost estimate. The UK government adopted the approach as mandatory policy for major public infrastructure business cases.

STORY CHECK

ESTABLISHED

Flyvbjerg's megaproject cost-overrun research and the UK government's adoption of reference class forecasting as official infrastructure planning policy are documented in his own peer-reviewed publications and UK government business case guidance documents.

POPULAR VERSION ADDS

Individual project overrun figures, such as the frequently cited Sydney Opera House statistic, are well documented for that specific case, though average overrun percentages across Flyvbjerg's broader dataset vary somewhat depending on the specific project type and time period being analyzed.

UNCERTAIN

The precise degree to which reference class forecasting has measurably improved actual project outcomes since its adoption, as opposed to simply improving the accuracy of initial estimates without changing execution itself, is an active and ongoing area of research.

VOCABULARY & PHRASAL VERBS

megaproject (n.) an extremely large-scale infrastructure or construction project. Flyvbjerg studied cost overruns across hundreds of megaprojects.

cost overrun (n.) the amount by which actual project cost exceeds the originally budgeted amount. Cost overruns were measured in the tens of percentage points on average.

empirically (adv.) based on observation or measured evidence rather than theory alone. The adjustment used empirically measured overrun data.

bottom-up (adj., planning) built by estimating individual components and summing them, as opposed to using aggregate historical data. The project's own bottom-up estimate excluded reference class data.

reference class forecasting (n.) a method of estimating outcomes based on the measured track record of comparable past cases. Reference class forecasting became mandatory UK government policy.

translate into (phrasal verb) to convert or apply an idea in a different, more concrete form. The academic finding was translated into binding government policy.

come in over (phrase, budget) to exceed a planned budget or cost. The Sydney Opera House came in more than 1,300 percent over budget.

structurally exclude (phrase) to leave something out as an inherent feature of a method's design, not by oversight. The inside view structurally excludes base-rate information.

binding (adj.) legally or formally obligatory, not merely advisory. The policy became a binding requirement for infrastructure business cases.

PRACTICE

Where the Story Breaks Down

Reference class forecasting requires a genuinely comparable reference class to exist and be identifiable; for a truly novel type of project with no clear historical precedent, the method has much less reliable data to draw on, and identifying what counts as "comparable" becomes itself a matter of judgment.

Use the Language

Describe a major project or decision your organization is planning, and identify a genuine reference class of comparable past cases, using reference class forecasting, cost overrun and empirically.

Discuss

What would it take to make reference class forecasting a mandatory, binding step in your own organization's project approval process, rather than an optional suggestion?

Write

In 100 words, identify a project your organization completed in the past that could serve as part of a reference class for a similar project currently being planned, and note what its actual outcome was.

SOURCES

A Flyvbjerg, B., "From Nobel Prize to Project Management: Getting Risks Right," Project Management Journal 37(3), 5–15 (2006).

C HM Treasury (UK Government), The Green Book: Central Government Guidance on Appraisal and Evaluation, including mandatory reference class forecasting guidance for major infrastructure business cases.

A Flyvbjerg, B., Bruzelius, N. & Rothengatter, W., Megaprojects and Risk: An Anatomy of Ambition, Cambridge University Press (2003).

THE CONTRAST

Story 51: Flyvbjerg vs. Story 49: Invert the Problem (Munger)

Story 51: Flyvbjerg: Corrects optimism by looking outward at a large dataset of comparable past projects, rather than relying on imagination at all.

Story 49: Invert the Problem (Munger): Corrects overconfidence by looking inward, imagining a single project's own specific failure modes in advance.

Is the more reliable check on an overconfident plan a look at how similar projects actually went, or a disciplined imagining of how this specific one could fail?

THIS CONNECTS TO

Story 8: The Patent Clerk and the Train (Einstein) CONTRAST

Einstein's thought experiment worked from pure reasoning, with no data at all. Flyvbjerg's method is the opposite — pure data, no imagination — which shows there's more than one legitimate route to correcting an assumption everyone else accepts.

Story 13: Solve Backward (Pólya) SAME MECHANISM

Both are disciplined methods for narrowing an otherwise open-ended planning problem, rather than trusting an unstructured forecast.

Story 50: The Aircraft That Met a Changing Threat (Avro Arrow) LIMITATION

The Arrow was cancelled due to a genuine shift in strategic circumstances. Flyvbjerg's chapter addresses a different, more common failure — projects that go over budget and behind schedule due to predictable planning optimism, not a change in the underlying threat.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation

Large infrastructure and technology projects were, across a huge dataset Bent

Flyvbjerg compiled, consistently going over budget and behind schedule, and each individual project's planners generally believed their own project would be the exception.

T — Task

Flyvbjerg needed a way to correct for the specific, well-documented optimism bias built into forward-looking, project-specific reasoning.

A — Action

He proposed reference class forecasting: instead of estimating a project's cost and timeline from its own internal details, compare it against the actual outcomes of a large set of genuinely similar past projects.

R — Result

His large-scale dataset confirmed the same optimistic inside-view bias appeared with striking consistency across hundreds of real megaprojects, and reference class forecasting has since been adopted by some governments as a required planning check.

Level 2 — Make It Your Story

> *Tell me about a time when you used previous experience or data from similar projects to challenge an overly optimistic plan.*

Use STAR to answer. Think about: a budget estimate you corrected, a timeline you challenged, a project comparison that changed a plan.

TAKE THE IDEA FURTHER

> *Before approving your organization's next major project budget, has anyone identified a genuine reference class of comparable past projects to check the estimate against?*

52

Design for the World Coming

DOCUMENTATION LEVEL 2 — MIXED

Some claims well documented, others contested — the chapter says which is which.

Richard Buckminster Fuller, an American architect, inventor and self-described "comprehensive anticipatory design scientist," spent much of a long and unconventional career pursuing a specific and consistently stated design philosophy: rather than designing primarily to solve the problems and satisfy the constraints of the present moment, Fuller argued that serious design should anticipate the resource constraints, population pressures and technological possibilities of decades into the future, deliberately designing for conditions that did not yet exist but that Fuller believed were reliably predictable in their general direction, even if their precise details remained uncertain. Fuller's geodesic dome, patented in 1954, is his best-known and most commercially successful realized design, built on a structural principle — distributing structural stress evenly across a framework of interconnected triangles — that allowed geodesic domes to enclose very large volumes of space using comparatively little structural material relative to the enclosed volume, a genuinely significant engineering efficiency for large-span structures, and one that has been applied successfully across a wide range of real-world contexts from radar installations to greenhouses to the well-known Spaceship Earth structure at Walt Disney World's EPCOT.

Not all of Fuller's specific proposals achieved comparable practical success, and an honest account of his legacy needs to include this directly rather than presenting him only through his most successful realized work. His Dymaxion car, a three-wheeled vehicle designed in the 1930s around principles of aerodynamic efficiency and interior space maximization, showed

real early promise but was ultimately abandoned as a commercial product following a fatal accident during a public demonstration and persistent, unresolved concerns about the vehicle's stability at speed — concerns that later engineering analysis has suggested were substantively founded rather than merely incidental bad luck. Fuller's proposed Dymaxion house, a mass-manufacturable, resource-efficient prefabricated home intended for widespread production, similarly never reached the large-scale commercial production Fuller had envisioned for it.

Fuller's broader and more durable influence operates less through any single specific realized product and more through a persistent set of design questions he insistently posed and that continue to shape environmentally and resource-conscious design thinking well beyond his own specific proposals: how much material does a structure actually require to perform its function, considered from first engineering principles rather than inherited convention; what resource and population conditions will a design actually need to function well under, decades after its construction, rather than merely at the moment of its completion; and whether a design is contributing to what Fuller termed "ephemeralization" — accomplishing more, functionally, with progressively less material and energy over time — a design ambition considerably easier to state clearly as a goal than to consistently achieve in practice, as Fuller's own genuinely mixed record of successes and failures across his long career itself demonstrates.

EXECUTIVE CLASS LENS

Connect the idea to your professional life, university, organization, team, or future career.

> *Is your organization designing primarily for the constraints and conditions of the present moment, or explicitly and deliberately for the*

> *resource and demand conditions you expect to exist a decade from now?*

KEY TAKEAWAYS

1. Fuller's geodesic dome achieved genuine, durable, real-world engineering and commercial success by distributing structural stress efficiently across a triangulated framework, using materials far more sparingly than conventional large-span structures.
2. Not all of Fuller's anticipatory design proposals succeeded — the Dymaxion car and Dymaxion house both failed to reach the commercial scale he envisioned, for reasons including a genuine, substantively founded safety concern.
3. Fuller's most durable influence operates through the design questions he consistently posed — material efficiency, decades-ahead resource conditions, doing more with less over time — rather than through any single specific realized product being universally successful.

SUMMARY

Buckminster Fuller pursued a design philosophy of anticipating future resource and population conditions rather than designing only for present-moment constraints, achieving genuine, durable success with the geodesic dome's efficient triangulated structure, while other proposals — the Dymaxion car and Dymaxion house — failed to reach comparable commercial success, including a documented, substantively founded safety concern with the car's stability. Fuller's more lasting legacy operates through the specific, still-relevant design questions he consistently raised about material efficiency and long-term resource conditions, rather than through uniform success across all of his individual proposals.

STORY CHECK

ESTABLISHED

The 1954 geodesic dome patent, its structural efficiency principle, and its wide range of successful real-world applications are documented in patent records and independent architectural and engineering history. The Dymaxion car's fatal 1933 demonstration accident is documented in period press coverage and subsequent historical accounts.

POPULAR VERSION ADDS

Fuller is sometimes presented in popularized retellings as an unqualified visionary genius whose every idea was ahead of its time; a more complete account includes his genuinely mixed record, with significant proposals that did not achieve their intended commercial or practical success.

UNCERTAIN

The precise causal contribution of the Dymaxion car's specific design features to the 1933 fatal accident, versus other contributing factors present at the time, has been debated by later engineering analysts, though concerns about the vehicle's high-speed stability are consistently documented as a contemporaneous concern independent of the specific accident's exact cause.

VOCABULARY & PHRASAL VERBS

anticipatory (adj.) relating to expectation or preparation for future events. Fuller described himself as an anticipatory design scientist.

geodesic (adj.) relating to a structure built from a framework of interconnected triangles distributing stress evenly. The geodesic dome distributes stress across a triangulated framework.

span (n., structural engineering) the distance a structure covers without intermediate support. Geodesic domes are efficient for large-span structures.

prefabricated (adj.) manufactured in standardized sections for rapid assembly elsewhere. The Dymaxion house was intended as a prefabricated, mass-produced home.

ephemeralization (n., Fuller's term) accomplishing more with progressively less material and energy over time. Fuller called this principle ephemeralization.

substantively founded (phrase) based on genuine, meaningful evidence rather than incidental or coincidental factors. The stability concerns were later judged substantively founded.

envision (v.) to imagine or plan a future possibility. He envisioned mass production the house never actually reached.

fall short of (phrase) to fail to reach an intended standard or goal. Several proposals fell short of Fuller's own commercial ambitions for them.

pose (a question) (v.) to formulate and present a question for consideration. Fuller consistently posed the same design questions across his career.

PRACTICE

Where the Story Breaks Down

Fuller's anticipatory design philosophy is easier to state as a general principle than to apply reliably in specific cases — his own record shows that anticipating future conditions correctly in the abstract (resource efficiency, doing more with less) did not guarantee that any particular specific design proposal would actually succeed commercially or practically.

Use the Language

Describe a product or process your organization is designing primarily for present conditions, and consider what anticipating future conditions would change, using anticipatory, envision and fall short of.

Discuss

What is the difference between genuinely useful anticipatory design and speculative design that simply guesses wrong about the future? How would you tell them apart before the outcome is known?

Write

In 100 words, apply Fuller's "ephemeralization" question — doing more with progressively less — to a real product, process or system in your own field, and describe what a more materially efficient version might look like.

SOURCES

A Fuller, R. B., US Patent 2,682,235, "Building Construction" (the geodesic dome patent, 1954).

B Baldwin, J., BuckyWorks: Buckminster Fuller's Ideas for Today, John Wiley & Sons (1996).

D Contemporary press coverage of the 1933 Dymaxion car demonstration accident; later engineering assessments of the vehicle's stability concerns are discussed with some variation across independent automotive history sources.

THE CONTRAST

Story 52: Buckminster Fuller vs. Story 48: Sacrificing the Present for the Future (Bezos)

Story 52: Buckminster Fuller: Designed explicitly for resource and population conditions decades in the future, accepting that some specific proposals would fail to reach commercial scale in the present.

Story 48: Sacrificing the Present for the Future (Bezos): Reframed the relevant measure of success from near-term profit to long-term market position, within a single ongoing company.

Is designing for the distant future best done as a personal design philosophy across many independent projects, or as a sustained strategy within one organization?

THIS CONNECTS TO

Story 38: The Farmer Who Did Less (Fukuoka) SAME MECHANISM
Both pursued an unconventional, long-horizon design philosophy that looked impractical by the standards of their era, and required sustained conviction well before mainstream validation.

Story 48: Sacrificing the Present for the Future (Bezos)
SAME MECHANISM
Both required judging success by a longer time horizon than the people around them were using, and accepting present-day skepticism as the cost of that horizon.

Story 43: Seeing the Smartphone Too Early (Nokia) LIMITATION
Fuller's anticipatory designs sometimes failed to reach commercial scale despite being right in principle. Nokia's chapter shows the organizational version of the same risk — being early is not the same as being rewarded for it.

INTERVIEW & WORKPLACE PRACTICE

Level 1 — Retell the Case

Imagine an interviewer asks you to explain this story. Use the STAR method.

S — Situation
Richard Buckminster Fuller believed that design should anticipate future

resource constraints and population pressures rather than simply solving the problems of the present moment.

T — Task

He needed to demonstrate this anticipatory design philosophy through real, engineered structures and products, not just through argument.

A — Action

He developed the geodesic dome, a structure distributing stress efficiently across a triangulated framework using far less material than conventional large-span construction, alongside other anticipatory designs including the Dymaxion car and house.

R — Result

The geodesic dome achieved genuine, durable commercial and engineering success; some of his other proposals, including the Dymaxion car, failed to reach commercial scale, but his underlying design questions about material efficiency continued to influence design long after his death.

Level 2 — Make It Your Story

> *Tell me about a time when you designed or planned for future conditions rather than just the constraints of the present moment.*

Use STAR to answer. Think about: a scalable system, a future-proofed design, a decision you made anticipating conditions years ahead.

TAKE THE IDEA FURTHER

> *Is your organization designing primarily for the constraints and conditions of the present moment, or explicitly and deliberately for the resource and market conditions of a decade from now?*

About the Author

Ron Taylor is a Canadian communication consultant with three decades of experience teaching English through individual, personalized lessons. Over the years he has taught students from around the world; today he works exclusively with Brazilian professionals building careers and lives in Canada.

His lessons are all one-on-one and online. They cover job interviews, the job search and the everyday English of Canadian workplaces. The case studies in this series grew out of those lessons: real decisions, discussed in plain business English, with the vocabulary professionals need to talk about them.

He is the founder of **Executive Class**, a private communication consulting practice for Brazilian professionals in Canada, and the author of a library of free ebooks, including *Executive Class*, *The Brazilian Edge* and *52 Hours to a Winning Interview*.

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